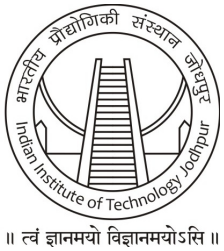
Content Based Analysis and Retrieval of Architectural Floor plans

A Thesis submitted by
Divya Sharma

in partial fulfillment of the requirements for the award of the degree of
Doctor of Philosophy



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Declaration

I hereby declare that the work presented in this Thesis titled *Content Based Analysis and Retrieval of Architectural Floor Plans* submitted to the Indian Institute of Technology Jodhpur in partial fulfillment of the requirements for the award of the degree of Doctor of Philosophy, is a bonafide record of the research work carried out under the supervision of Dr. Chiranjoy Chattopadhyay. The contents of this thesis in full or in parts, have not been submitted to, and will not be submitted by me to, any other Institute or University in India or abroad for the award of any degree or diploma.

Divya Sharma
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Certificate

This is to certify that the thesis titled *Content Based Analysis and Retrieval of Architectural Floor Plans*, submitted by *Divya Sharma (P15VSS002)* to the Indian Institute of Technology Jodhpur for the award of the degree of *Doctor of Philosophy*, is a bonafide record of the research work done by her under my supervision. To the best of my knowledge, the contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Chiranjoy Chattopadhyay
Ph.D. Thesis Supervisor

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List of Symbols

N	Total number of samples in a dataset
$\mathscr{I}$	Input floor plan image
$\oplus$	Dilation
$\ominus$	Erosion
S_a	Structuring Element
$\mathscr{I}_w$	Input floor plan image without walls
F	Furniture template signature
$\mathscr{C}$	Furniture count
$\mathscr{T}$	Furniture type set
$\mathscr{B}$	Blobs obtained through morphological fill operation
$\mathbf{CC()}$	Connected Component Operator
$\lvert\cdot\rvert$	Cardinality
C	Connected components of a blob
$\mathscr{A}$	Set containing areas of the connected components in a blob
$S()$	Signature
G_k	Topology graph representing adjacencies in k^{th} floor plan
V_k	Vertex set of a graph representing k^{th} floor plan
E_k	Edge set of a graph representing k^{th} floor plan
λ_k	Eigen values
$\vec{\phi}_k$	Eigen vectors
Adj_k	Adjacency matrix of the k^{th} graph
$\vec{F}_k$	Feature vector
Cv	Covariance Matrix
$cost_{qm}$	Match cost between two floor plan samples q and m
$\mathscr{C}^u$	Unique furniture inside each floor plan
Ar	Carpet Area of a room
L_k	k^{th} layout
$\lvert.\rvert$	Absolute value of a function
$A(\cdot)$	Carpet Area Ratio of a room corresponding to a layout
$e(\cdot,\cdot)$	Edit Distance
$\rho(\cdot)$	Room adjacency string matching score
$\psi(\cdot)$	Carpet Area Ratio matching score
$\phi(\cdot)$	Difference in number of furniture
$\theta(\cdot)$	Difference in type of furniture
F^+	Cumulative match score of a feature be it θ, ψ, ϕ, ρ
η	Weight Coefficients
I	Image domain
S	Sketch domain
f_I^i	i^{th} floor plan image sample
f_S^j	j^{th} floor plan sketch sample from S-ROBIN
$\mathbf{F(.)}$	Mapping function mapping images to sketches
$\mathbf{G(.)}$	Mapping function mapping sketches to images
D_S	Adversarial Discriminator for sketches
D_I	Adversarial Discriminator for images

$f_S \rightarrow p(f_S)$	Probability distribution of a sample $f_S \in S$
$f_I \rightarrow p(f_I)$	Probability distribution of a sample $f_I \in I$
$S2I_A(.)$	Adversarial loss function for mapping a floor plan sketch to an image
$I2S_A(.)$	Adversarial loss function for mapping a floor plan image to a sketch
$CCL(.)$	Cycle consistency loss
$X(.)$	Final objective function in Cycle GAN
α	Hyperparameter in Cycle GAN network
$\mathscr{E}$	Encoder
$\mathscr{D}$	Decoder

List of Abbreviations

ARD	Adjacent Room Detection
ARG	Attributed Relational Graphs
CAD	Computer Aided Design
CAR	Carpet Area Ratio
CBIR	Content Based Image Retrieval
CMS	Cumulative Match Score
CNN	Convolutional Neural Network
CVC-FP	Computer Vision Center- FLoor Plan
DAR	Document Analysis and Retrieval
DIA	Document Image Analysis
DNN	Deep Neural Networks
D-SIFT	Dense Scale Invariant Feature Transform
FCR	Furniture Composition Record
GAN	Generative Adversarial Network
GI	Generated Images
GS	Generated Sketches
GUI	Graphical User Interface
HOG	Histogram of Gradients
MAP	Mean Average Precision
OASIS	Online Algorithm for Scalable Image Similarity
PR	Precision Recall
RAS	Room Adjacency String
RDM	Room Decor Matching
ReLU	Rectified Linear Unit
RLH	Run Length Histogram
RLM	Room Layout Matching
ROBIN	Repository Of BuildIng plaNs
SESYD	Systems Evaluation SYnthetic Documents
SIFT	Scale Invariant Feature Transform
SOA	State of the Art
S-ROBIN	Sketched Repository Of BuildIng plaNs

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Abstract

Architects often refer to existing layouts, while designing new projects. This process aids in providing insight into how such similar architectural situations were handled in the past. By studying one or several previous reference projects the architect tries to derive a solution for the current problem. The manual look-up process for such similar projects through the layouts can be rather cumbersome. With several architectural projects archived in digital form, the research and development of fast automatic retrieval techniques in floor plans is the need of the hour.

Floor plan analysis is a special case of document image understanding. It aims at extracting semantic and structural details of an architectural layout by analysis of the 2D image of the floor plan. Symbol spotting and retrieval in architectural layouts have been solved as individual problems in the past. Moreover, both the image and sketch have been used as modalities for the symbol spotting task. Thus, upon performing a keen analysis related to the existing work in the area of architectural floor plans, it was concluded that retrieval in this particular domain is a challenging yet less researched area and has a variety of applications in today's digital scenario. Specific requirements by buyers during property rent/sale can be met using a composite, automated framework that takes into account semantics, as well as the content inside a floor plan for retrieval. Existing floor plan datasets like Systems Evaluation SYnthetic Documents (SESYD) and Computer Vision Center- Floor Plan (CVC-FP) dataset are not fit for a retrieval task as they don't suffice in number and variation in the floor plan samples for the task of floor plan retrieval. Keeping in mind the current status of floor plan analysis research, two publicly available benchmark datasets Repository Of BuildIng plaNs (ROBIN) and Sketched-Repository Of BuildIng plaNs (S-ROBIN) are proposed in this thesis that will aid research for the community in the area of floor plan analysis and retrieval.

In this thesis, techniques to analyse architectural floor plans and extraction of different features to aid in content based retrieval in floor plans is proposed. In one such attempt, retrieval in floor plans under the query by example paradigm, based on similar overall layout designs as well as the arrangement of the room decor present in the layouts, is proposed. Here query is taken as a floor plan image. A room layout segmentation and adjacent room detection algorithm is presented to represent layouts as an undirected graph. The vertices of the graph represent the rooms, while the edges represent the connectivity between them. Also a novel graph spectral embedding feature is proposed to uniquely represent the layout of the architectural floor plan. This helps in effective and efficient matching of the room layouts. To match the semantic similarity between a pair of floor plans, a two stage matching technique is proposed and high retrieval accuracy is obtained. An interactive graphical user interface to aid users to select, analyse and retrieve similar floor plans is also proposed in this thesis.

In another attempt a Convolutional Neural Network (CNN) framework to extract both low and high level semantic features is proposed for floor plan retrieval. Experiments were conducted on publicly available datasets as well as ROBIN. The key contributions in the proposed approach are, a novel deep learning framework to retrieve similar floor plan layouts from repository. Also, in this part of the work the effect of the individual deep convolutional neural network layers for floor plan retrieval task is analysed.

Experiments have shown that the deep learning frameworks work very well when the target image itself has a lot of features. Examples include natural images, textual documents, etc. On the other hand for images, which are not that feature rich, the off-the-shelf CNNs are not that effective.

Moreover, deep features mostly capture the global similarity in an image. It was envisaged that effective combination of domain specific features may give superior results. The effect of combining various extracted features in a weighted manner to aid in giving preference to a certain feature while retrieval is also proposed. A novel end-to-end framework for extracting high level semantic features like area and room-wise decor arrangement for the task of fine grained retrieval is proposed. Further, a technique to perform feature fusion to aggregate high-level semantic features extracted is also proposed. Weighted feature fusion helps in setting preferences to particular characteristics of the floor plan while retrieval and satisfying specific user demands.

In an attempt to explore other query modes than query by example in the form of floor plan image, sketch based retrieval is proposed. Sketch based retrieval comes with its own set of challenges in terms of both representation and recognition. However, this mode of query can aid in better correspondence while capturing the user's intent in a query. A composite network comprising of Cyclic Generative Adversarial Networks (Cycle-GAN) along with CNN is proposed to bridge the gap between sketch and image domains while retrieval. An improved approach using autencoders in conjunction with Cyclic Generative Adversarial Networks is proposed, which outperforms all other state-of-the-art techniques for sketch based floor plan retrieval by using an efficient domain mapping approach.

1
Introduction

In today's time, despite the increasing trend in the creation and processing of digital data, handwritten or paper based storage is still active and growing [Hilbert and López, 2011]. This has in turn led to a growth in the research and applications in Document Image Analysis (DIA) systems. The primary purpose of any DIA system is, automated storage, extraction, and processing of the information present in documents. The research in DIA lies at a very interesting intersection of individual fields like pattern recognition, image processing, machine learning, artificial intelligence, storage systems and linguistics and encompasses several sub research areas, like document categorization and understanding, hand-written and typed text recognition, graphics recognition and graphics retrieval. Figure 1.1 depicts a typical example and various types of tasks [Kasturi *et al.*, 2002] that fall under the umbrella of DIA.

Graphics recognition is a sub-domain in Document Image Analysis which aims at analyzing the content of graphical documents. A graphical document can be defined as a virtual or a physical object created to convey a piece of information. This piece can manifest itself in a picture or a diagrammatic form, conforming to a graphical language. Examples of graphical documents can be engineering drawings, flowcharts, maps, diagrams, architectural floor plans, etc. Graphical languages are human-made communication systems enabling the expression of complex semantic concepts via graphical information. In graphical languages, the vocabulary comprises of a symbol set, and the grammatical rules are context based relations between the symbols in the symbol set which provides the complete meaning to the given graphics in the document. Thus, the understanding of a graphical document comprises of both parsing of, the unique, isolated symbols as well as the relationship between their context. Still, graphical languages, have variations in terms of, the nature of reader, the difficulty level in the comprehension of the enclosed information, design and creativity matters, etc. Apart from recognition of graphical symbols in a graphical document, the next important thing is retrieval of such documents based on the semantic content derived from the symbols identified in those documents.

The amount of data available in digital format has grown exponentially in recent years. Data, specifically in the form of images are generated in large quantities and stored every day, and this results in a massive, distributed, and in most cases unstructured multimedia information. There are mainly two ways using which one can access images from a database; they are:

1. Text based retrieval: In this case, all the images are tagged using manual or semi-automatic tools during the cataloging stage. Indexes are formed over the database based on these tags. After that a suitable data structure is created that suits the retrieval task. During the retrieval stage, traditional text based retrieval techniques are applied to get the result based on the user query.

2. Content-based retrieval: In this type of retrieval method the search is performed by specifying the actual content of the image data in terms of query image, sketch etc. For example, a Content Based Retrieval System, which is described as the process of retrieving desired images from a large collection on the basis of syntactical image features. Such a system uses

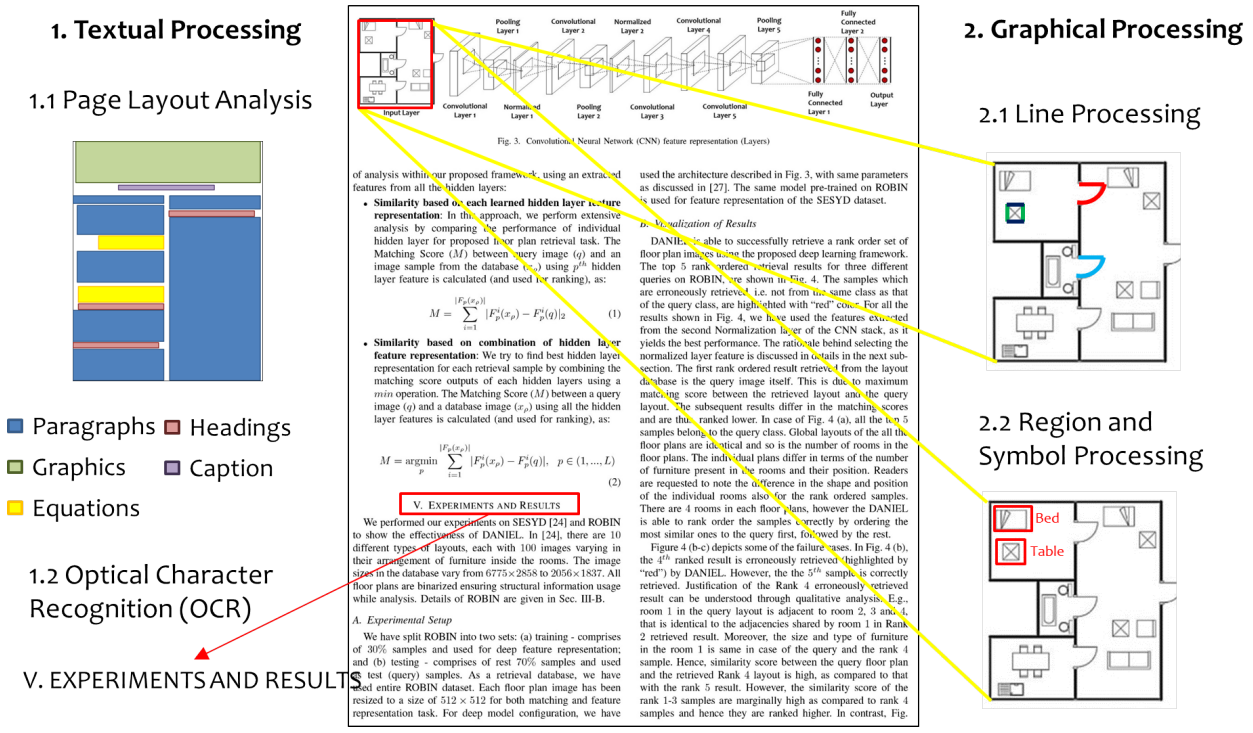


Figure 1.1. : Illustration of various subareas of document image analysis (DIA) on a sample scanned document [Sharma *et al.*, 2017].

different modes of query and assesses a wide variety of of content from the given query for retrieval purposes.

Design of a Content Based Image Retrieval (CBIR) system (refer Fig. 1.2) is a complex task and includes concepts from various academic disciplines such as image processing, computer vision, pattern recognition, machine learning, visual perception etc. Content-based means that the search analyzes the contents of the image rather than the metadata such as keywords, tags, or descriptions associated with the image. The term "content" in this context might refer to colors, shapes, textures, or any other information that can be derived from the image itself. CBIR is desirable because searches that rely purely on metadata are dependent on annotation quality and completeness. Having humans manually annotate images by entering keywords or metadata in a large database can be time consuming and may not capture the keywords desired to describe the image. The process of CBIR involves retrieving similar images in response to user's search queries, using the inherent properties (features) extracted from the image itself. These features are mapped onto a feature space and the images with the most closely placed features in the feature space (depends on a certain metric) are deemed similar and retrieved from the database. An illustration of the same is shown in Fig. 1.2, where the context is chosen to be natural images. One more key aspect that plays an important role in a CBIR system is the modality of the query to the system. As illustrated in Fig. 1.3, query can be in the form of text, image, sketch, colour layout which depicts the colour combination that a user wants to look in the retrieved images or a concept layout where images that contain multiple related objects as specified in the query needs to be retrieved. However, the quality of the retrieval result of any CBIR system depends on the quality of content representation in the feature database. In a content based retrieval system, given a query and a database to search, the following questions need to be addressed:

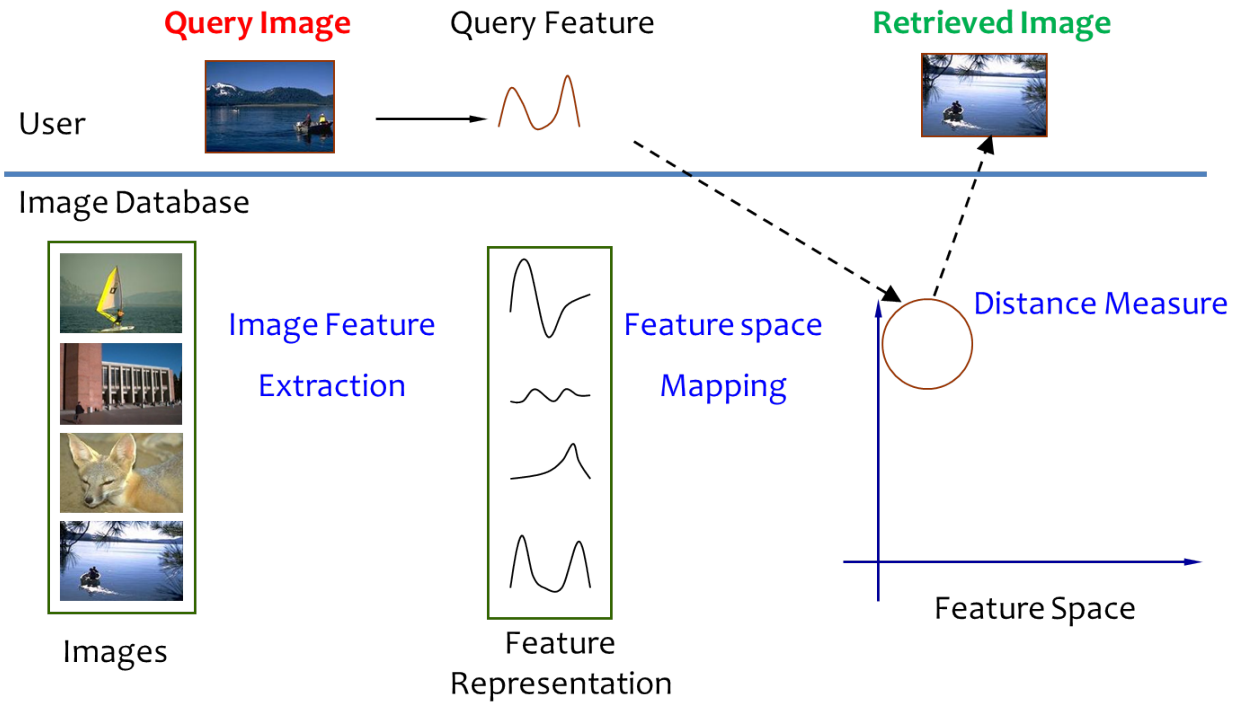


Figure 1.2. : An illustration of a basic Content Based Image Retrieval system.

1. What are the kinds of databases? This pertains to what kind of images are there in the database and what are the categories that the images have been classified into in the database.

2. What are the kind of queries? As illustrated in Fig. 1.3, the mode of the query can be an image, sketch, text, colour layout or a concept layout. Since each query mode has its own challenge, choice of modality is a critical research issue.

3. In which category does this image belong? What constitutes a category in the database and how one generates the feature set which represents each category present in the database distinctively.

4. In the given category, what are the "most similar" images in the database, with respect to the given query? Development of an efficient matching scheme, that helps evaluate the feature level similarity between the query as well as the categorized database images efficiently.

5. How does one make such searches efficient? What indexing mechanisms and efficient matching schemes can be used to make searching across databases fast and accurate.

Existing CBIR systems have been used for retrieval and have found applications in Art collections, Crime prevention, Geographical information, Remote sensing systems, Medical diagnosis, Retail catalogs, Face Finding, etc. Document image retrieval is a branch of CBIR that deals with extracting meaningful content out of document images such as, bank forms, cheques, scanned documents and then retrieving them for various applications. These days a great number of documents are scanned and archived in the form of digital images in digital libraries to make them available and accessible to the web. Therefore, a potential application of document image retrieval can be to make an online digital library where users would be able to do a search on a set of keywords, and get a list of relevant articles across the web, for viewing or printing. Architectural

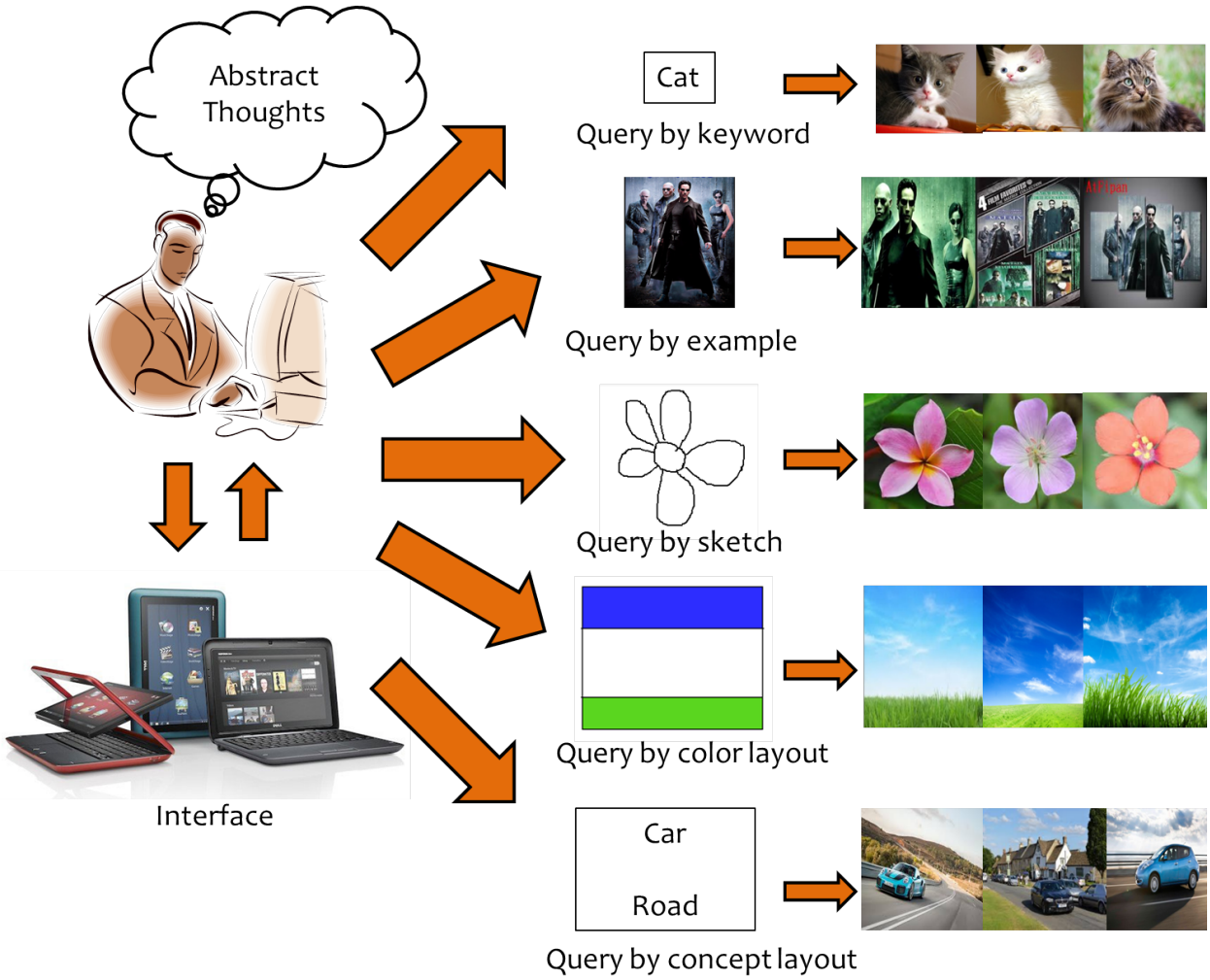


Figure 1.3. : Illustration of various query modes along with their corresponding retrieval results

floor plans are also a special type of documents, which have yet not been analysed with respect to its content and then retrieved based on similarity. Representing the inherent content present in a floor plan image/sketch automatically and identifying similar images from the database poses one of the fundamental challenge for the retrieval task. How to overcome the inherent challenges and come up with an efficient content based retrieval system specifically designed to represent content in architectural floor plans and retrieve thereafter, is a point which this thesis will reinforce.

1.1 MOTIVATION OF THE PROBLEM

The motivation behind the problem of retrieving similar floor plans by analysing meaningful content originated from the current needs and demands of the real-estate industry. To build a dream house in reality, customers visit architects, and specify the requirement "in words". An architect manually studies previous reference projects and suggests the best house and also searches for inspiration or a solution without having the final design at hand. This is an adhoc and tedious process. With the increase in number of online platforms, fast and automatic techniques for retrieval of similar projects are need of the hour. However, "*How to specify the requirements*

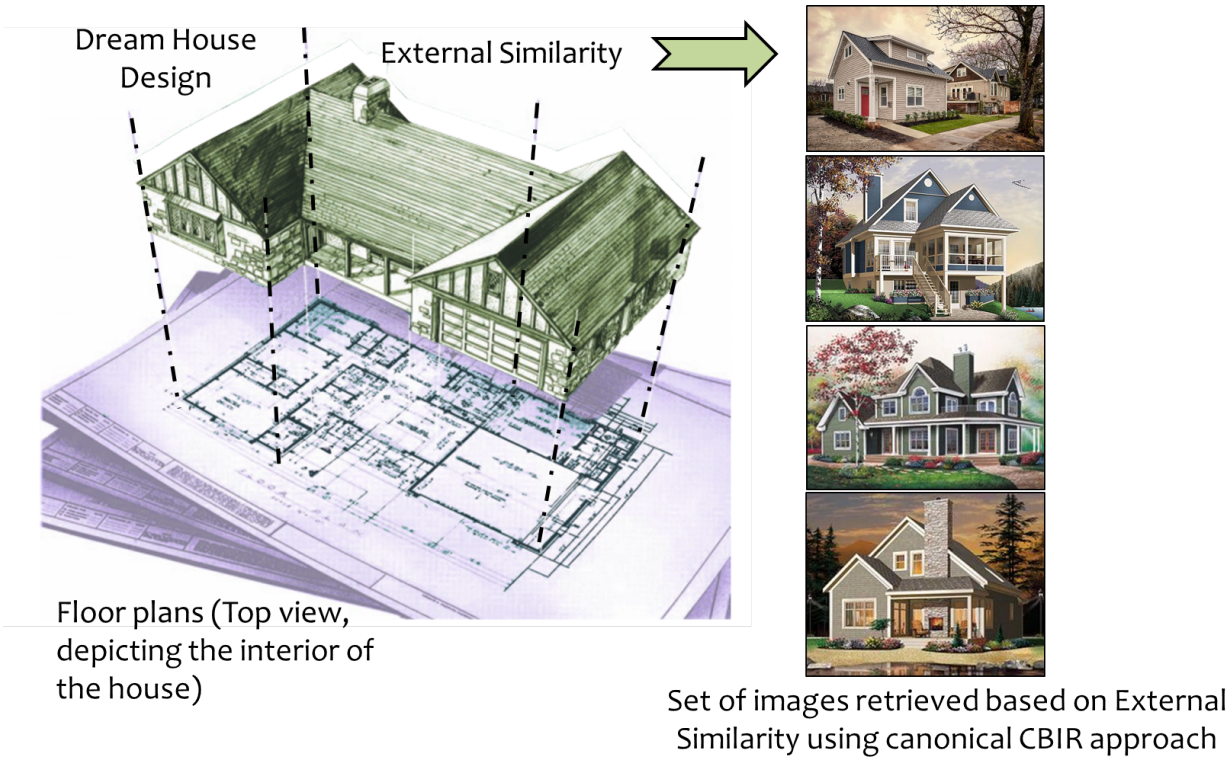


Figure 1.4. : Motivation of the problem.

of a dream house?" is still unclear. A naive solution is to follow the Content Based Image Retrieval (CBIR) approach to cater external similarity. Figure 1.4 shows such a scenario. It can be observed that the retrieved images of houses from the database are externally similar to that of the query image. However, there is no guarantee that these houses would match in their internal similarity as highlighted by the projection line and the floor plan.

However, the requirements of a user are usually pointing towards the internal structure of the home and the decors to be placed within each room. Consider a typical example where the requirement is specified as, "*the house must have* 4 *rooms, with* 2 *bedrooms with attached bathroom,* 1 *drawing room and* 1 *kitchen. The bedrooms must face towards west,...*", then interior similarities along with external structure similarities are also to be matched. Instead of using textual (subjective) query, a pictorial representation is always a better option. A floor plan is a fundamental form of architectural diagram, that gives a view of a house equivalent to a 2D horizontal cross section along with details of the interiors. Figure 1.5 depicts some examples of floor plan images corresponding to various types of houses.

Even though, floor plans are "*graphical language of communication*", however, investigation is required to answer "*how to match a pair of floor plans?*". In literature, researchers have carried out floor plan analysis in terms of symbol spotting and semantic segmentation. Symbol spotting in architectural floor plans deals with retrieving symbols from a database of floor plans, given a query symbol. Here the symbol set contains images of decor items like table, chair, bed, etc. It also finds the specific locations where the query image of the symbol is likely to be found. The other fields researched are floor plan analysis where existing works range from simple techniques for interpreting a hand-sketched floor plan for creating CAD models and automatically interpreting map drawings to analysing differentiation between thick, medium, and thin lines for segmenting

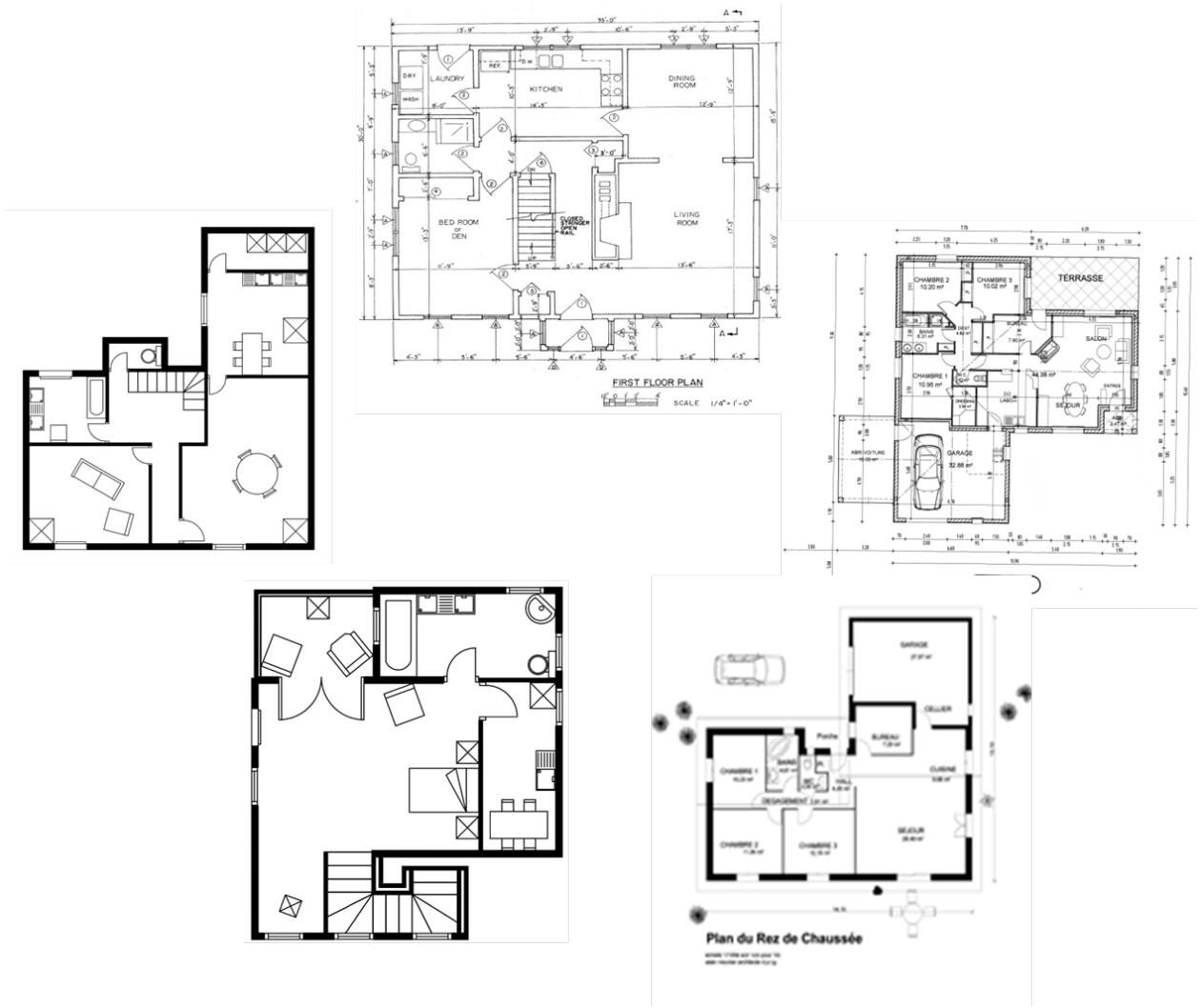


Figure 1.5. : Floor plans, showing 2D cross sections of buildings and displaying heterogeneity in design.

the room layouts in an efficient manner. Thus, upon doing a keen analysis of the existing work in the area of architectural floor plans, it was observed that retrieval in this particular domain is a challenging yet less researched area and has a lot of applications in today's digital scenario.

1.2 PROBLEM STATEMENT

The goal of the work proposed in this thesis is to design a content based retrieval system to perform the following task:

Given a query image/sketch of architectural floor plans retrieve alike floor plans from the database, rank-ordered by similarity in content (refer Fig. 1.6).

As proposed in this thesis, the input to the system would be a floor plan sketch or a floor plan image and the expected output would be similar floor plans as retrieved according to the specified content by the user. Content specification and appropriate feature set to represent the mentioned content plays a pivotal role in success of the retrieval system. Efficient techniques for

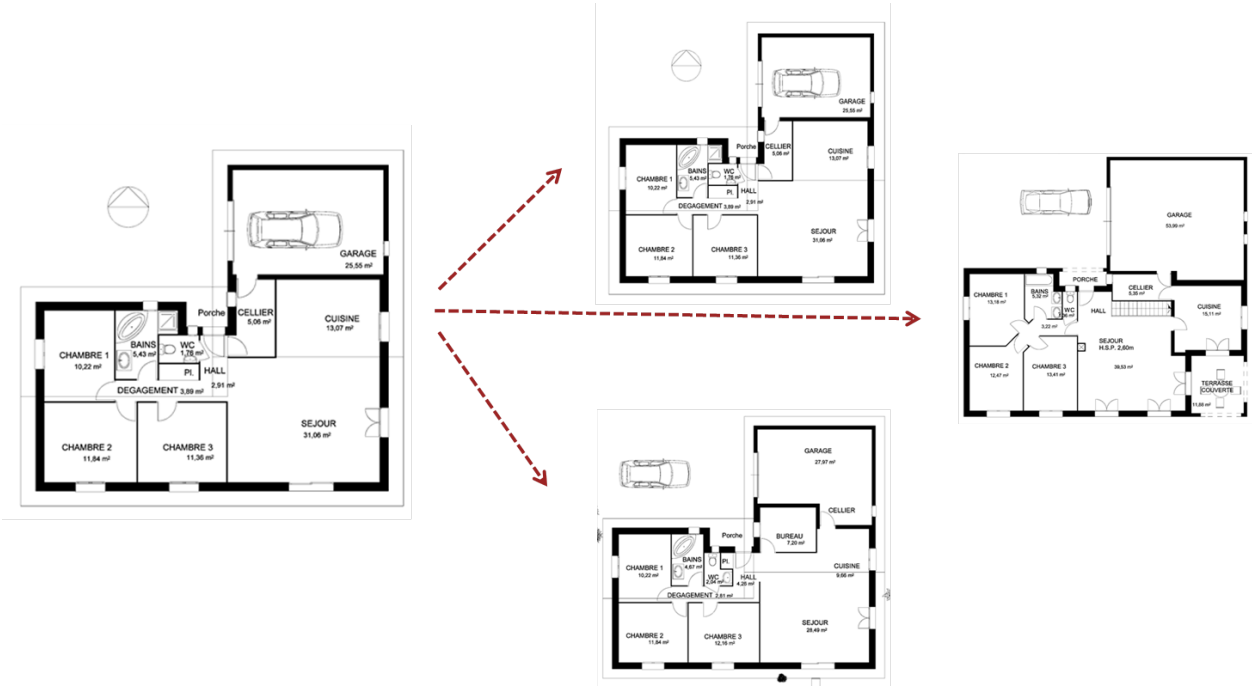


Figure 1.6. : Retrieval in floor plans by extracting meaningful content and looking for similarity across floor plan databases.

matching and retrieval are also proposed alongside capturing appropriate content from the floor plans.

1.2.1 Brief description of the Work Done

In Fig. 1.7, the problem of floor plan retrieval is represented alongwith its various possible solutions depicted in the form of a cube, containing a modality-axis, a feature-axis and an approach-axis. As shown in Fig. 1.7, the modality-axis of the cube corresponds to the type of modalities while querying. A content based retrieval system can have different modalities of representing the input query floor plan, for e.g. query can be an image (I) which represents floor plan images, a sketch (S), which can prove to be a convenient choice. Using pen and paper to scribble down ideas has always been a conventional way to query. The user looking for floor plans can easily draw an abstract floor plan with some interior details on a hand-held device and the sketched floor plan obtained thereafter can be used for querying. The third method can be a hybrid combination (H) of sketch and image representing a unified framework to capture the buyer's intent perfectly. Given a sketch or an image as query the framework should be able to retrieve similar floor plan images from the database. The term hybrid query mode is coined because in this mode the user can make online changes to the query, i.e. image or sketch, by adding or removing components in an existing floor plan. This mode currently serves as a future scope for this thesis. The feature-axis in Fig. 1.7 corresponds to the type of features extracted in the framework. The query and the existing layouts in the repository can be analysed for structural features as well as semantic features. Structural features correspond to the layout features that pertain to the room level analysis being used as content while extraction. These features use the topology and placement of rooms as matching criteria. Whereas, semantic features represent high-level features representing interior room details, sizes of the rooms, room-wise decor placement etc. It is to be noted that both of these features can be hand-crafted, which are typically

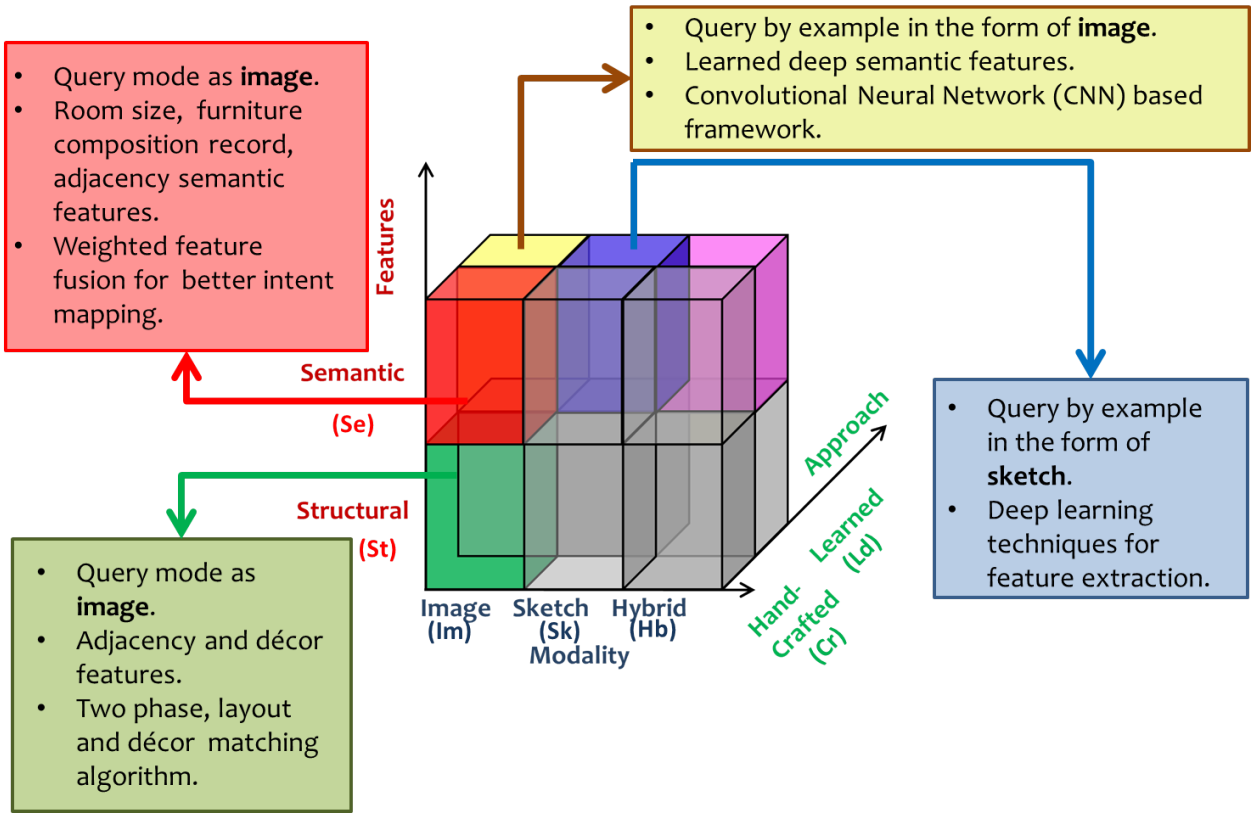


Figure 1.7. : Analysing floor plans for novel content aimed at retrieving similar layouts from a large repository.

chosen to represent the content in a floor plan, as well as, these features can be learned using a sophisticated learning framework, where explicit mentioning of the features is not necessary. The third axis of the cube in Fig. 1.7, represents such a scenario, where these two different approaches, i.e. the learned/ hand-crafted feature approaches are mapped. Thus, it can be inferred that such an amalgamation of various attributes while retrieval gives a great array of challenging problems to be dealt with while representation, analysis and finally matching and retrieval of floor plans. In this thesis, attempts were made to propose solutions to problems involving the following combinations (Im-St-Cr), (Im-Se-Ld), (Im-Se-Cr), (Sk-Se-Ld) as shown in Fig. 1.7.

1.3 RESEARCH ISSUES

Some of the key challenges encountered in the task of Content based retrieval and specifically floor plan analysis and retrieval are highlighted below:

1. In most cases, images/sketches require several stages of processing such as segmentation, smoothing, thresholding. The choice of an appropriate feature to correctly represent the content is in itself a big challenge.

2. As floor plans are 2D cross-sections of buildings, they are generally black and white and carry pixel information in a binary form. Moreover, there exists significant amount of whitespaces in the floor plan. Thus, throughout the image the information content is sparse in nature

which makes feature engineering a challenging task.

3. There is a dearth of publicly available datasets as architects refrain from sharing their designs. Thus, for implementation of an algorithm there is lack of publicly available sufficient data for the task of floor plan analysis and retrieval.

4. There is a huge-variation in the shapes and sizes of floor plans. This makes training a network, for instance, a deep learning framework, pretty challenging as resizing varied shaped floor plans leads to loss in information.

5. Sketch based query mode is in itself challenging to work with. Sketches incorporate a lot of user-variability and noise and require careful pre-processsing before analysis.

6. Representing image/sketch content using features demands a careful development of both hand-crafted and deep-learning features. The representation should ensure high intra-class similarity, at the same time high interclass dissimilarity. The representation should also be able to distinguish between overlapping classes. Moreover, the similarity criteria used for computing the match cost between the query and the model feature representation should also be unbiased and efficient.

Assumptions

The problem of content based retrieval in floor plans is solved under the following assumptions:

1. All the images are binarized.

2. Sketched drawings are scanned into image form for processing.

3. The floor plan images are digitally synthesised, i.e. they are not scanned and thus, free from any transformation like rotation, shear, scaling etc.

The focus of this thesis is on representation of floor plans and design of efficient similarity measures, under realistic situations. At the end of the thesis, an attempt has been made to bridge the gap between the two query modes, floor plan sketches and images using domain adaptation approach.

1.3.1 Organization of the Thesis

In this chapter, the importance of the problem of CBIR is introduced. Along with it a brief overview of the motivation behind a Content based retrieval for architectural floor plans is highlighted. The inherent challenges while proposing an efficient solution for this problem are listed. A brief outline of the approach to solve the problem is provided and the objectives that are trying to be achieved are listed.

In Chapter 2, a review of the existing work in the area of floor plan analysis is given, from which the motivation for this work is derived. Various aspects of architectural floor plan analysis dealing with symbol spotting, room level analysis and sketch based retrieval in floor plans are discussed. In the same Chapter, the datasets available in the floor plan analysis community are discussed. Also, basic image feature extraction strategies are reviewed. Some of the deep learning methodologies for feature extraction and domain adaptation are also discussed in Chapter 2, which serve as the motivation for the proposed multi-modal deep learning network.

Chapter 3 proposes a framework for the matching and retrieval of similar architectural floor plans under the query by example paradigm. Here query is taken as a floor plan image. A room layout segmentation and adjacent room detection algorithm is presented to represent layouts as an undirected graph. Also a novel graph spectral embedding feature is proposed to uniquely represent the layout of the architectural floor plan. This graphical approach helps in effective and efficient matching of the room layouts. Room semantics in terms of both the room structures and room decor is used to retrieve similar floor plans from the repository. To match the semantic similarity between a pair of floor plans, a two stage matching technique is proposed and high retrieval accuracy is obtained.

In Chapter 4, a deep learning paradigm to extract both low and high level semantic features from a layout image is proposed. The key contributions in the proposed approach are, a novel deep learning framework to retrieve similar floor plan layouts from repository and analysing the effect of individual deep convolutional neural network layers for floor plan retrieval task.

In Chapter 5, a novel end-to-end framework for extracting high level semantic features like area and room-wise decor arrangement for the task of fine grained retrieval is proposed. Further, a technique to perform feature fusion to aggregate high-level semantic features extracted is also proposed. Weighted feature fusion helps in setting preferences to particular features while retrieval and satisfying specific user demands.

In Chapter 6, a novel system which uses sketch based query mode to retrieve corresponding similar floor plans from the repository is proposed. Two approaches are discussed, one with a system using Cyclic Generative Adversarial Networks (Cyclic GAN) for mapping between sketch and image domain. The key contributions here being a novel sketch based floor plan retrieval framework using an intuitive and convenient sketch query mode and conjunction of Cyclic GANs and Convolution Neural Networks (CNNs) for the task of hand-drawn floor plan image retrieval. The second approach being the usage of autoencoders along with Cyclic GANs for improving the performance of the retrieval framework and helping in better domain mapping between sketch and image domains.

Finally, in Chapter 8, the contributions of this thesis in the field of content based retrieval in architectural floor plans are summarized and a few possible extensions and future prospects for the work are identified.

Two appendices are also provided at the end of this thesis, where, Appendix 1 details out the creation of the floor plan datasets, Repository Of BuildIng floor plaNs (ROBIN) and Sketched-Repository Of BuildIng floor plaNs (S-ROBIN) that have been specifically designed to aid the analysis and retrieval in floor plans. Appendix 2 contains the description of a graphical user interface, designed for the task of interactive floor plan retrieval.

2
Background and Related Work

2.1 INTRODUCTION

In this chapter, the existing literature regarding the research areas discussed in this thesis is presented, followed by a brief review of the existing datasets in the area of floor plan analysis. Document Image Analysis (DIA) tasks involving Graphics Recognition specifically have evolved mainly due to the growing trend of not only storing digital documents these days, but also due to extraction, classification, and indexing of them according to the information conveyed by the documents. This gained popularity after the emergence of scanners that increased the managing and handling of digitized documents. Application of Graphics Recognition to domains like engineering drawings, electrical diagrams, flowcharts, maps, mathematical and chemical formulae, and architectural plans has been done from the very start of the evolution of this field. After one recognizes the graphics in a particular document the next task is to derive something meaningful out of it. As this thesis pertains with engineering drawings, specifically floor plans therefore, in the next section the existing work related to symbol spotting, floor plan analysis and retrieval in floor plans, the commonly used datasets in floor plan analysis and generic feature representation and deep learning retrieval methods is discussed.

2.2 STATE OF THE ART IN FLOOR PLAN ANALYSIS

Floor plan analysis is one of the most trendy applications in graphical document understanding due to the actual demand of real world applications. Thus, several systems dealing with heterogeneous architectural inputs such as sketch drawings, machine generated documents, and CAD files have been presented in the recent years. Nevertheless, floor plan interpretation problem has not yet been solved for two main reasons: the inherent difficulty of the problem i.e. the visual language may differ from plan to plan, and the lack of reusability and improvement of the existing approaches. In the following paragraphs, the most recent techniques for floor plan analysis are described in detail. In Tab. 2.1, some key approaches are listed with their salient features.

2.2.1 3D reconstruction of floor plans

The work proposed in [Ah-Soon and Tombre, 1997], [Dosch and Masini, 1999], and [Dosch *et al.*, 2000] deals with floor plan 3D reconstruction. In these works, the input is scanned floor plans. First, a preprocessing step is followed to separate both text and graphics information. Then the thick and thin lines present in the graphical layer are segregated and vectorized. Thick lines depict walls whereas, the thin ones depict the rest of the symbols, including windows and doors. In this process, two variations of walls are considered: first being the walls represented by thick parallel lines and the second case being walls represented by a single thick line. Arcs in the symbols represent doors, windows are detected using small closed loops, and rooms are generally

Table 2.1. : Reviews of approaches for floor plan analysis

Paper Title	**Technique**	**References**
A system to understand hand-drawn floor plans using subgraph isomorphism and Hough transform	Attributed graph structure, Straight line Hough transform for speeding matching	Lladós *et al.* [1997]
A Prototype System for Interpreting Hand-Sketched Floor Plans	Automatically converts hand-sketched floor-plans into the CAD formation, Template matching of closed regions	Aoki *et al.* [1996]
A complete system for the analysis of architectural drawings	Converts the drawing into a description in terms of basic architectural entities, Reconstructing in 3D	Dosch *et al.* [2000]
Robust and Accurate Vectorization of Line Drawings	Vectorizing the graphical parts of line drawings, Compute feasibility domains for accurate vectorization	Hilaire and Tombre [2006]
The Room Connectivity Graph: Shape Retrieval in the Architectural Domain	Characterize 3D architectural models on underlying arrangement of their rooms, Retrieval of 3D models.	Wessel *et al.* [2008]
A System to Detect Rooms in Architectural Floor Plan Images	Hough transform, Decomposing rooms to near convex regions	Macé *et al.* [2010]

a composition of bigger connected components. Finally, a single level in a floor plan or several floors in the same building are identified and their 3D reconstruction is performed [Ah-Soon and Tombre, 1997] by finding special symbols as beams, staircases, etc. [Dosch and Masini, 1999]. Both in [Dosch and Masini, 1999] and [Dosch *et al.*, 2000], emphasis is made on the necessity of human feedback while working with complex plans. Or *et al.* [2005] focus on 3D model generation from a 2D plan. Rendek *et al.* [2004] preprocess the image by firstly segregating the text and graphics and then converting the graphical layer to a vector form. They further, manually delete other noisy lines and symbols to finally detect the plan structure. Once the remaining structure consists of only lines belonging to walls, doors, and windows, a set of polygons using each polyline of the vectorized image is generated. At the end, blocks are represented by polygons; thick lines represent outer walls, rectangles represent windows inside the walls, and arcs represent doors. This system is able to generate the 3D model corresponding to single story buildings. Again, this approach is plan specific and slight change in the floor plan leads to modification of the technique. Some works such as proposed by Lu *et al.* [2007] for 3D reconstruction contain a CAD file as input which consists of lines of the architectural design and real non-distorted original polylines. First, T, L and X shapes are identified by extracting parallel overlapped lines. Then, the 3D reconstruction of the plan is

retrieved by identifying connections between the lines. After that, segmentation of the graphical symbols such as furniture and staircases is done by deleting the structural lines. This method is primarily based on symbol recognition using geometrical features extracted from the image.

2.2.2 Semantic analysis of floor plans

Cherneff *et al.* [1992] present a knowledge-based interpretation method for architectural drawings. Their aim is to extract the complete structure of the plan, which comprises of the recognition of walls, doors, windows, rooms, and the relations between them. The input is a floor plan vectorized in order to obtain unique symbols for doors. Then, the system is composed of two models: the semantic model and the structural model. The semantic model represents the contextual composition of the architectural elements. The structural model entails the geometrical configuration of the plan, including the 2D spatial indexing of primitives. Walls are defined as parallel segments incidenting to doors or windows at their ending points. This fact places a constraint on the possibilities of interpretation, since it is based on the assumption that walls are always straight, however in real world scenario walls might be curved or even not be modeled using parallel lines. The work presented by Cherneff *et al.* [1992] focuses on room segmentation. A proximity map based semi-automatic approach is used for searching for regions in machine printed floor plan images. This method serves as an extension of the area-filling approach which mainly deals with splitting rooms when there is a lack of physical separation. However, the drawback here is that this method retrieves many false positives in the form of all those objects that are modeled as closed regions, for example, doors, tables, and staircases. [Macé *et al.*, 2010] focus on the extraction of the building structure from scanned images. Similarly to [Dosch *et al.*, 2000], [Dosch and Masini, 1999] they firstly separate textual from graphical information. Then, on the graphical layer, they recognize walls by detecting parallel lines at the contours of the thicker lines. Thick lines are earlier separated from the thin ones morphologically. Finally, rooms are recognized by a recursive decomposition of the images into convex regions. In this method, the wall detector strongly depends on the wall notation, which encompasses a redefinition in order to be able to deal with a variety of floor plan models. Work proposed by Ahmed *et al.* [2012] also starts with the classical text/graphics separation. After that, they morphologically separate the graphical components into thin, medium, and thick lines. Medium and thick lines are considered to be walls whereas small lines model the rest of the architectural elements. Then, doors and windows are detected by means of a patch-based description of the image using SURF [Bay *et al.*, 2006]. Rooms are recognized by finding closed areas enclosed by walls, doors, and windows. At the end, the textual labelling inside the room areas are used to verify and correct the final segmentation. This whole method focuses on a single graphical notation, specially in the process of wall extraction. Also, the work of Zhi *et al.* [2003] takes input in the form of a CAD file. It then automatically extracts the topological and geometrical information from a 2D architectural drawing and builds a simulator that helps in building evacuation in case of crisis situations. Firstly, filtering of redundant information such as dimensions, text, and furniture, is carried out in a semi-automatic fashion and only the remaining necessary objects like doors, windows and walls are kept. Then, the plan is transformed into an attributed graph and loops are searched for in the floor plan. Further, according to their respective attributes, loops are classified into different categories such as rooms and corridors are represented by spatial loops, walls and columns through physical loops, window loops, door loops and unrecognized loops. Finally, the compartments in the floor plans are identified and the system is integrated in a simulation model that aids in difficult evacuation strategies by connecting rooms to their nearest exits/staircases in complex buildings.

2.2.3 Structural Analysis

Works as proposed by Aoki *et al.* [1996] and Lladós *et al.* [1997] analyze hand-sketched floor plans. On the one hand, [Aoki *et al.*, 1996] transforms the sketches into CAD files. In this particular work the lines that model the building structure are extracted, which are basically handwritten on a preprinted paper with a grid of lines. This method again describes line elements in the form of windows and walls, and closed region elements here represent doors. On the other hand, [Lladós *et al.*, 1997] recognizes building elements and their respective topological properties using Hough transform and subgraph isomorphism. Subgraph isomorphism aids in symbol recognition whereas Hough transform helps in detecting walls modeled using hatched patterns. It should be noted that in both, [Aoki *et al.*, 1996] and [Lladós *et al.*, 1997]], the conventions to be used while drawing are prescribed beforehand. Floor plan structural retrieval has gathered the attention of architects recently. The works of [Ahmed *et al.*, 2014], [Weber *et al.*, 2010] and [Wessel *et al.*, 2008] are three examples of this application. In the case of the work proposed by Weber *et al.* [2010], the query is specified in the form of a sketch drawn online by a user. Their system allows the user to sketch an abstract schematic representation of a floor plan and uses this representation to search for structurally similar documents. The sketch is modelled as a graph, which encloses the structure of the plan, and it is compared to the repository representations using subgraph matching algorithms. As proposed by Wessel *et al.* [2008], the specified input is in the form of a polygon representing a 3D floor plan, so they do not need to vectorize the plan. They construct a graph where attributed nodes are rooms and attributed edges are connections between them: doors or windows. Based on this connectivity graph, fast and efficient shape retrieval is achieved.

Table 2.2. : Reviews of approaches for symbol spotting

Paper Title	**Technique**	**References**
Statistical Grouping For Segmenting Symbols Parts From Line Drawings, With Application To Symbol Spotting	Groups line drawings into shapes, Identifying ROI and aiding matching process	Nayef and Breuel [2011]
Symbol Spotting in Line Drawings Through Graph Paths Hashing	Hashing the shape descriptors of graph paths, Local sensitivity hashing, Spatial voting scheme	Dutta *et al.* [2011]
Combination of Product Graph and Random Walk Kernel for Symbol Spotting in Graphical Documents	Path similarity using product graph, Random Walk Kernel for calculating distance measures	Dutta *et al.* [2012]
An integer linear program for substitution-tolerant subgraph isomorphism and its use for symbol spotting in technical drawings	Integer linear formulation, Substitution-tolerant subgraph isomorphism.	Le Bodic *et al.* [2012]
A symbol spotting approach in graphical documents by hashing serialized graphs	Hashing the shape descriptors of graph paths, Local sensitivity hashing, Spatial voting scheme	Dutta *et al.* [2013]
Object Recognition in Floor Plans by Graphs of White Connected Components	Region Adjacency Graph, White connected components as nodes	Barducci and Marinai [2012]

2.3 STATE OF THE ART IN SYMBOL SPOTTING

Symbol spotting in floor plans deals with given an image of a query symbol, location of the query symbol and associated documents are retrieved. There are two approaches to symbol spotting. The first pertains to a region-based approach, which holds its basis on searching for regions of interest inside drawings, using various descriptors to describe them, and finally using a query symbol to index them. The second approach for spotting deals with isolating the symbols using segmentation to get the symbols segregated from their background, and then performing their recognition. Generally the spotting methods mentioned in the literature can be categorized into the region-based approach. However, this approach has the following inherent problems: (1) It is difficult to locate regions of interest in a scaled down drawing; (2) Rotation invariance in such type of drawings is hard to achieve; (3) As the patterns in line segments are simplistic in nature, special descriptors need to be developed to represent line drawings, instead of using the regular textual based descriptors and, (4) Indexing techniques like hashing fail to scale well when one brings large databases into the picture. Due to these problems, the methods implementing the region based approach result in low recall and precision rates. Inspite of the drawbacks, a primary reason to follow this approach, rather than the segmentation-based approach, is the incapability of the current segmentation techniques to peform while dealing with complicated technical drawings. Table 2.2 lists some important works related to symbol spotting.

2.3.1 Region based symbol spotting

In this spotting approach, various methods for detection and description of regions of interest within drawings are employed by the state-of-the-art methods. Authors in [Rusinol and Lladós, 2007] and [Rusiñol *et al.*, 2010], propose the use of different spotting methods working with detecting closed regions to further identify the regions of interest. In [Rusinol and Lladós, 2007], a bipolar coordinate system is used to depict a region of interest and for indexing purposes a hash table is used. In [Rusiñol *et al.*, 2010], attributed strings are used to represent regions of interest. A polygonal approximation of the contour of the closed regions is computed and association of adjacent segments is done in the form of polylines. These polylines are encoded as attributed strings, where attributes consist of, the length of the segment and the angle between the current and the previous segment. Similar or alike strings are then clustered together in a look-up table and a set of median strings represent the indexing keys to index into the table. String matching is used for comparison of the query to regions in the image. Finally, a Hough voting scheme is used for the verification of the initially found matches. In another work as proposed in [Ah-Soon and Tombre, 2001], overlapping sliding windows are used to detect regions. The regions here are described as feature vectors depicting the relations between the connections of the region segments, and graphs are used to describe the descriptions. The symbol search is then ensured by calculating the similarity of the relationship between the segments in the network.

2.3.2 Segmentation and recognition of isolated symbols for symbol spotting

Graph representations carry their fair share of popularity in isolated symbol recognition methods for symbol spotting. In [Qureshi *et al.*, 2007], authors propose the use of attributed relational graphs (ARG) for the representation of line primitives in the form of vectors, to represent floor plan drawings. The relationship between the identified vectorial primitives paves way for the detection of regions inside the floor plans. Matching of the query to the regions is done using a graph description and matching method proposed by Qureshi *et al.* [2006]. A combination of graph parsing and matching for representing line drawings was proposed by Llados and Sanchez

[2004]. In this paper, the authors present a mechanism for graph indexing to spot known symbols in the drawings. Locteau *et al.* [2007] again used a graph based approach for representation of geometrical shapes which collectively represent geometrical primitives in line drawings. Detection of the regions is done based on open or closed curves. Subsequent query matching to the regions is performed using edit distance metric. Tabbone *et al.* [2003] presented a spotting method that uses the segmentation-based spotting approach. In this particular approach authors propose the need of a segmentation method to segregate text and graphics along with user feedback to further perform symbol segmentation in simplistic line drawings. The descriptor used here is a discrete version of the F-signature which is the histogram of forces. For matching the query to the segmented symbols, the sum of absolute differences is used.

It can be safely stated for the spotting methods following the region-based approach, that if one is able to accurately and precisely identify the regions in the line drawings, then the subsequent steps of matching and description can achieve good results. On the other hand, methods employing isolated symbol spotting approach, rely on efficient prior segmentation, which makes this approach a bit challenging. It is observed that there is an increase in the number of spotting methods which project good resultant recall rates, but usually have a drawback, that they give low precision rates.

The online process for symbol spotting is faster and efficient and usually deals with the following steps: (1) locating regions of interest, (2) clustering (3) creating an efficient indexing data structure from the obtained clusters. During the time of retrieval, the query description is compared with the entries already stored apriori in the data structure. Approaches mentioned in [Nguyen *et al.*, 2009], [Luqman *et al.*, 2010], [Kong *et al.*, 2011] and [Rusiñol *et al.*, 2010] follow this particular technique for symbol spotting. To summarize, the issue with using these methods is that they are lacking in projection of efficient steps for content analysis, which further results in a not-so-good representation of the floor plan drawings, and thus, inadvertently affects the recall and precision rates of retrieval, reducing them to low values.

2.4 STATE OF THE ART IN GENERIC FEATURES FOR RETRIEVAL

So far the existing techniques involved in floor plan analysis and retrieval are discussed. In the latter chapters of the thesis, comparisons with generic retrieval based approaches are done. This section presents a review of features used for retrieval tasks. Several book chapters and papers in the literature present a thorough review of the advances in the feature descriptors [Doermann *et al.*, 2014; Zhang and Lu, 2004]. The primary goals of these description techniques can be listed down as: (1) minimizing the intra-class distances and maximizing the inter-class distances, (2) minimizing the spatial dimensionality to enhance the time complexity and the classification efficiency, and (3) dealing with noise, transformations and distortions in the images. Some popular techniques involve adapting existing feature descriptors from popular research fields like pattern recognition, e.g SIFT [Lowe, 2004], SURF [Bay *et al.*, 2006], and HOG [Dalal and Triggs, 2005]. In HOG [Dalal and Triggs, 2005], the idea is that the shape and appearance of an object within an image can be described by a histogram of intensity gradients. This descriptor is built by dividing the image into small regions, and computing a histogram of the edge orientations for each region. The combination of these histograms form the descriptor for the image. This method has been used for human detection in images. SIFT [Lowe, 2004] has been one of the most widely used feature descriptor in object recognition. Lowe [2004] first proposed this method of representing an image as a collection of local feature vectors which are invariant to translation, scaling, and rotation. These features are initially detected using a Harris-Laplace corner detector in scale space. Once corners are detected, the scale space at which they were detected are stored. Then for each of these keypoints, the dominant gradient is computed over a neighborhood of 16×16. This is also stored

and after which, the descriptor for this keypoint is computed. This keypoint descriptor is described by a 128 dimensional vector. This vector is obtained by selecting a 16×16 neighborhood around the keypoint. This is then partitioned into 4×4 regions with each region containing 4×4 pixels. For each region, a histogram with 8 bins is computed from gradient magnitudes in that region. Thus, the 128-dimensional vector is formed. SIFT has also many variants. The first of which is SURF or Speed Up Robust Features [Bay *et al.*, 2006]. The key difference is that this uses integral images to speed up convolutions. Other variants include RGB-SIFT, HSV-SIFT, and Opponent-SIFT. These mainly use SIFT on different color spaces since SIFT was primarily made for grayscale images. RGB-SIFT operates on the RGB channels, HSV-SIFT on HSV channels, and C-SIFT on opponent channels. One final variant is called dense SIFT (or D-SIFT) where descriptors are computed for every pixel. In certain cases, describing the symbol characteristics requires strategies which are highly domain specific, e.g. zoning [Escalera *et al.*, 2009], geometric moments [El Rube *et al.*, 2006], and histogram-oriented [Yang, 2005] techniques. Finally, classification techniques on extracting the features are highly benefited by mathematical techniques such as the computation of sums, products and distances. Thus, several effective algorithms for classification tasks, such as K-nearest neighbour, SVM, boosting and neural network learning have been widely employed in the literature. Another technique used for extracting features is Run Length Histograms (RLH) [de las Heras *et al.*, 2013] approach where, the style of an architectural drawing is characterized by the perception of lines, shapes and texture. Authors propose runlength histograms extracted in vertical, horizontal and diagonal directions as a characterization of line and space properties in floor plans that further estimate a description of walls and room structure. Another efficient technique is an Online Algorithm for Scalable Image Similarity learning (OASIS) [Chechik *et al.*, 2010], which learns a bilinear similarity measure over sparse representations. OASIS is an online dual approach using the passive-aggressive family of learning algorithms that employ an efficient hinge loss cost.

2.5 STATE OF THE ART IN DEEP LEARNING FRAMEWORKS FOR IMAGE AND SKETCH RETRIEVAL

With the extensive research on deep neural networks (DNN) [Krizhevsky and Hinton, 2011; Szegedy *et al.*, 2015; Simonyan and Zisserman, 2014], the success and efficiency of the learning-based features have been observed in multiple areas. High-level abstractions providing an approximation to the human cognition system can be learned [Bengio *et al.*, 2009] using deep architectures. As a result, DNN can be used to extract semantically oriented features using the activation in different layers in the networks. In the work proposed by Hörster and Lienhart [2008], feature extraction is carried out using local patches employing a deep restricted Boltzmann machine. Convolutional neural networks (CNN) form a sub-component of the DNNs. Authors in [Krizhevsky *et al.*, 2012b] have demonstrated a highly superior performance in various tasks on image recognition and retrieval [Razavian *et al.*, 2014]. In the paper proposed by Wan *et al.* [2014], detailed studies are conducted on the capability of learned visual features using deep CNN for a variety of applications which include content based image retrieval as well. Razavian *et al.* [2014] study the VGG-Net and Alex-Net and further try to tap the response of the last convolutional layers along with max pooling for image representation for the final task of image retrieval [Razavian *et al.*, 2016]. In the work of Zheng *et al.* [2015], the sixth layer of the Alex-Net [Krizhevsky *et al.*, 2012b] is activated and the activations are taken out as a DNN feature for each image, which is integrated in the image similarity match score level along side traditional visual features that include HSV histogram, SIFT-based BoW feature, and GIST. The state-of-the-art sketch based deep retrieval models [Sangkloy *et al.*, 2016] are deep models that aim to close the domain gap by learning a joint feature embedding for the two domains. Multibranch deep convolutional neural networks (CNNs) are employed where each branch corresponds to one domain and the final shared layer defines the

embedding space which is subject to various discriminative losses such as pairwise contrastive loss or triplet ranking loss. These losses are designed to pull matching pairs of photos and sketches close and push mismatched pairs away. These models thus indirectly align the two domains. However, with limited training data and by focusing only on discriminative losses, these models struggle to capture all the domain-invariant information and thus generalise poorly to test data where the domain discrepancies and misalignments could be different from those in the training data. Alternatively, Generative Adversarial Networks (GANs) have also been employed for generic sketch retrieval [Creswell and Bharath, 2016] and [Chen and Hays, 2018] but have not yet been explored for application in floor plan retrieval. GANs combine the discriminative model with a generative component. One of the typical cases is to use source images, noise vectors or both to generate simulated samples that are similar to the target samples and preserve the annotation information of the source domain. One more approach that is used is the reconstruction-based domain adaptation approach, which assumes that the data reconstruction of the source or target samples can be helpful for improving the performance of domain adaptation. The reconstructor can ensure both specificity of intra-domain representations and indistinguishability of inter-domain representations by using stacked autoencoders (SAEs), that combine the encoder network for representation learning with a decoder network for data reconstruction [Ghifary *et al.*, 2016],[Zhuang *et al.*, 2015].

2.6 FLOOR PLAN DATASETS

Table 2.3. : Details of publicly available existing floor plan datasets

Floor plan Datasets		
Dataset	Count	Remarks
CVC-FP [de las Heras *et al.*, 2015]	122	4 Sub-Categories, varying in wall textures, to study graphical notations in floor plans
FPLAN-POLY [Rusiñol *et al.*, 2010]	42	Used for floor plan analysis and room analysis
SESYD [Delalandre *et al.*, 2007]	1000	100 layouts/ class, differ in arrangement of symbols, used for symbol spotting tasks

Performance evaluation of symbol spotting tasks and layout analysis on floor plans requires datasets that are publicly available to aid the research in this area. Table 2.3 contains the details of some publicly available floor plan datasets. One of the widely popular databases for symbol recognition and spotting tasks is the Systems Evaluation SYnthetic Documents (SESYD) database [Delalandre *et al.*, 2007]. It is a collection of labelled synthetic images. It consists of electrical and architectural drawings, designed specifically for symbol spotting, recognition and retrieval. The whole dataset consists of 5 document collections containing $11,100$ images with $128,700$ symbols. It contains 1000 floor plans divided into 10 broad categories, each containing 100 samples. Inter-class similarity is very low as layouts across each category have a huge difference. Intra-class similarity is very high as floor plans within a class only differ in the furniture/symbol arrangement and orientation in each plan.

Another floor plan dataset FPLAN-POLY proposed in [Rusiñol *et al.*, 2010] contains 42 floor plan vectorial images in dxf format which can be viewed with the common CAD softwares. The main goal of this database also is to provide a framework for the evaluation of different symbol spotting methods in vectorized graphic documents. CVC-FP dataset proposed in [de las Heras *et al.*, 2015] is annotated for the architectural objects and their structural relations. It contains 122 floor plans spread over 4 different layout categories. This dataset aims at aiding in learning and

better interpretation about how the elements are structurally arranged in a floor plan.

2.7 SUMMARY

In this Chapter, various research areas such as floor plan analysis, symbol spotting, retrieval, feature extraction and deep learning models for analysis are discussed. Although the literature in the above mentioned areas is quite rich, there does not exist a composite framework in the area of floor plan analysis that takes meaningful information extracted out of floor plans to perform content based retrieval. Moreover, the publicly available datasets in the area of floor plans are mainly suited for symbol detection and structural analysis in floor plans. Neither are the existing datasets diverse enough to offer varied content for retrieval task nor are the samples in them adequate in number to implement deep learning approaches for efficient analysis and retrieval. Thus, in the consequent chapters of this thesis techniques for content based retrieval in floor plans are proposed to aid users/architects to help find their dream property according to their specific requirements.

3

Topological features for representation and retrieval of similar floor plans

As observed by the detailed literature survey given in Chapter 2, researchers in the area of analysis of architectural floor plans have mainly explored areas like:

- *Symbol spotting*: Symbols are described as a set of logically related graphical parts. These parts are generally visual primitives such as vectors, lines, and arcs or a certain combination of all of these. These primitives in a certain context denote a symbol instance. In the task of symbol spotting thus, all the decor symbols present inside a floor plan are localized and recognized.

- *Floor plan Analysis*: This pertains to structural and semantic analysis of floor plans. Examples of various categories of problems addressed in this type of task are: detecting the various type of walls in a floor plan, segmentation of various rooms in a floor plan, understanding the relationship between the rooms, identifying various functionalities of various parts of a floor plan.

From the literature it can be inferred that symbol spotting and analysis in architectural floor plans have been active research areas in the past. However, a composite framework comprising of analysing a floor plan structurally as well as semantically, and further retrieving floor plans based on certain content or categories is a relatively novel problem. Upon closer observation it was found that a floor plan has various types of meaningful information (content) that can be explored while retrieval. At the same time, inter-class dissimilarity and intra-class similarity among the samples of a floor plan dataset are also to be taken into consideration. Some examples for a meaningful content are, layout information, inner components or decor information, accessibility or approachability of each room from the entry or exit or from each other, aesthetics information or proper placement of the rooms inside the layout, and so on.

In this chapter, the primary focus is on the first two types of content namely, the layout information and the decor information to distinguish between a pair of floor plans. A baseline global framework comprising of match scores computed based on both the topological and decor features is proposed, which paves way for a novel framework for the introduction of retrieval in the area of architectural floor plans. In the subsequent chapters of this thesis, improvement over this baseline is discussed. The Chapter organization is as follows: Sec. 3.1 gives a brief insight into the proposed approach. Subsequent sections detail out the methodology of the proposed approach. In Sec. 3.6 the experimental setup, qualitative and quantitative results of the proposed technique are presented. Section 3.7 provides an analysis of the results obtained and finally Sec. 3.8 concludes the Chapter.

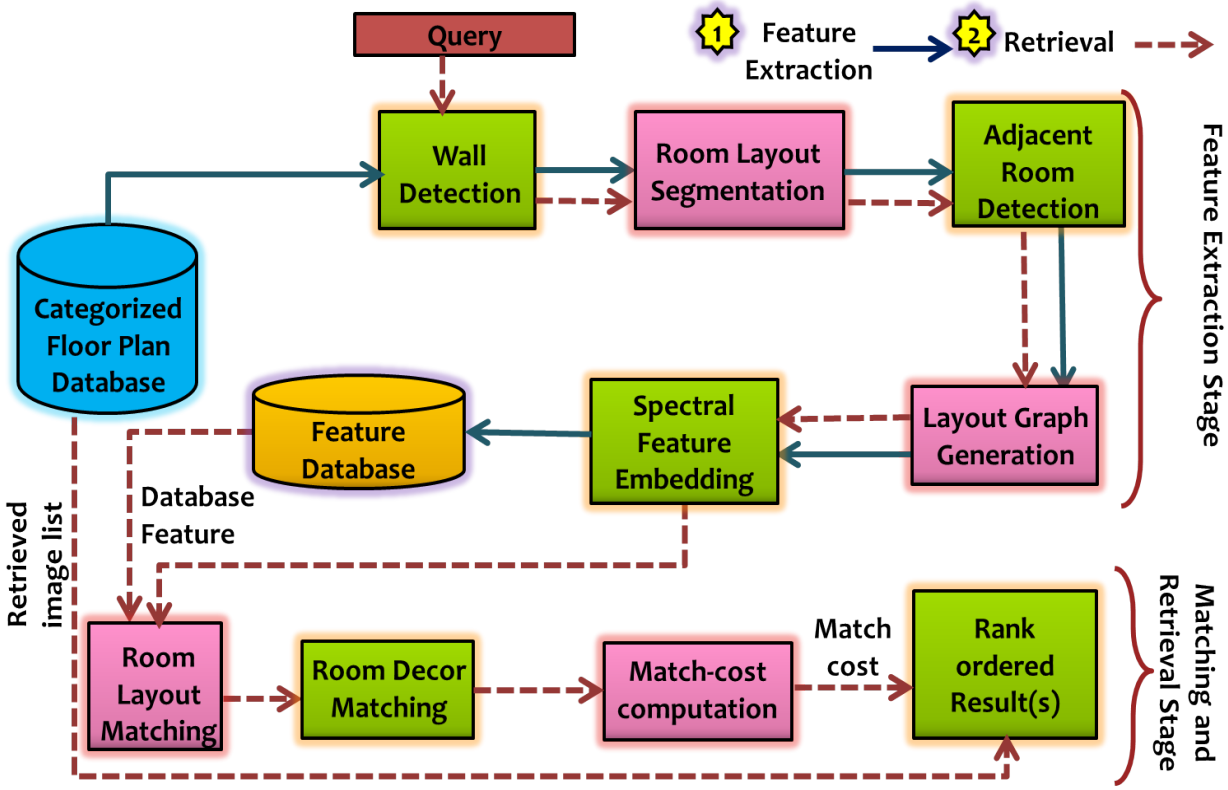


Figure 3.1. : Framework diagram for our structural retrieval based on topology

3.1 BRIEF OVERVIEW

A query by example paradigm has been followed to retrieve similar floor plan images. Figure 3.1 depicts the complete framework of the proposed technique of semantic based architectural floor plan retrieval. The framework has two phases : A) Feature extraction, and B) Matching and Retrieval phase. The solid arrows in the framework diagram (Fig. 3.1) correspond to the feature extraction phase, whereas, the dashed arrows correspond to the retrieval phase. The various components of the framework, in order of operation, are:

1. Floor plan repository, which in this case are the architectural floor plan databases SESYD [Delalandre *et al.*, 2010a] and ROBIN [Sharma *et al.*, 2017].

2. The Wall Detection block, which focuses on delineating a floor plan and detecting room boundaries.

3. Room Layout Segmentation block, which segments the layout into meaningful rooms discarding all the inner decor components present inside the rooms.

4. Adjacent Room Detection (ARD) block wherein the relationship between the segmented rooms is established. Here, the relationship pertains to the topological information of the layout.

5. The output of the ARD block is sent to the Layout Graph generation step where, the topological information of a layout is modelled into a graph for feature representation.

Considering the adjacencies a graph of the floor plan is generated using the topology graph idea proposed in [da Fonseca, 2004].

6. The next block is the Spectral Feature Embedding block where, the topology (neighbourhood) information is embedded into a pattern space as a feature vector. The adjacency matrix obtained from this layout graph is used to extract the spectral features.

7. The next component, the feature database, stores all the feature vectors extracted from the floor plans and serves as a feature repository. This procedure is carried out apriori for all the images present in the floor plan database.

The blocks mentioned above constitute the Feature Extraction Stage. As soon as the query is fired, first the features are extracted from the floor plans using the Stage 1 blocks and then the Matching and Retrieval Stage (Stage 2) comes into picture. The main blocks in Stage 2 are:

1. Room Layout Matching (RLM) block, where, the dissimilarity between the query feature vector and each of the model feature vectors is computed using Euclidean distance metric. The model corresponding to the minimum distance of its feature vector from the query feature vector is chosen as the best match.

2. Room Decor Matching (RDM) block, where the layout from the database to which the query layout matches the best is then chosen and further in the images of that matching layout category, the arrangement of furniture is analyzed. A match cost is computed by matching the room decor of each image of the layout category with the query image and this match cost is then used to rank order the results for obtaining the best matching floor plan image to the query floor plan image.

3. Match-cost Computation block helps in calculating the cumulative match cost obtained from both the Room Layout Matching block and the Room Decor Matching block.

4. Finally, using this cumulative match cost floor plans are fetched from the repository and rank-ordered on the basis of their similarity with the query floor plan.

The main contributions towards the first attempt for designing a baseline framework for floor plan retrieval are as follows:

1. A novel approach to perform floor plan retrieval incorporating both symbol spotting and room level analysis.

2. Performing two level matching for floor plans based on semantics such as layout and decor.

3. Graph spectral embedding of layouts into a pattern space for efficient and fast matching.

All these steps finally complete the retrieval framework for similar floor plans. The details of each step mentioned in the framework diagram are further discussed in the following sections.

3.2 LAYOUT SEGMENTATION

The first step in the proposed framework is layout image segmentation, which consists of structural and semantic analysis to segment the floor plans into respective rooms. Figure 3.2 depicts the various stages of layout segmentation and analysis. Boundary extraction is performed where

Figure 3.2. : Various stages of structural and semantic analysis on an image from SESYD dataset [Delalandre *et al.*, 2010a]: (a) unsegmented layout, results of (b) morphological operation and wall, boundary extraction, (c) Detection of doors and windows (d) Closing gaps in walls and (e) labelling and segmenting the room structure

the boundary extraction algorithm and morphological operations are carried out detecting the walls in the layout. Let the input floor image be denoted by $\mathscr{I}$. This floor plan image is first dilated ($\oplus$) by a disk type structuring element (S_a) of unit radius followed by erosion operation ($\ominus$) by the same element, according to the following equation:

$$\mathscr{I} = \mathscr{I} \oplus S_a \;\; ; \;\; \mathscr{I} = \mathscr{I} \ominus S_a \tag{3.1}$$

These operations lead to segregation of the layout boundary from small unnecessary designs present inside the layout (Fig. 3.2(b)). This detects the walls in the layout to represent the room structures and thus, delineates the rooms within the layout from each other. Next gap closing is performed at the location of doors and window in the floor plan. Floor plans contain doors and windows that connect each room, therefore, are important for semantic analysis. However, the presence of these doors and windows doesn't influence the overall structure of the building. Hence, these two components are removed for structural analysis. The doors and windows inside the layout are present in various orientations and their elimination leads to production of gaps in the walls at their location (refer Fig. 3.2 (c)). The first task therefore, is to detect the doors and windows in the floor plan and close the gaps at their locations so as to construct definite boundaries for the rooms (Fig. 3.2(d)). To detect the doors and windows Harris corner detector [Harris and Stephens, 1988] is employed.

It is observed that the door symbol has an arc like structure (triangular structure) and thus, Harris corner detector detects three corners at the three triangle edges for the door symbol (refer Fig. 3.3(a)). It is to be noted that only the door symbol out of all the decor symbols in the symbol library has this type of a unique triangle-bound 3-corner structure. Through this unique

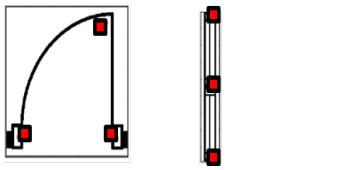


Figure 3.3. : Detecting doors and windows using Harris corner detector

structure the doors stand out as compared to the rest of the symbols in the symbol library. To detect whether the obtained corners form a triangle or not, the three corners identified through Harris corner detector for the door symbol are joined with their adjacent corner point, creating a closed triangle. Further, length of each side (a,b,c) of this triangle is computed and using the triangular inequality : "sum of any two sides should be greater than the third side ($a+b>c$, $a+c>b$, $b+c>a$)", the closed structure is determined to be a triangle. Thus, using this approach doors are detected. The next symbol that needs to be detected in a floor plan for gap closing, is the window. The windows are observed to have collinear corners (refer Fig. 3.3(b)) and thus, get detected through the detection algorithm as the only symbol with collinear corners. Again to identify whether the obtained corners are collinear or not, the co-ordinates of the corner points are noted (x_1, y_1, x_2, y_2 and x_3, y_3). Further, slope of the line between each pair of points is calculated as:

$$slope = \frac{y_1 - y_2}{x_1 - x_2}; \qquad \frac{y_2 - y_3}{x_2 - x_3}; \qquad \frac{y_3 - y_1}{x_3 - x_1} \tag{3.2}$$

If the slope is the same between any pair of corner points, then they are identified to be collinear, hence belonging to the window structure. Thus, uptil now the door and the window locations have been identified.

Upon identifying and eliminating the doors and windows from the floor plan, the next task is to close the gaps in the walls (as shown in Fig. 3.2 (c)). Closing of the gaps is achieved by joining the detected co-ordinates of the doors (in this case the co-ordinates forming 0° angle with each other) and windows setting the values of the pixel location between them as 0, with a thickness of about 40 pixels. Hence, creating a closed wall-structure as in Fig. 3.2 (d). Next, connected component analysis using 8-neighbourhood criterion is performed, to label each room in the layout separately. By this step, the rooms are segmented and are surrounded with walls separating each one of them 3.2(e)). The segmented room layout is sent to the Adjacent Room Detection block for adjacency graph creation.

3.3 ADJACENT ROOM DETECTION

In this step the segmented image is passed through an Adjacent Room Detection (ARD) module to determine the adjacencies between the rooms in the layout. As the segmented rooms are still divided by walls surrounding them, a threshold corresponding to the wall width of about 40 pixels is taken along both vertical and horizontal directions to detect the differently labelled pixels in the image corresponding to the adjacent room. By determining the adjacencies the topology graph

for the layout is created, according to the technique mentioned in [da Fonseca, 2004]. Figure 3.4(c) depicts an example of a topology graph created from the room adjacencies (Fig. 3.4(b)) determined from a given floor plan (Fig. 3.4(a)). The parent node (the one at the centre) represents the layout and the child nodes represent the rooms in the layout. The solid edges between the parent node and the child nodes represent inclusion and dashed edges between the child nodes represents adjacencies. This graph creation further, aids in constructing an adjacency matrix for the rooms detected in the layout which is then used for feature extraction.

However, topology graphs are not directly used for searching similar drawings, since graph matching is a NP-complete problem. Instead of that, the corresponding graph spectrum is used. For each topology graph to be indexed in a database, the descriptors are computed on the basis of its spectrum. In this way, the problem of isomorphism between topology graphs is reduced to computing distances between descriptors. The next section details out the steps to identify the decor components inside the floor plan which are also used for matching purposes.

3.4 FURNITURE DETECTION AND CATEGORIZATION

Algorithm 1 depicts the steps involved in furniture categorization. Firstly, the walls are detected in the floor plan image using a morphological operation as walls are thicker than the other objects inside a floor plan. Then both, the original floor plan image and the image containing only walls are subtracted to obtain only the furniture present inside the floor plan. The decor present inside the whole floor plan is counted on the basis of the application of morphological fill operation applied to the image. Blobs obtained are taken as the decor components inside the floor plan. Further, to categorize the obtained blobs, area-ratio of the individual connected components inside the blob is taken and a unique signature for each furniture component is established by considering the area-ratio of three largest components in each blob. Prior to this, such an area-ratio feature taking the three largest connected components in a furniture symbol is obtained for all the 12 symbols in the symbol library and stored. Signature obtained of all the components inside a floor plan image is finally compared with the signature of the components present in the symbol library and thus, the type of the furniture is assigned to each decor item in the floor plan. Figure 3.5 shows how a furniture in the furniture set is assigned a signature according to Algo. 1.

(a) Architectural Floor plan (b) Results of proposed room layout segmentation technique (c) Layout graph of (a)

Figure 3.4. : Topological graph for the segmented room layout

Algorithm 1 Furniture detection and categorization

Input: Room image without outer walls ($\mathscr{I}_w$), Furniture Template Signatures (F)
Output: Furniture Count (C), Furniture Type Set ($\mathscr{T}$)

C=0, $\mathscr{T}$={}
$\mathscr{B}$=Morphological fill operation($\mathscr{I}_w$)
$C = |\mathscr{B}|$ ▷ $|\cdot|$: *Cardinality*
for $j = 1$ to C **do**
  $C = CC(\mathscr{B}_j)$ ▷ CC()=Connected Component
  for $k = 1$ to $|C|$ **do**
    $\mathscr{A}_k = Area(c_k)$, where $c_k \in C$
  end for
  $A = Sort_{desc}(\mathscr{A})$
  $S(\mathscr{B}_j) = \{(A_1/A_3), (A_2/A_3), 1)\}$ ▷ $S(\cdot)$: Signature
  for $k = 1$ to $|F|$ **do**
    if $S(\mathscr{B}_j) == F_k$ **then**
      $T(\mathscr{B}_j) = T(F_k)$ ▷ $T(\cdot)$: Type of Furniture
    end if
  end for
  $\mathscr{T} = \{\mathscr{T} \cup (T(\mathscr{B}_j)\}$
end for

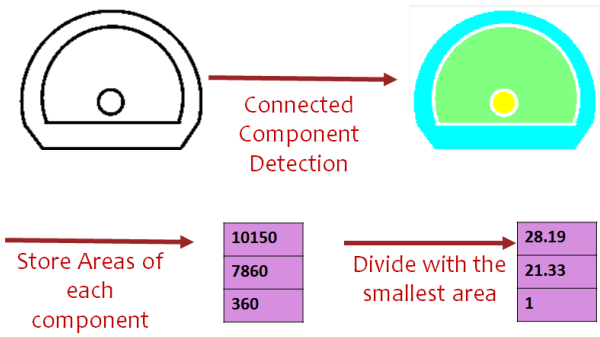
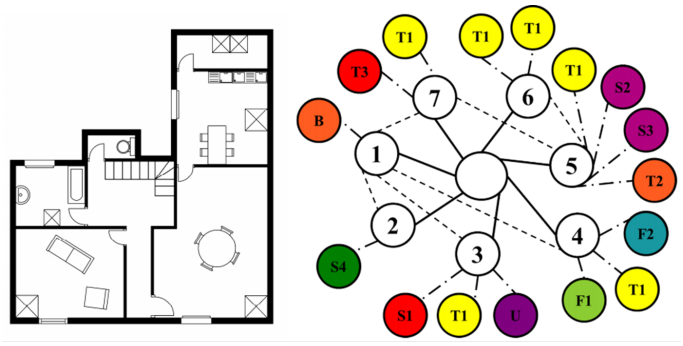


Figure 3.5. : Example showing how a sink symbol in the furniture set is assigned a unique signature.

Figure 3.6(b) depicts an example of the semantic layout graph, consisting of both room adjacencies as well as, room decor representation, corresponding to a layout shown in Fig. 3.6(a) from the SESYD [Delalandre *et al.*, 2010a] dataset. The uncoloured nodes labelled with numbers correspond to the number of rooms present in the layout. The adjacencies between these rooms are depicted by dashed edges. The coloured nodes in the graph correspond to the furniture inside each room. A 360° colour wheel is quantized into uniform bins of value 30° each. According to the orientation of the furniture inside the rooms, a corresponding colour value from the colour wheel is assigned to the furnitures. After obtaining the graph depicting the adjacencies between the rooms and also the decor components, the next task is graph spectral embedding that helps in mapping all graphs to pattern space for retrieval purposes.



(a) Architectural Floor plan (b) Semantic graph of (a)

Figure 3.6. : Room layout and corresponding graph consisting of both room adjacencies as well as room decor representation

3.5 GRAPH SPECTRAL EMBEDDING

Graph Spectral Embedding [Luo *et al.*, 2003] approach is adopted to represent the room layout graphs created in the previous stage in a pattern-space. Such a spectral embedding helps to map similar floor plans closer in the pattern space and thus while retrieval all similar plans are clustered together for efficient matching. Using this technique one can map graphs containing a number of nodes to a feature vector of uniform size for all the floor plans in the database. To overcome the problem of how to map the structure of a graph onto a vector of fixed length, the graph-spectral decomposition methods are explored. Leading Eigenvectors of the graph adjacency matrix are used to define Eigenmodes of the adjacency matrix. Thereafter, the spectral decomposition of the adjacency matrix is performed. The use of graph spectrum as an indexing method is beneficial as (1) it captures local topology, (2) it is invariant to subgraph re-order, and (3) it is stable, since small changes in the graph produce little changes in its spectrum. The graph obtained in the previous step is further represented as an adjacency matrix for spectral decomposition as shown in the next subsection.

3.5.1 Spectral Feature representation

Given N images in the database, their graphs are represented as $G_1, G_2 ..., G_N$. The k^{th} graph is denoted as $G_k = (V_k, E_k)$, where V_k is the set of vertices and $E_k \in V_k \times V_k$ is the edge-set. For each graph G_k, an adjacency matrix A_k of size $|V_k| \times |V_k|$ is computed, where $|\cdot|$ denotes the size of the set.

From the calculated adjacency matrices $A_k, k = 1 \ldots N$, where, N is the total number of graphs for the images in the database, the Eigenvalues λ_k are computed by solving the equation

$$|A_k - \lambda_k I| = 0 \tag{3.3}$$

and the associated Eigenvectors $\vec{\phi}_k$ by solving the system of equations,

$$A_k \phi_k^i = \lambda_k^i \phi_k^i \tag{3.4}$$

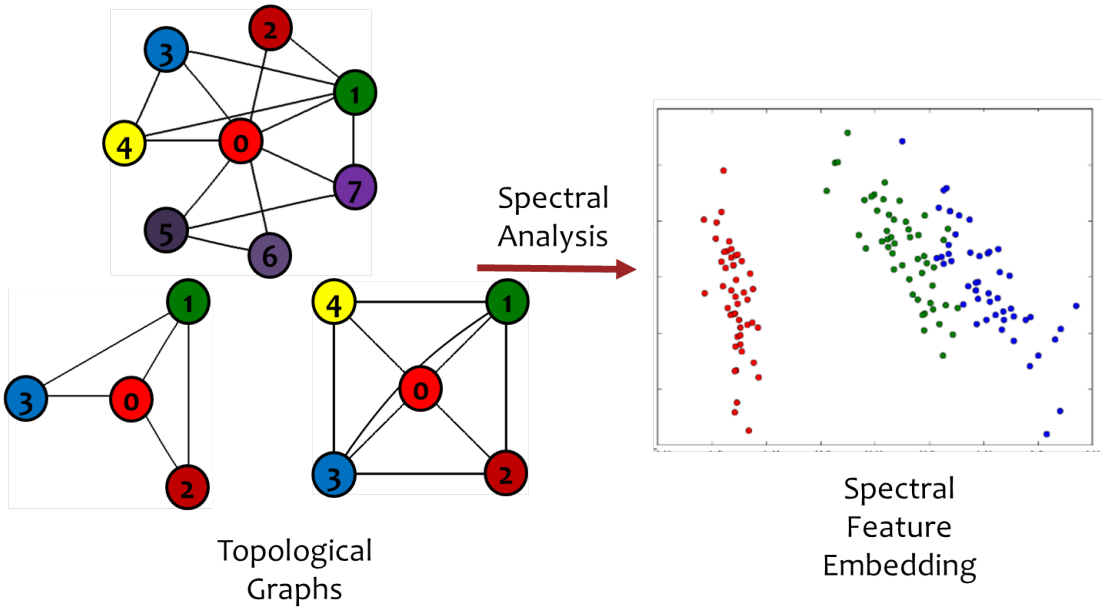


Figure 3.7. : Embedding features from the topological graphs into spectral space. Adjacency matrices with varying sizes are reduced to fixed length vectors in the pattern space. Kindly note, in the right hand side, similar vectors are clustered closely in the feature space.

, where, i is the Eigenmode index.

Spectral feature vector representing the spectrum of the graph G_k is constructed from the top n Eigenvalues of A_k taken in decreasing order. For the k^{th} graph, this vector is represented as :

$$\vec{F}_k = (\lambda_k^1, \lambda_k^2 .... \lambda_k^n)^T. \tag{3.5}$$

Figure 3.8 (a) represents a floor plan image and its corresponding segmentation into rooms. Figure 3.8 (b) represents the topological graph for this segmented layout depicting adjacencies and relationships between the rooms. Node colours correspond to the rooms in the floor plan. Figure 3.8 (c) is an adjacency matrix representation of this topological graph. This adjacency matrix is spectrally analysed and broken down into it's leading Eigen values arranged in a decreasing order as shown in Fig. 3.8 (d). This further forms the feature vector for each layout.

3.5.2 Spectral feature embedding

Once a feature vector is obtained as mentioned in Sec. 3.5.1 the next task is to analyse this feature vector and reduce it into a uniform size vector representing the layout information from each floor plan (refer Fig. 3.7, where due to feature extraction all graphs have been projected on a space of uniform dimension). Uniformity in the feature size is needed as floor plans in the dataset have different number of rooms leading to different sizes of adjacency matrices. Hence, to compare all the adjacency matrices for similarity there was a need to reduce them to a same size feature vector. To achieve this, principal components analysis is performed, following the parametric Eigenspace idea of [Murase and Nayar, 1994]. Rationale behind using this approach is to organise graphs into a pattern-space in which similar structures are close to one-another, and dissimilar structures are far apart. The graphs extracted from each image are vectorised in the way outlined in the Sec. 3.5.1. Principal components analysis (PCA) of the feature vector $\vec{F}_k$ is performed next. For the N

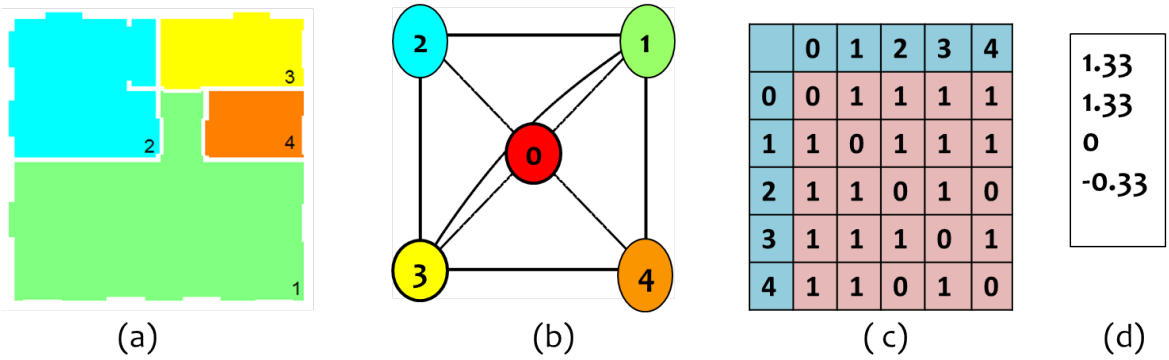


Figure 3.8. : Topological feature extraction from a room layout in a floor plan

different graph representations of the layouts in the database their spectra are arranged in a matrix as:

$$R = [\vec{F}_1, \vec{F}_2, \ldots, \vec{F}_N]. \tag{3.6}$$

The Covariance matrix, Cv for R is computed as,

$$Cv = RR^T. \tag{3.7}$$

A spectral decomposition of Cv results in Eigenvalues ρ upon solving the Eigenvalue equation

$$|Cv - \rho I| = 0 \tag{3.8}$$

and the corresponding Eigenvectors $\vec{\psi}$ are found by solving the Eigenvector equation.

$$Cv\vec{\psi}_i = \rho_i \vec{\psi}_i. \tag{3.9}$$

The Principal components directions are obtained by using the first three leading Eigenvectors of C. The three leading Eigenvectors give a wholesome estimation of the adjacencies in the layout [Luo *et al.*, 2003]. Three orthogonal vectors span the co-ordinate system of the Eigenspace as $\Phi = (\vec{\psi}_1, \vec{\psi}_2, \vec{\psi}_3)$. This aids in projecting the individual graphs represented by the vectors $\vec{F}_k; k = 1, 2, \ldots, N$ on the pattern space as

$$\vec{x}_k = \Phi^T \vec{F}_k. \tag{3.10}$$

Hence, each graph obtained from the layouts is represented as a three-component vector $\vec{x}_k$ in the Eigenspace as $\vec{x}_k = (x'_k, x''_k, x'''_k)^T$. After the computation of the features, in the next section, the matching technique is described.

3.5.3 Feature Matching and Retrieval

A two phase matching and retrieval, Room Layout Matching (RLM) and Room Decor Matching (RDM) is proposed. Feature matching and retrieval of similar layouts (RLM) is performed by determining the proximities between all the layouts. The distance (d) between query graph's feature vector $\vec{x}_q = (x'_q, x''_q, x'''_q)^T$ and that of the model graph's feature vector $\vec{x}_m = (x'_m, x''_m, x'''_m)^T$ is calculated as:

$$d = \sqrt{(x'_q - x'_m)^2 + (x''_q - x''_m)^2 + (x'''_q - x'''_m)^2} \tag{3.11}$$

.

The smaller the distance more similar the graphs are. Thus, this similarity measure helps us to order the architectural layouts represented as graphs. Rank ordering of the images helps in generating the most similar images to the query input and can efficiently retrieve exactly matched or approximately similar layouts from the database to suit the application.

After the first level matching of the layouts the next level match to find the semantic difference between the layouts (RDM) is done. An algorithm to compare the number and type of the objects inside the room (see Alg.2) is proposed, where $\mathscr{T}$ represents type of furniture, $\mathscr{C}^u$ represents number of unique furniture in the whole floor plan. After determining the closest layout, the unique furnitures present in each room are identified. To compute the semantic difference, between a pair of layouts, the number and type of the objects present inside the rooms of the query layout is compared with the model layout. Further a matching cost ($cost_{qm}$) is assigned which increments if there is an object level matching between the layouts and decrements otherwise. The final $cost_{qm}$ obtained from Alg.2 is used to rank order retrieved results based on semantic difference. A flowchart for Room Decor Matching is also provided in Fig. 3.9 to depict the various steps involved in the RDM framework.

Algorithm 2 Calculate semantic difference between layouts

Input: Query Image ($\mathscr{I}_q$), Model Image ($\mathscr{I}_m$)
Output: Matching Cost ($cost_{qm}$)

$\mathscr{T}_q = \{\mathscr{T}_q^i\}; 1 \leq i \leq |\mathscr{C}_q^u|$ ▷ $\mathscr{C}^u$: No. of unique items
$\mathscr{T}_m = \{\mathscr{T}_m^j\}; 1 \leq j \leq |\mathscr{C}_m^u|$ ▷ $\mathscr{T}$: type of furniture
$cost_{qm} = 0$
if $(|\mathscr{C}_q| \neq |\mathscr{C}_m|)$ **then** ▷ $\mathscr{C}$: No. of items
    $cost_{qm} = -1$
else
    for all i, j **do**
        if $\mathscr{T}_q^i = \mathscr{T}_m^j$ **then**
            $cost_{qm} \leftarrow cost_{qm} + 1$
        else
            $cost_{qm} \leftarrow cost_{qm} - 1$
        end if
    end for
end if
    return $cost_{qm}$

3.6 EXPERIMENTS AND RESULTS

The experiments are performed on the SESYD dataset [Delalandre *et al.*, 2010a] and the ROBIN dataset [Sharma *et al.*, 2017]. The SESYD dataset has 10 different types of layouts, each with 100 images varying in their arrangement of furnitures inside the rooms. The image sizes in the database vary from 6775×2858 to 2056×1837. On the other hand, the ROBIN dataset has about 510 floor plans each divided into 3 broad categories based on the number of rooms present inside them and varying in the global structure. All floor plans are binarized to ensure that only structural information of the floor plans is used for the analysis.

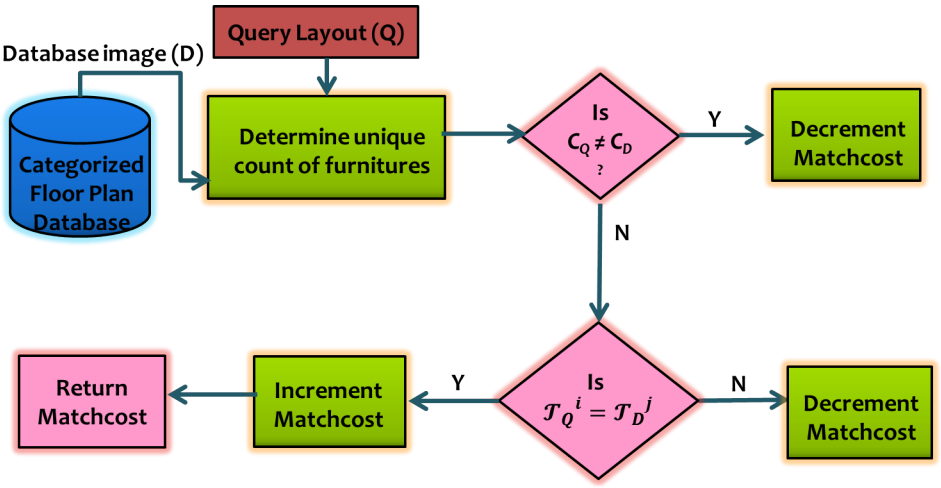


Figure 3.9. : Framework for Room Decor Matching

Figure 3.10. : Top 5 retrieved results corresponding to a query from the SESYD dataset.

3.6.1 Qualitative Results

The framework starts off with segmenting the architectural layouts into rooms followed by, detecting the various furnitures present in each room. A semantic representation of the layout graph (see Sec. 3.3) is obtained, by connecting the furniture nodes with the room node that they belong to using dash-dot edges.

The proposed RLM and RDM techniques, helped in rank ordering of the matched layouts based on the semantic difference i.e. differences in the arrangement and number of furniture inside the rooms. Figure 3.10, shows qualitative retrieval result corresponding to a query from the SESYD dataset. The red bounding boxes depict a change of decor between the query layout and the retrieved layouts, thus leading to dissimilar layouts being retrieved later than the similar ones

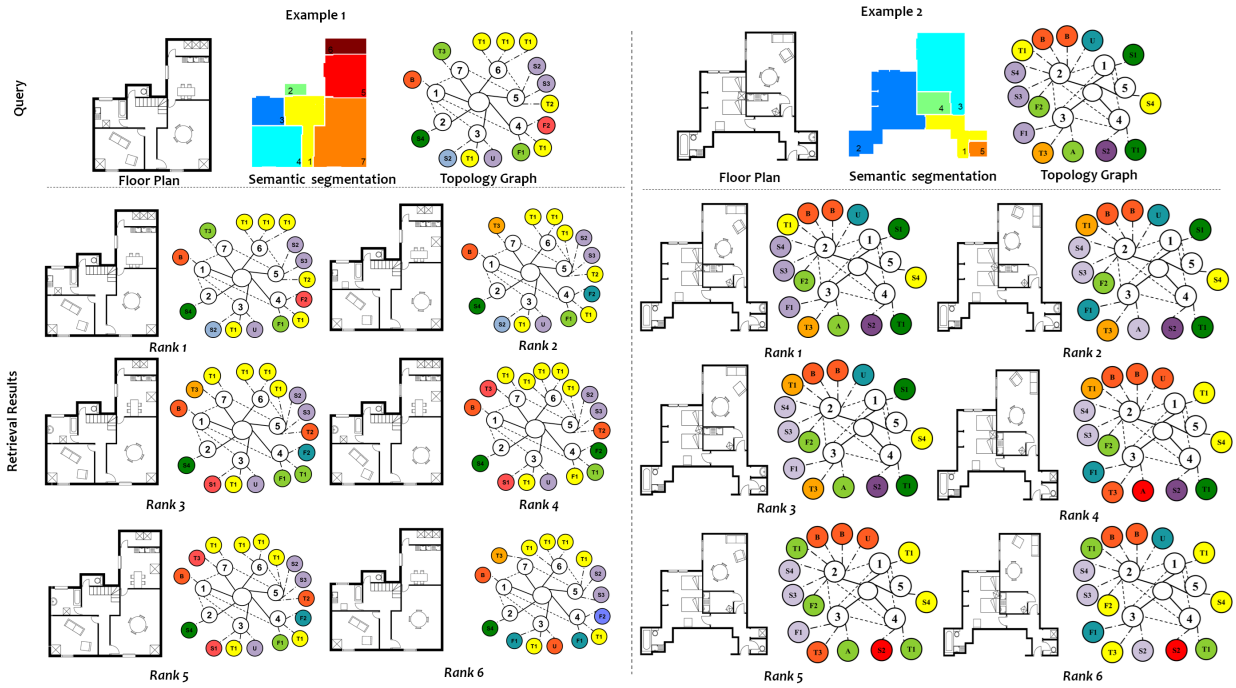


Figure 3.11. : Retrieval results for two different query architectural floor plans taken from the SESYD dataset. The top row shows the query image, the corresponding room layout and the semantic graph generated from the floor plan image. For the given query images only the top six retrieval results are shown.

during matching. As the result is demonstrated on SESYD dataset and it is observed that within a category in SESYD dataset only the decor arrangement differs, and the layout remains the same. Hence, the first rank ordered result is the query layout itself. The subsequent results differ in the placement, position, type and number of the furniture present inside the floor plan. For example, the Rank 5 result differs from the query floor plan in terms of a table's orientation as well as lacks a small table in comparison to the query. Hence, is retrieved later. To elaborate further, the rank ordered retrieval results along with their corresponding semantic graphs for two such queries, are shown in Fig. 3.11. The first row in Fig. 3.11 represents the query image, its room level segmentation, and the corresponding semantic graph. The following rows correspond to the top six rank ordered results along with their semantic graphs. It can be observed that the layout category in the rank ordered results is the same as that of the query and thus, ensures high retrieval accuracy. The first rank ordered result obtained from the layout database is the query image itself due to minimum distance from the query. The subsequent results differ in the type, number and orientation of furnitures inside each room.

After explanation of how the retrieval results differ in their layout/ decor arrangement, further, more qualitative results on both SESYD dataset and ROBIN dataset are shown in Fig. 3.12. In Fig. 3.12, the subfigures 1. (a) and (b) correspond to the query given from SESYD dataset and the subsequent rank-ordered results. While, 2. (a) and (b) correspond to the query given from ROBIN dataset. On closer observation it can be seen that retrieval framework performs well while layout matching and all layouts from the same category are retrieved as can be seen form the same global shapes of the layouts in the query as well as the retrieved results.

Figure 3.12. : Retrieval results for two different query architectural floor plans taken from the SESYD dataset and the ROBIN dataset. 1. (a) and (b) correspond to the query given from SESYD dataset and the subsequent rank-ordered results. 2. (a) and (b) correspond to the query given from ROBIN dataset

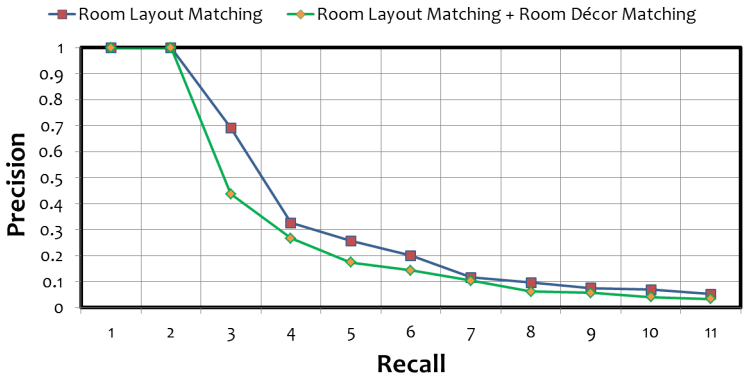


Figure 3.13. : Quantitative result comparing the RLM and the RDM approach on the ROBIN dataset

3.6.2 Quantitative Results

Quantitative analysis of the proposed algorithm was also done to observe the results. Precision (P) and Recall (R) are calculated using the equations mentioned below:

$$Precision(P) = \frac{|\{\text{relevant floor plan images}\} \cap \{\text{retrieved floor plan images}\}|}{|\{\text{retrieved floor plan images}\}|} \tag{3.12}$$

$$Recall(R) = \frac{|\{\text{relevant floor plan images}\} \cap \{\text{retrieved floor plan images}\}|}{|\{\text{relevant floor plan images}\}|} \tag{3.13}$$

Figure 3.13 shows the PR plot obtained by taking query from the ROBIN dataset and retrieving based on layout matching and a combination of layout and decor matching separately. In Fig. 3.13, the blue line marked with red squares denotes the PR plot obtained through layout matching (RLM) and the green line with yellow diamond markers depicts PR plot obtained with combination of both RLM and RDM. This study helped in observing how well the algorithm behaves in case of both matching strategies taken exclusively. The initial Recall value was obtained as 1 due to the first rank ordered retrieved result to be the same as the query. The subsequent results had some differences in the layout and decor. It was observed that taking the global layout, i.e. the placement of rooms inside a floor plan was a broader way of classifying the layouts, hence, the area under the PR plot with layout matching was obtained to be greater. This also owes to the fact that as more detailed decor characterization and matching is looked into, it leads to a slightly poor categorization of the layouts because of the specificity while retrieval.

Similarly, quantitative analysis of retrieval performance on the SESYD dataset was also performed. As discussed before, SESYD dataset has large differences between the layouts across its 10 categories but inside a category, all the layout shapes are the same, only differing in their decor arrangement. The values of the distance metric depicting the similarity between layouts from SESYD dataset are shown in Figure 3.14. In Fig. 3.14, the x-axis corresponds to the ten different categories of layouts present in the database, whereas, the y-axis corresponds to the distance value between two layout categories in terms of the distance between the feature vectors representing them. The nodes depict the distance of each category from the 10 categories in the database. For example, as shown in Fig. 3.14, yellow line with triangles corresponds to the distances with respect to layout category 3 in the SESYD database. Note that the distance is 0 with respect to category 3 and highest with respect to category 9. This is due to the very large difference in the layout characteristics between category 3 and 9. It is evident from the plot that our proposed RLM technique (see Sec. 3.5.3) yields high intra-category similarity and low inter-category similarity between layouts, as expected. To illustrate further, one can see in Fig. 3.15 for layout 8 and 4 that the first rank retrieved result is layout 8 itself due to exact matching between the query and the Rank 1 result. Further, Rank 2 result has similar number of rooms and adjacency information as the query. Other retrieved results follow suit. Due to high inter-class dissimilarity between floor plans in the SESYD dataset, a perfect Precision value was obtained for all the recall values while retrieval.

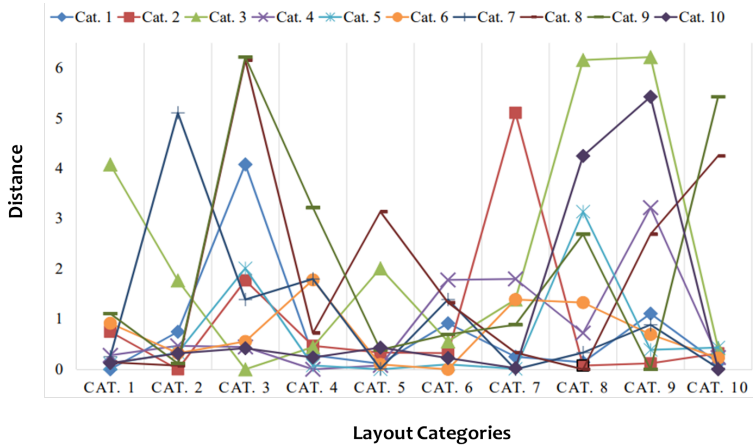


Figure 3.14. : Plot of intra-class and inter-class distances between layouts using our feature matching technique on the SESYD datset.

3.7 DISCUSSION

The framework discussed in this Chapter efficiently performs the various processing steps starting from, wall detection to room segmentation, graph creation of the layouts, feature computation, graph matching between the layouts, object level matching between the layouts and finally rank ordering the results. Another important aspect to be analyzed here is the search time. To compute the search time of the proposed approach time taken in seconds is computed at every step for the SESYD dataset. This time is averaged over 1000 samples belonging to a single class in the SESYD dataset. Figure 3.16 shows the class-wise computation time-ratio for the layouts in the dataset. Colour coding signifies various components of the proposed framework and helps establishing correspondence to understand which step performs computationally efficient in the proposed framework.

It can be observed that the wall detection step took an average time of 4.481 secs. for all the layout categories. Segmentation into rooms was a little computationally expensive step owing to a number of intermediate processing steps. The average time taken by the segmentation step was 18.047 secs. Graph generation and spectral feature embedding of the graphs took an average time of 5.513 secs. and 5.292 secs. respectively. Figure 3.16 shows this analysis, where, values shown in the bar chart denote the ratio of time taken in seconds for each step to the total time taken for execution.

As discussed in Sec. 3.3, graph matching is a computationally expensive process. To avoid this spectral graph embedding is proposed in this Chapter, which creates uniform three component vectors for representing all the graphs, which helps in clustering similar graphs closer in the pattern space. Therefore, this technique considerably reduced the computation time while matching the layouts. The average time taken at this step is 7.569 secs.

Analysis revealed that the furniture level matching inside the layouts was the most computationally expensive step in the proposed approach. This step took the maximum average time of 154.036 secs for every layout because of 1) iteratively searching for the arrangement of furniture inside each layout, and 2) large dimension (6775×2858) of the layout images. An efficient indexing method or hashing technique for the second level match could yield better results.

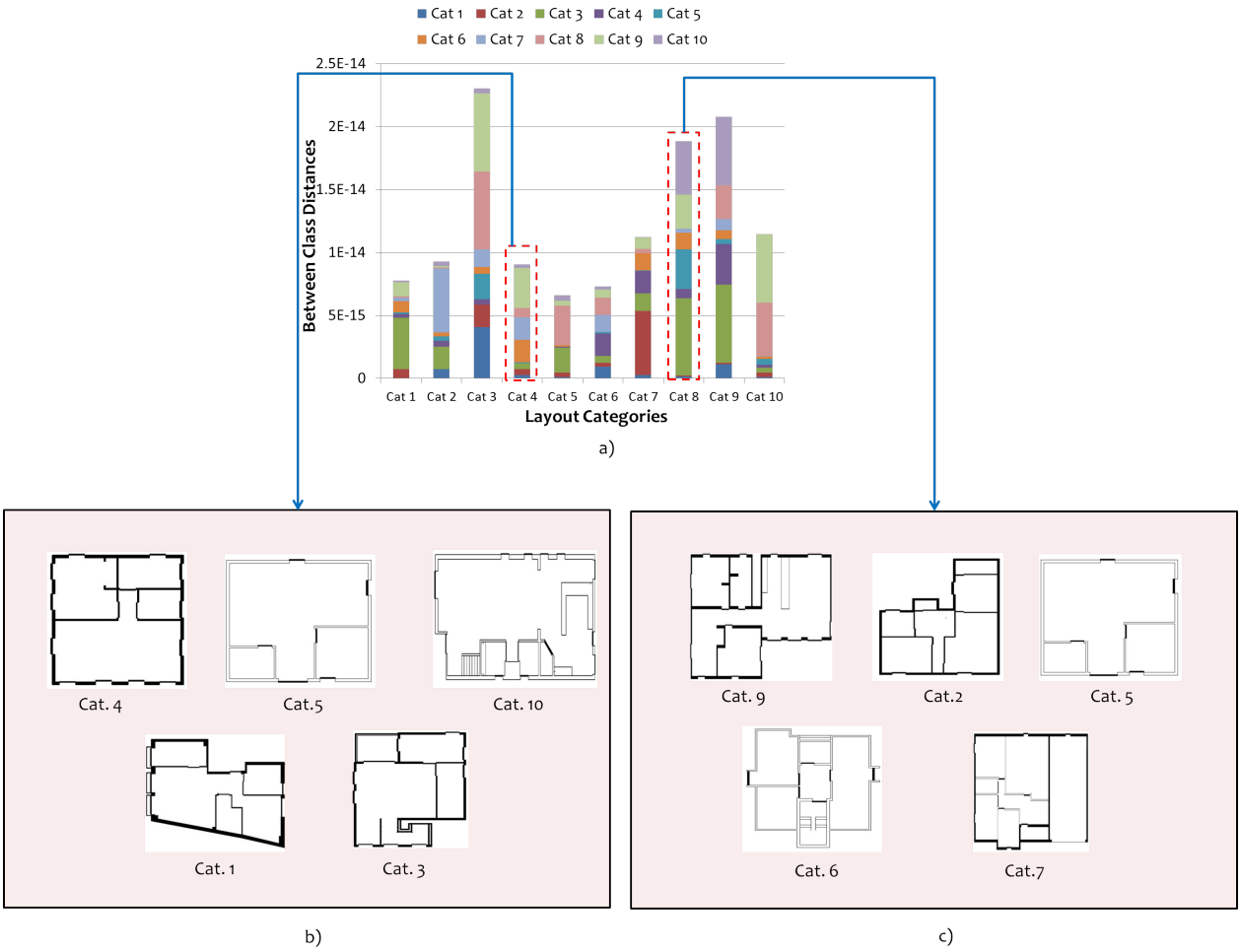


Figure 3.15. : Layout matching from floor plans taken from SESYD dataset. Two such results from Category 8 and 4 are highlighted with a red bounding box.

Similar analysis was also performed for the ROBIN dataset. As the classes in ROBIN dataset are 51 in number, therefore, it is not feasible to show class-wise search time graph for the ROBIN dataset. Alternatively, in Fig. 3.17 the search time taken in seconds averaged over all the 510 samples in the ROBIN dataset, at every step of the framework is represented as a bar-graph. It was observed that the most computationally step was the decor matching step again taking an average time of about 7.34 secs., which results from the fact that, iteratively searching for the arrangement of furniture inside each layout is computationally expensive. On the other hand, the most computationally efficient step is the Automatic Room Detection step.

As this was a novel baseline framework proposed in this Chapter, therefore, due to a lack of availability of the implementation of such a composite framework for retrieval and matching of floor plans in the literature, there is an inability to compare the results with other approaches. Although, in Fig. 3.18 the comparison of segmentation step to the one proposed in [de las Heras *et al.*, 2014] is shown. Where, segmentation results on two images from CVC-FP [de las Heras *et al.*, 2015] dataset are shown, with, the left hand side being (a) the input floor plan and (b) and (c) correspond to the technique proposed in [de las Heras *et al.*, 2014] and the proposed approach respectively. It can be observed that the proposed segmentation technique performs at par with the technique proposed in [de las Heras *et al.*, 2014].

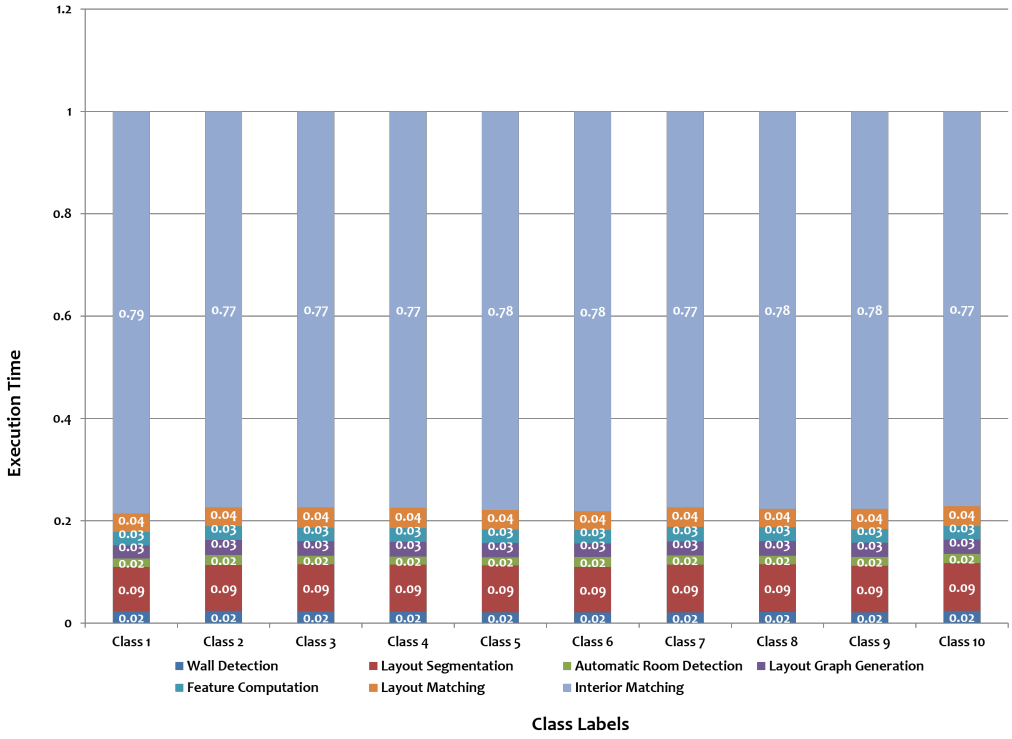


Figure 3.16. : Execution time for the intermediate stages of processing of the framework for different categories of floor plans on the SESYD dataset. Values shown in the bar chart denote the ratio of time taken in seconds for each step to the total time taken for execution.

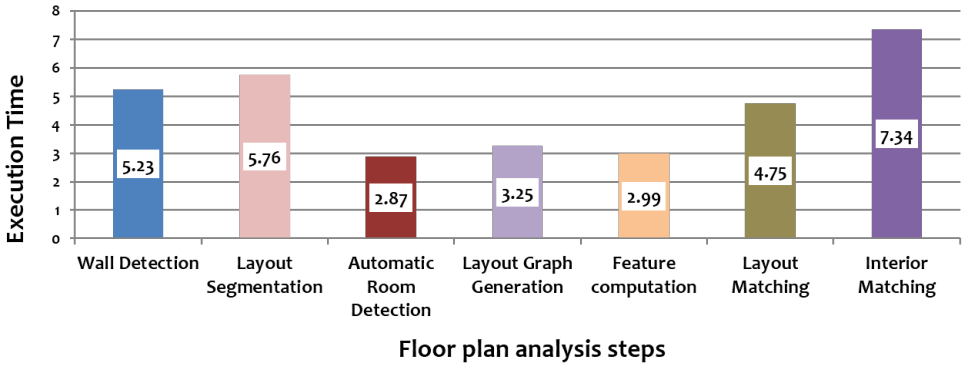


Figure 3.17. : Execution time for the intermediate stages of processing of the framework averaged over all 51 categories of floor plans in the ROBIN dataset. Values shown in the bar chart denote the time taken in seconds.

The approach proposed in [Weber *et al.*, 2010] taking sketch as a query to retrieve floor plans is partly related to the work proposed in this Chapter. However, the key difference is that the authors in [Weber *et al.*, 2010] represent query layouts as sketches using only primitive shapes

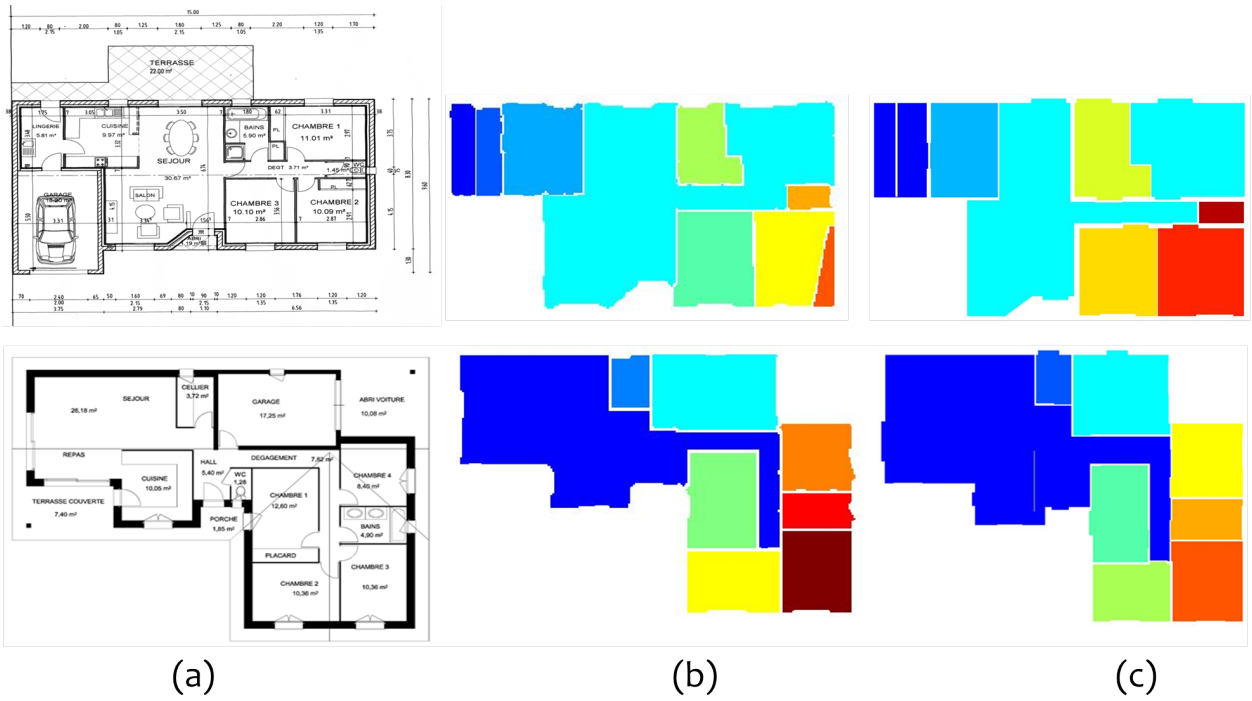


Figure 3.18. : Segmentation results on two images from CVC-FP [de las Heras *et al.*, 2015] dataset (a) Input floor plan, (b) Technique proposed in [de las Heras *et al.*, 2014], (c) Our approach

such as rectangles. In contrast, in this Chapter the room decor represented by complex geometric primitives is considered. Moreover, in [Weber *et al.*, 2010] the adjacencies between the rooms represented in the layouts, in sketch form were very simplistic in nature which is quite complex in our case. Furthermore, they have ignored the representation and matching of the room decor between the layouts.

3.8 SUMMARY

In this Chapter, a framework for searching floor plans by structural analysis, as well as, semantic similarity is proposed. Inclusion of the room decor arrangement during retrieval is a novel idea that can find application in scenarios where the user has specific choices in terms of the fully furnished floor plans. Furthermore, a spectral embedding approach to represent graphs obtained from the layout is proposed in the baseline retrieval framework proposed in this Chapter. This technique helped in reducing the computation time taken during the graph matching stage to a considerable level.

In this Chapter, handcrafted topological features were proposed to be extracted from the floor plans. Topology and room decor features although help in distinguishing two floor plans and retrieving similar ones, but inclusively are not a good enough measure to analyse a floor plan completely. It is very difficult to curate handcrafted features powerful enough to extract all the details from a floor plan. On the other hand, deep neural networks are able to analyse a floor plan in a minute manner extracting all the relevant features at once. In the literature, deep neural network based features have shown promising results for the retrieval of natural images. However their application to floor plan images has not been researched. Therefore, in the next chapter, deep

neural networks are explored to extract structural and semantic information from a floor plan for the retrieval task.

4

Deep convolutional neural networks for Floor plan Analysis and Retrieval

In Chapter 3, a baseline retrieval framework was explored by extracting topological features for understanding the layout in a floor plan. Additionally the decor features were extracted, to understand the room semantics better. During retrieval, these two features are combined and the floor plans are rank ordered based on the feature similarity. For efficient and accurate characterization of a given floor plan, as well as coming up with a feature representation that is invariant to various perturbation, feature engineering is a key component. Broadly the feature engineering approaches are classified into two categories, namely: hand-crafted features and machine learned features (deep features). The features discussed in Chapter 3, are hand-crafted. For hand-crafted features the pipeline for feature extraction is designed such that it is particularly suitable for the underlying application, and sufficient to meet the requirements. These features are robust and give satisfactory retrieval performance. However, the hand-crafted features have the following deficiencies:

1. The hand-crafted features mostly reflect the low-level identities of the objects present in the scene or the image under consideration. As a result these features are not directly suitable for representing the mid-level or even high-level depiction of the objects and require further processing or integration with additional knowledge.

2. During the feature extraction process the image level information is ignored, i.e. the features extracted are independent of the content present in the image.

This limits the matching algorithm to identify the similarities between two given floor plans. Recently, Deep Neural Networks (DNNs) have demonstrated their great power in computer vision and have achieved better performance for many tasks. For example, Convolutional Neural Networks (CNNs) have achieved good performance for ImageNet image classification [Russakovsky *et al.*, 2015]. The CNNs automatically learn the image representation from the images. The image labels are used to make the image representations separable. Such CNN learned features are termed as the machine learned features. These machine learned features capture the distribution information about the underlying data, and such information is especially important to the image representation. Hence, in this Chapter a deep learning framework is proposed for the analysis and retrieval of similar floor plans, with query by example paradigm, where the queries are floor plan images.

The rest of the chapter is organized as follows: Section 4.1, gives a brief description of the proposed technique. Section 4.2 and 4.3 give an insight into the proposed deep architecture for floor plan feature extraction and retrieval. Section 4.4 analyses the qualitative and quantitative results obtained through the extensive experiments that were carried out. Finally, Sec. 4.5 concludes the Chapter.

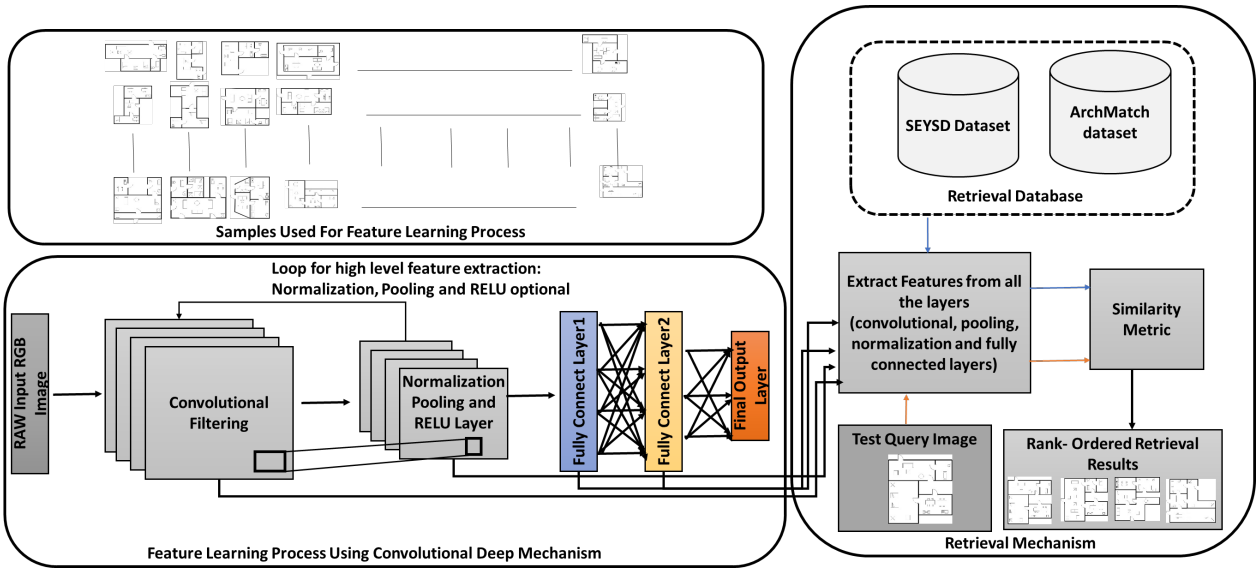


Figure 4.1. : Deep Learning Model based Framework for Floor Plan Retrieval

4.1 BRIEF OVERVIEW

Figure 4.1 depicts the complete framework of the proposed deep learning technique. A brief overview of the proposed framework for similar floor plan retrieval is as follows. The entire framework has been divided into two main phases. They are: (1) deep feature representation; and (2) matching and retrieval. In the deep feature representation task, deep neural network layers like Convolution, Normalization, Pooling and ReLU are used. In this work an additional Normalization layer is used which basically is useful when ReLU neurons are being dealt with. This is due to the fact that ReLU neurons have unbounded output and thus need to be normalized. If there is a normalization around the local neighborhood of the excited neuron, it becomes even more sensitive as compared to its neighbors. At the same time, it dampens the responses that are uniformly large in any given local neighborhood. If all the values are large, then normalizing those values diminishes all of them. Therefore, the normalization layer encourages some kind of inhibition and boosts the neurons with relatively larger activations. The output of all the layers mentioned above is passed through a network of fully connected layers to finally give the feature vector for an input image. The framework first learns the deep feature representation for floor plan retrieval task using a few samples from the ROBIN dataset. The learned deep representation helps to extract deep features from the samples of ROBIN, and stores it in the feature database. During the retrieval stage, the framework extracts the deep features from query image also using the same deep representation, where query image is both from the SESYD and the ROBIN dataset. The extracted deep features are then matched with the deep features in the database. A similarity score for a particular layout is thus calculated for all the query samples. To understand the effectiveness of the individual layers or feature representation of the deep learning network, the similarity is calculated using the features at the outputs of all the layers.

In this Chapter, the goal is to come up with new effective deep learning based feature representation and to investigate the effectiveness of learned features on floor plan retrieval tasks. In particular, two open issues are addressed:

- How to learn a new efficient deep learning feature representation for floor plan retrieval task?

- How the individual deep feature layers affect the performance of the retrieval system?

The proposed architecture has two main components, (1) Feature Representation and (2) Matching and Retrieval, which are discussed in detail in the subsequent subsections:

4.2 FEATURE REPRESENTATION

Traditionally, feature extractors are manually engineered and optimised through a laborious trial-and-error cycle involving re-learning the classifiers. In this work, the representation is learned using a CNN instead, by jointly optimising the performance of the features, as well as, of the classifiers. Existing literature also validates that deep features are a better way to represent an image. However, it is also shown that improper training of deep layers may lead to over-fitting or under-fitting of a model which can be a disadvantage.

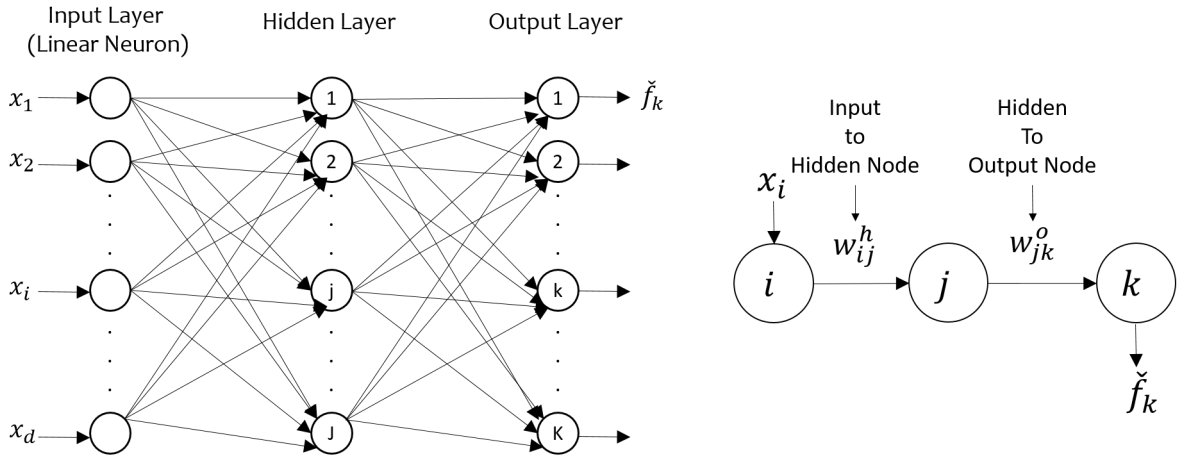
Convolutional Neural Networks (CNNs) have shown promising results for various computer vision tasks like classification and detection. Convolutional neural networks are similar to feed-forward neural networks. They are made up of neurons that have learnable weights and biases. Each neuron receives some inputs, performs a dot product and generally follows it with a non-linearity. The entire network expresses a single differentiable score function: from the raw image pixels on one end to class scores at the other end. There is a loss function on the final fully-connected layer. The difference that convolutional neural networks have from feed-forward networks is that, these architectures make the explicit assumption that the inputs are images. This assumption then leads to a vast reduction in the number of parameters in the network. Convolutional neural networks have a variety of layers serving different purposes. Using the approach of CNNs, many classification related tasks have been done in the past. It was observed that researchers have not extensively explored the CNNs for floor plan retrieval tasks. The work proposed in this Chapter uses a combination of convolution, pooling, normalization, ReLU, and fully connected layers along with dropout regularization technique (see Fig. 4.1). Before going into the detailed description of the proposed architecture, in the next sub-section, for the sake of completeness, a brief description of multilayer feed forward neural network and its relation to CNNs is presented.

4.2.1 Multilayer feed-forward neural network Architecture

Multilayer feed-forward neural networks (MLFFNN) receive an input (a single vector), and transform it through a series of hidden layers. Figure 4.2 depicts an architecture of a typical MLFFNN. Each hidden layer has a number of neurons. Each neuron is connected to all neurons in the previous layer (Fig. 4.2(a)). Neurons in a single layer do not share any connections. The final fully-connected layer called the output layer represents the class scores. Let, the input feature vector $\bar{x}$ be of d dimension and there are k output classes (in a k-class classification problem). According to the Universal Approximation theorem, if $\Phi(.)$ is a non-constant, bounded, monotonically increasing continuous function, then there exists an integer J such that:

$$\tilde{f}(\bar{x}) = \sum_{j=1}^{J} \alpha_j \Phi_j \left(\sum_{i=1}^{d} w_{ij} x_i + w_{jo} \right) \tag{4.1}$$

is an approximate realization of the desired function $f(\bar{x})$. Here, J is the number of neurons in the hidden layer, and w_{ij}s are the synaptic weights between the pre-synaptic neuron i, and post-synaptic neuron j. The theorem guarantees that we will be able to design an artificial neural network with J



(a) Illustration of a typical multilayer feed-forward neural network

(b) Illustration of a connection between pre- and post-synaptic neurons, interconnected with synaptic weights

Figure 4.2. : Architecture of feed-forward neural network, with the input layer, a hidden layer and the output layer

hidden nodes and synaptic connections, which will be able to represent the input feature vector to the best possible extent. In Fig. 4.2(b), the synaptic weights between input to hidden and hidden to output layer are explicitly shown. To determine the class label for a given test sample, "winner takes all" decision logic is followed.

In case of MLFFNN, the input vector dimension plays an important role in the scalability of the entire solution. An increase in the value of d leads to increase in the number of synopsis, and hence the synaptic weights. As a result, there are huge number of parameters to be learned. For example, a given image of size, say, $32 \times 32 \times 3$, requires 3072 weights for each neuron. For a bigger image that is 200×200, each neuron in the first layer would require 120000 parameters. This network with a huge number of parameters, would not only take long to train, but may also be prone to overfitting.

On the other hand, the convolutional neural network has its origin deeply rooted at the seminal work by Hubel and Wiesel (Nobel prize awarded in the 1980's). Hubel and Wiesel's "simple" and "complex" cell's computational properties were later well-described by linear models and rectifiers. The key components incorporated in convolutional neural networks, which are similar to the various components of the visual cortex are:

- CNNs are spatially organized, so that nearby cells act on nearby parts of the input image (similar to the Retinotopic map)
- CNNs use spatially localized linear filters, which are followed by thresholding (resembling the simple cells)
- CNNs use pooling units to incorporate invariance to shifts of the position of the feature (mimics the complex cells)

Some recent work has shown that artificial neurons in convolutional neural networks have similar hierarchical structure to the visual pathway. CNN (refer Fig. 4.3) takes advantage of the fact

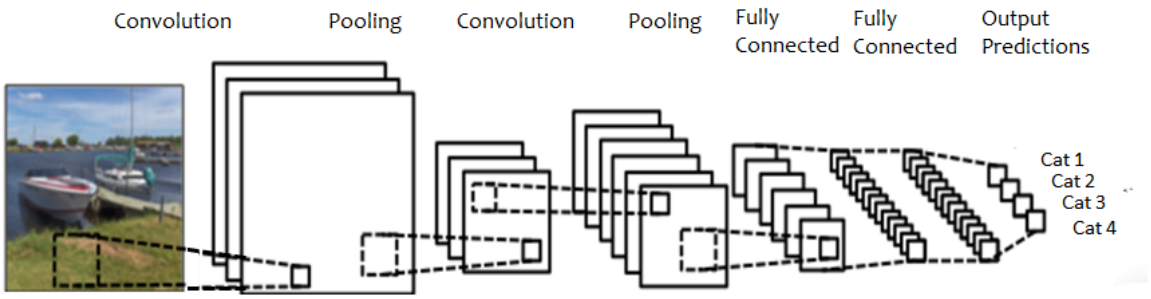


Figure 4.3. : Architecture of a Convolutional Neural Network Model for classification of an image [LeCun *et al.*, 2015]

that the input are images and they constrain the architecture. In particular, the layers of a CNN have neurons arranged in 3 dimensions: width, height and depth, where, depth refers to the third dimension of an activation volume.

4.2.2 Proposed CNN architecture

The Convolutional Neural Network (CNN) proposed in this Chapter for the classification of floor plans has an architecture comprising of an Input layer, a Convolution layer, a ReLU layer, a Pooling layer, a Normalization layer and a Fully Connected layer. Details of each layer are as follows:

- The Input layer holds the raw pixel values of the floor plan image.
- Convolutional layer computes the output of neurons that are connected to local regions in the input, each neuron computing a dot product (weighted sum) between its weights and a small region it is connected to in the input image. Figure 4.4 depicts an example of such a weighted sum between an input image and a filter. The convolutional layer defines a collection of filters (or activation maps), each with the same dimension as the input. Let, the input be of dimensionality (w,h,d), and the filter dimensionality be (w_f,h_f,d), typically $w_f < w$. The depths being equal means that the output of this convolution operation is 2D. The output slices of the convolution operations with each filter are composed together to form a $(w+w_f-1,h+h_f-1,n_f)$ tensor, where n_f is the number of filters. A convolutional layer

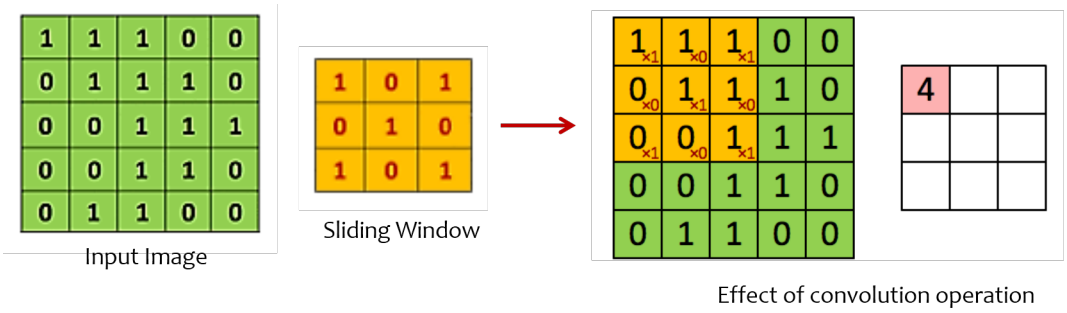


Figure 4.4. : Example of how convolution operation is performed on an image matrix.

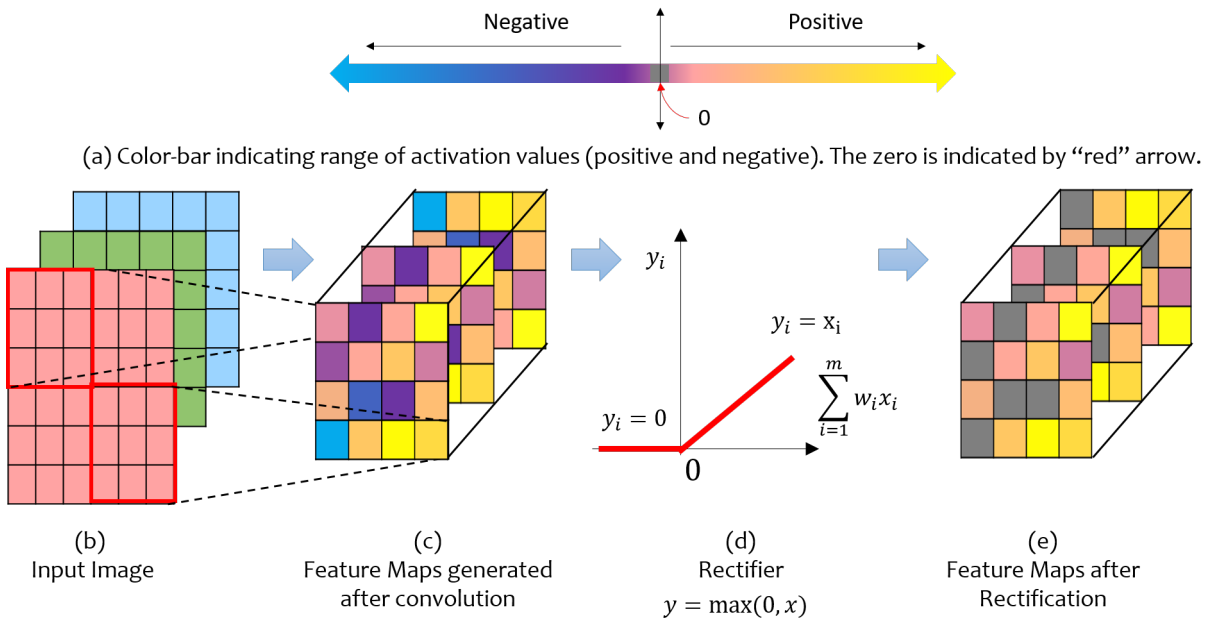


Figure 4.5. : Example of how a ReLU Operation works.

finally learns a set of n_f filters, $Filt = \{filt_1, filt_2 .... filt_{n_f}\}$ with input floor plan image I to produce a set of 2D feature maps:

$$z_k = filt_k \circledast I \tag{4.2}$$

where $\circledast$ is the convolution operator. Sharing of weights is done over the entire image which in turn results in the reduction of the number of parameters. Also, if there are m inputs and n outputs in a hidden layer, a MLFFNN would require $O(mn)$ operations to compute the output. Whereas, in CNN each output is connected to only $k = w_f h_f d$ inputs, and thus requires $O(kn)$ operations to compute the output, thereby reducing the computation time.

- In the network proposed in this Chapter, ReLU (Rectified Linear Unit) layer is used to induce non-linearity. The ReLU clips negative values to zero while keeping positve values unchanged. Figure 4.5 depicts how ReLU works. The popularity of ReLU is due to the fact that it highly enhances the acceleration of the convergence of stochastic gradient descent as opposed to the sigmoid/tanh functions due to its linear non-saturating form. It also avoids and rectifies vanishing gradient problem [Krizhevsky *et al.*, 2012b]. This is due to the fact that as seen in Eq. 4.3 the gradient maintains a constant value. An added advantage is that it is computationally less expensive. ReLU has the following mathematical form:

$$y = max(0, x) \tag{4.3}$$

It applies an element-wise activation function, in the form of $max(0,x)$ with thresholding at zero which leaves the size of the volume unchanged.

- In the proposed architecture, a pooling layer is used for reducing the size of the feature maps and to provide invariance between floor plan images with very little differences. Max pooling is usually preferred as it prevents blurring of the activations and gradients throughout the network because the gradient is placed in a single location during back propagation. It also avoids cancellation of negative elements (refer Fig. 4.6) [Cadène *et al.*, 2016; Yin *et al.*, 2016]. Pooling layer performs a down-sampling operation along the spatial dimensions.

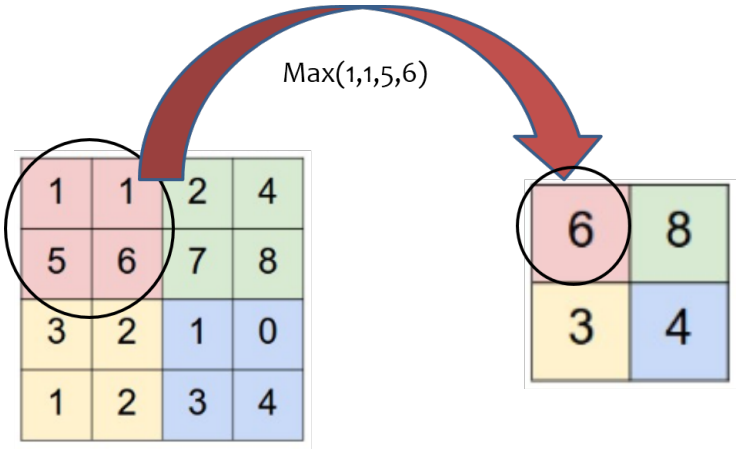


Figure 4.6. : Example of how a Max-Pooling Operation behaves on an image matrix.

- The next layer in the proposed architecture is the Batch normalization layer, which is widely popular as it converges faster [Ioffe and Szegedy, 2015]. The Batch normalization layer leads to an addition of a normalization step which helps in shifting the inputs to zero-mean and unit variance [Cadène *et al.*, 2016]. It allows activation functions to not get stuck in the saturation mode (gradient equal to 0).
- The fully-connected layer computes the class scores. Classification into categories is done by this layer. Neurons in a fully connected layer have connections to all activations in the previous layer.

In this way, the proposed Convolutional Neural Network transforms the original floor plan image, layer by layer from the original pixel values to the final class scores. In particular, the Convolutional/Fully Connected layers perform transformations that are a function of not only the activations in the input volume, but also of the parameters (the weights and biases of the neurons). On the other hand, the RELU/Pooling layers implement a fixed function. The parameters in the Convolutional/Fully Connected layers are trained with gradient descent so that the class scores that the Convolutional Neural Network computes are consistent with the labels in the training set for each image.

This proposed architecture takes inspiration from the methodology described in the work AlexNet [Krizhevsky *et al.*, 2012a], to obtain an effective feature representation from floor plan images (refer Fig. 4.7).

The basic highlights of the AlexNet framework [Krizhevsky *et al.*, 2012a] are:

- ReLU is used instead of Tanh to add non-linearity. This layer accelerates the speed by 6 times at the same accuracy for classification tasks.
- Dropout is used instead of regularisation to deal with overfitting.
- Pooling is overlapped to reduce the size of network. It reduces the top-1 and top-5 error rates by 0.4% and 0.3%, respectively.
  The AlexNet framework contains 5 convolutional layers and 3 fully connected layers. ReLU is applied after every convolutional and fully connected layer. Dropout is applied before the first and the second fully connected layer. Through Dropout, at each training stage, individual nodes are dropped out of the net to ensure that there is no overfitting in the network.

However, the model proposed in this Chapter is different from AlexNet in terms of two

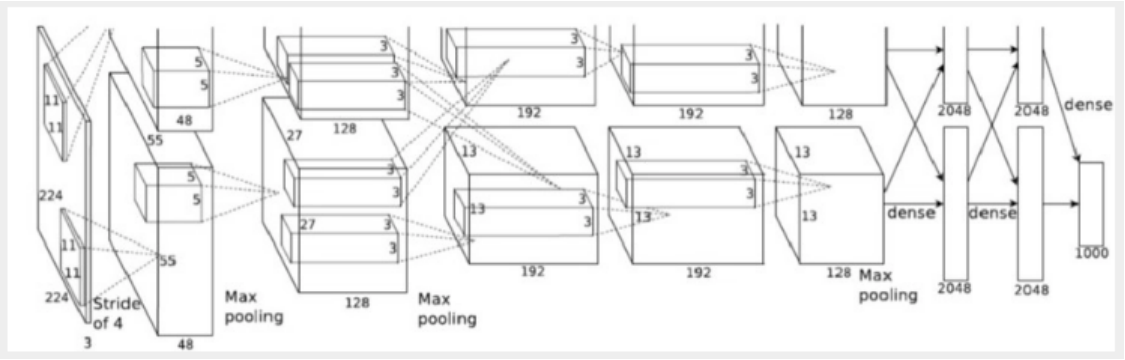


Figure 4.7. : Sequence of layers in AlexNet Framework [Krizhevsky *et al.*, 2012a]

properties:

- The network is not trained with data-augmentation.
- The order of pooling and normalization layers is switched (pooling is done before normalization) [Jia *et al.*, 2014].

The values of the parameters used in the framework such as, number of layers, type of layers, channels, filter sizes, convolution strides, pooling size, pooling stride and the padding size are listed out in Tab. 4.1. The parameters left blank in the table are the ones which do not have any role to play in that particular layer.

As an example taking the first layer, which is a combination of a convolution layer, a maxpool layer and a normalization layer. The size of the output image after passing through a convolution layer can be computed as:

Let, O_c = Size of the output image, I_c = Size of input image, K_c = Size of channels used in the convolution layer, S_c = Stride of the convolution operation and P_c = Padding Size. The size of the output image O_c is calculated using the equation below:

$$O_c = \frac{I_c - K_c + 2P_c}{S_c} + 1 \tag{4.4}$$

Now as, our input size I_c is 512×512. The first convolutional layer has 96 kernels of size 11×11. The stride is 4 and the padding is 0, therefore, the output image size is:

$$O_c = \frac{512 - 11 + 2 \times 0}{4} + 1 = 126 \tag{4.5}$$

The output image size is : $126 \times 126 \times 96$, corresponding to the number of channels. This image is then sent to the maxpool layer.

Let O_m be the size of the output image, I_m be the size of the input image to maxpool layer, S_m be the stride of the pooling operation and P_m be the pool size. The size of the output image is

Layers	1	2	3	4	5	6
Type	conv1+ maxpool1+ norm1	conv2+ maxpool2+ norm2	conv3	conv4	conv5+ maxpool3	full (fc6,fc7)
Channels	96	256	384	384	256	4096
Filter Size	11×11	5×5	3×3	3×3	3×3	-
Convolution Stride	4×4	1×1	1×1	1×1	1×1	-
Pooling Size	3×3	3×3	-	-	3×3	-
Pooling Stride	2×2	2×2	-	-	2×2	-
Padding Size	-	1×1	1×1	1×1	1×1	-

Table 4.1. : Network Parameters for the proposed framework

calculated using the following equation:

$$O_m = \frac{I_m - P_m}{S_m} + 1 \tag{4.6}$$

As per this equation, the size of the output image after passing through the first maxpool layer is:

$$O_m = \frac{126 - 3}{2} + 1 = 62 \tag{4.7}$$

So, the output image is of size $62 \times 62 \times 96$. After passing through the normalization layer, the size of the output remains unchanged, i.e. $62 \times 62 \times 96$. In a similar manner according to the parameters, sizes of the output across each of the layers are calculated.

For learning a representation of CNN (deep) layers, a few samples from the proposed floor plan dataset are used. Unlike the networks mentioned in the literature, the activations of all the

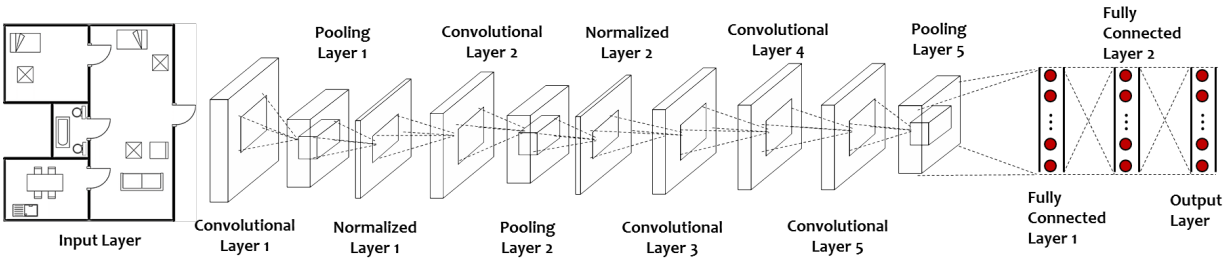


Figure 4.8. : Architecture of the Convolutional Neural Network (CNN) model used for feature extraction from the floor plan images.

layers (convolution, pooling, norm, and fully connected) for the task of floor plan retrieval are taken to investigate the effectiveness of each individual layer (see Fig. 4.8). The major motivation behind using lower convolution neural network layers is to understand the effectiveness of low level features as compared to higher-level feature representation for floor plan retrieval tasks.

To obtain the feature representation from query as well as the database samples, the images are directly fed into the input layer of the learned CNN model, and then the activation values from all the layers are taken (see Fig. 4.8). For conv, norm, and pool layers, the activation output is vectorized. Since computation from the feed-forward network based on the matrix multiplication is only needed once, the whole scheme is highly time efficient. In the next stage, i.e. matching and retrieval, query samples are matched with database samples to effectively retrieve the rank-ordered samples from the database.

4.3 MATCHING AND RETRIEVAL

Floor plan layout matching using the extracted features is essential for retrieving similar layouts to the query layout. In the rest of this section, the procedure to calculate matching score M for a pair (query and database) of layouts is described, corresponding to learned deep features.

Let N be the total number of images in the form of (x_ρ, y_ρ), $\rho = 1,2...,N$ where, x_ρ is the observed variable and y_ρ is the corresponding class label in a retrieval database. Let $F_H(x_\rho) : \{F_{H1}(x_\rho), F_{H2}(x_\rho), .., F_{HL}(x_\rho)\}$ be the set of hidden layer features extracted for x_ρ sample, where $F_{Hp}(x_\rho)$ represents p^{th} hidden layer feature extracted from x_ρ sample in database and L represents the total number of hidden layers in deep feature hierarchy. For experimentation purpose, features are extracted from 12 hidden layers of CNN model (see Fig. 4.8). For each sample in the database, features from each hidden layer ($F_{Hp}(x_\rho)$) are extracted and stored in the feature database as discussed. Let $F_H(q) : \{F_{H1}(q), F_{H2}(q), .., F_{HL}(q)\}$ be the the set of hidden layer features extracted from a query image (q). In this Chapter, analysis of the proposed framework is done using features features extracted from all the hidden layers:

- **Similarity based on each learned hidden layer feature representation**: In this approach, extensive experimental analysis was performed by comparing the performance of individual hidden layer for proposed floor plan retrieval task. The Matching Score (M) between query image (q) and an image sample from the database (x_ρ) using p^{th} hidden layer feature is calculated (and used for ranking), as:

$$M = \sum_{i=1}^{|F_{Hp}(x_\rho)|} |F^i{}_{Hp}(x_\rho) - F^i{}_{Hp}(q)|_2 \tag{4.8}$$

- **Similarity based on combination of hidden layer feature representation**: On the other hand, in this approach, the focus is on identification of the best performing hidden layer representation for each retrieval sample. This was achieved by combining the matching score outputs of each hidden layers using a *min* operation. The Matching score (M) between a query image (q) and a database image (x_ρ) using all the hidden layer features is calculated (and

Figure 4.9. : Rank ordered retrieval result of the proposed framework for five different query floor plans from the ROBIN dataset. Here, top five rank ordered floor plans are shown for each query.

used for ranking), as:

$$M = argmin_p \sum_{i=1}^{|F_{Hp}(x_\rho)|} |F^i{}_{Hp}(x_\rho) - F^i{}_{Hp}(q)|_2, \quad p \in (1,...,L) \tag{4.9}$$

The next section describes the qualitative and quantitative assessment of the proposed framework.

4.4 EXPERIMENTS AND RESULTS

The experiments were performed on the SESYD [Delalandre *et al.*, 2010b] and the ROBIN dataset to show the effectiveness of the framework. As already illustrated the SESYD dataset [Delalandre *et al.*, 2010b], contains 10 different types of layouts, each with 100 images varying in their arrangement of furniture inside the rooms. The image sizes in the database vary from

6775×2858 to 2056×1837. All floor plans are binarized to ensure use of only structural information while analysis. On the other hand, ROBIN dataset contains 510 floor plans divided into 3 broad categories based on the global layout shape.

The ROBIN dataset is split into two sets: (a) training - comprises of 30% samples and used for deep feature representation; and (b) testing - comprises of rest 70% samples and used as test (query) samples. As a retrieval database, the entire ROBIN dataset was used. Each floor plan image has been resized to a size of 512×512 for both matching and feature representation task. For deep model configuration, the architecture described in Fig. 4.8 is used, with same parameters as discussed in [Krizhevsky *et al.*, 2012a]. The same model pretrained on the ROBIN dataset is used for feature representation of the SESYD dataset.

4.4.1 Qualitative Results

The proposed framework is able to successfully retrieve a rank order set of floor plan images using the proposed deep learning framework. The top 5 rank ordered retrieval results for five different queries on ROBIN, are shown in Fig. 4.9. The arrow underneath the diagram depicts the decreasing order of similarity as one goes from left to right in Fig. 4.9 with green colour signifying high similarity and red signifying low similarity. The samples which are erroneously retrieved, i.e. not from the same class as that of the query class, are highlighted with "red" color. For all the results shown in Fig. 4.9, features extracted from the second normalization layer of the CNN stack were used. The rationale behind this choice was that the second normalizaton layer yields the best performance. Details analysis of such a selection is discussed in details in the next sub-section.

The first rank ordered result retrieved from the layout database is the query image itself. This is due to maximum matching score between the retrieved layout and the query layout. The subsequent results differ in the matching scores and are thus, ranked lower. In case of Fig. 4.9 (a), all the top 5 samples belong to the query class. Global layouts of the all the floor plans are identical and so is the number of rooms in the floor plans. The individual plans differ in terms of the number of furniture present in the rooms and their position. Readers are requested to note the difference in the shape and position of the individual rooms also for the rank ordered samples. There are 4 rooms in each floor plans, however the framework is able to rank order the samples correctly by ordering the most similar ones to the query first, followed by the rest.

Figure 4.9 (b-c) depict some of the failure cases. In Fig. 4.9 (b), the 4^{th} ranked result is erroneously retrieved (highlighted by "red") by the framework. However, the the 5^{th} sample is correctly retrieved. Justification of the rank 4 erroneously retrieved result can be understood through qualitative analysis. For example, room 1 in the query layout is adjacent to room 2, 3 and 4, that is identical to the adjacencies shared by room 1 in rank 2 retrieved result. Moreover, the size and type of furniture in the room 1 is same in case of the query and the rank 4 sample. Hence, similarity score between the query floor plan and the retrieved rank 4 layout is high, as compared to that with the rank 5 result. However, the similarity score of the rank 1-3 samples are marginally high as compared to rank 4 samples and hence they are ranked higher. In contrast, Fig. 4.9 (c) shows an instance where two out of top five retrieved results are incorrect. The rank 4 image has only three rooms, while the the rank 5 floor plan has four rooms in it. In this case, the framework is unable to capture the detailed features of the floor plans correctly. Even though, the number of rooms in a floor plan is different than the query, the framework ranks it higher.

Layer	conv1	pool1	norm1	conv2	pool2	norm2	conv3	conv4	conv5	pool5	fc6	fc7
MAP	0.503	0.527	0.554	0.537	0.557	0.561	0.539	0.537	0.541	0.536	0.472	0.465

Table 4.2. : Performance Comparison (MAP) of CNN layers used for floor plan retrieval task (on ROBIN dataset).

4.4.2 Quantitative Results

To quantify the performance of the framework the Precision (P) and Recall (R) metric is used. The precision values are averaged over all the queries for the particular recall values. Given a query floor plan image, retrieved layouts should belong to the same sub-category of layouts as the query, keeping in mind the preference set by the property seeker during querying. Remember that the global shape of the layout is the criteria for a given floor plan to belong to a certain class. The PR plot is shown for only the proposed *ROBIN* Dataset, as for SESYD dataset, the proposed and some of the existing state-of-the-art techniques [Sharma *et al.*, 2016a,b; Dalal and Triggs, 2005] yielded a flat PR curve (Precision value 1 for all Recall values). The reason behind such a flat curve for SESYD dataset is the simplicity of the dataset.

To understand the effectiveness of the proposed CNN layers, the Mean Average Precision (MAP) values were calculated for floor plan retrieval task using each layer separately (see Fig. 4.10 and Tab. 4.2). From Fig. 4.10 and Tab. 4.2, it can be concluded that: (1) it is not always true that the final layer i.e. the fully connected layer will always work best; and (2) normalization and pooling layers are more powerful as compared to convolutional layer for floor plan retrieval task.

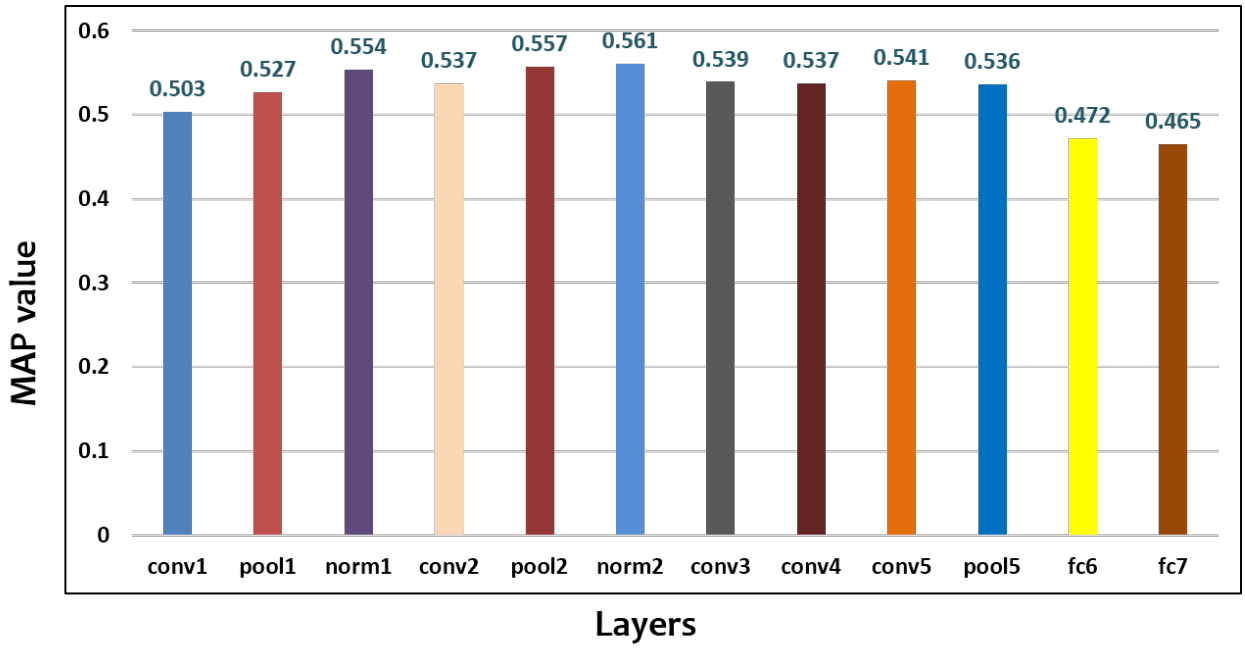


Figure 4.10. : Mean Average Precision values, comparing result of the framework on ROBIN dataset, using features from all the hidden layers of the CNN.

To show the effectiveness of the proposed framework, the technique proposed in this chapter

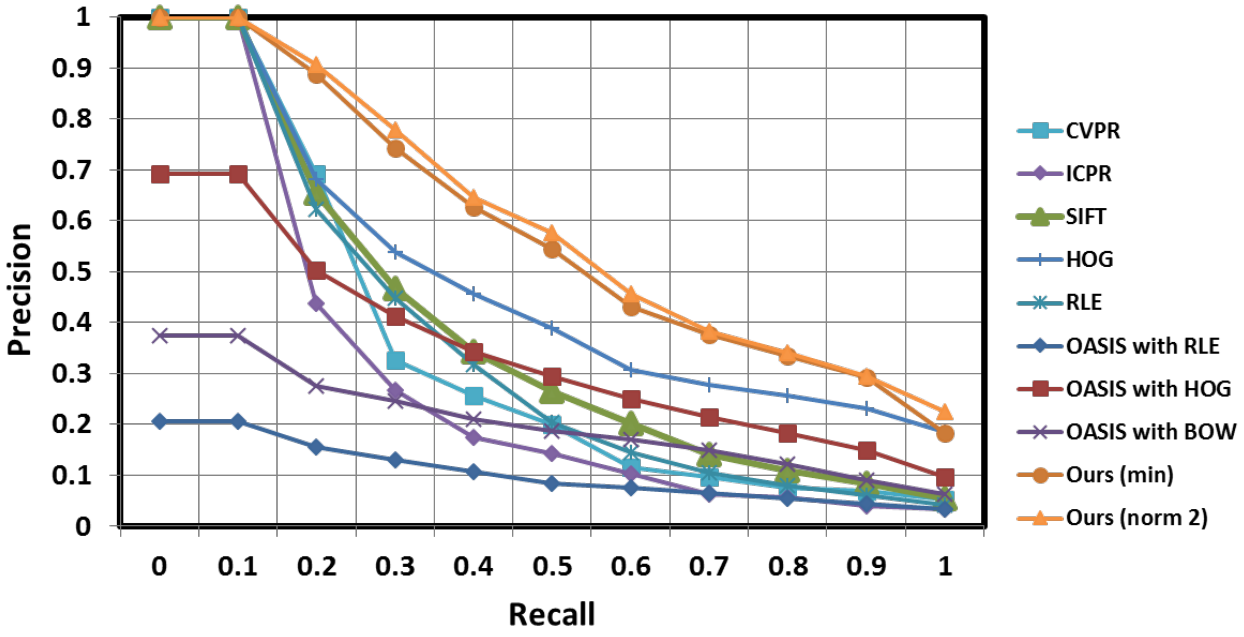


Figure 4.11. : Precision and Recall (PR) plot, comparing state-of-art methods with the proposed framework on ROBIN dataset.

is also compared with several state-of-the-art (SOA) techniques. Figure 4.11 depicts the PR plot comparing result of the framework on ROBIN dataset, with the other SOA techniques. It can be observed that the framework with features extracted from the Normalized layer 2 (see Fig. 4.10) is giving the best retrieval performance. It can also be noted from Fig. 4.11 that the proposed min based approach (which selects best layer for each retrieval sample, refer Eq. 4.9) is also outperforming all the state-of-the-art techniques. There is a significant improvement in the performance while using deep features as compared to other features or techniques.

As shown in Fig. 4.9 (a), given a query sample the rank 1 result is the query floor plan itself. This leads to the highest precision value of 1 during the initial recall. With further retrieval the average precision value decreases due to some incorrectly retrieved results belonging to other categories of layouts as compared to the query layout. It is to be noted that area under the PR curve is the highest for proposed deep learning based retrieval framework, as compared to other techniques applied for floor plan retrieval task as can be noted through the orange marker on the PR plot (Fig. 4.11).

The SOA techniques with which the proposed approach is compared with are standard image retrieval techniques that use hand-crafted features such as HOG [Dalal and Triggs, 2005], SIFT [Lazebnik *et al.*, 2006a], Run-Length Histogram and Online Algorithm for Scalable Image Similarity (OASIS) [Chechik *et al.*, 2009]) but are not able to capture well the complete content of the floor plan images. Also, reason behind the under performance of OASIS [Chechik *et al.*, 2009] is that, it focuses on the difference of similarity values between relevant and irrelevant image pairs, while ignoring the similarity value between highly similar images. Hence, when the same features (HOG, SIFT, RLH) are used under a canonical CBIR paradigm, better results are obtained in comparison to using them in conjuction with OASIS (see Tab. 4.3).

Techniques	**MAP**
CVPR [Sharma *et al.*, 2016b]	0.29
ICPR [Sharma *et al.*, 2016a]	0.23
HOG [Dalal and Triggs, 2005] based CBIR	0.40
SIFT [Lazebnik *et al.*, 2006a] based CBIR	0.33
RLH based CBIR	0.30
OASIS [Chechik *et al.*, 2009] with HOG	0.31
OASIS [Chechik *et al.*, 2009] with BOW	0.19
OASIS [Chechik *et al.*, 2009] with RLH	0.01
Ours (min)	0.54
Ours (norm2)	**0.56**

Table 4.3. : Performance Comparison in terms of MAP on *ROBIN* dataset.

4.5 SUMMARY

In this Chapter a deep learning framework is proposed to extract semantic features from floor plans and those features are used for matching and retrieval of similar floor plan images. Extensive experimentation was conducted and performance was compared with eight different state-of-the-art methods to demonstrate the edge that the deep learning framework has, as compared to the hand-crafted feature based retrieval. Although, deep learning is convenient and bypasses the need of curating features manually, the issue lies in the fact that the layers in the deep learning framework extract features implicitly. This implicit feature extraction at each layer does not give the user, the ability to give priorities to certain aspects or features while retrieval. For example, if a user wants to give higher preference to the feature "size of the rooms" while retrieval, then in the deep learning framework there is no such provision of giving additional weights to a particular feature. This, makes the application of the framework a little less intuitive. Hence, in Chapter 5, high level semantic analysis of floor plans and their subsequent retrieval is proposed. Additionally, a provision to adjust the weights of each feature while retrieval is given in the proposed technique. Such a provision can prove to be more practical to users while looking for similar floor plans sufficing their specific preferences.

5

Requirement driven feature integration for floor plan retrieval

In Chapter 4, a deep learning architecture for floor plan retrieval is proposed which was the first attempt of applying deep learning to the task of floor plan retrieval. A naive approach was introduced to learn a deep learning feature representation and study how the individual deep feature layers affect the performance of the system during retrieval. In the deep learning framework, a combination of convolutional, pooling, normalization, ReLU and fully connected layers in the Convolutional Neural Network (CNN) framework were used to obtain an effective feature representation from floor plan images. Quantitatively this framework gave encouraging results with a mean average precision of 0.56. However, certain key issues were observed during the retrieval process:

- The CNN framework is a simple network that mainly distinguishes by the global shape of a floor plan. It would fail while doing fine grain categorization of furniture components in the layouts. It would also not perform well if two layouts from different classes have almost similar shapes.

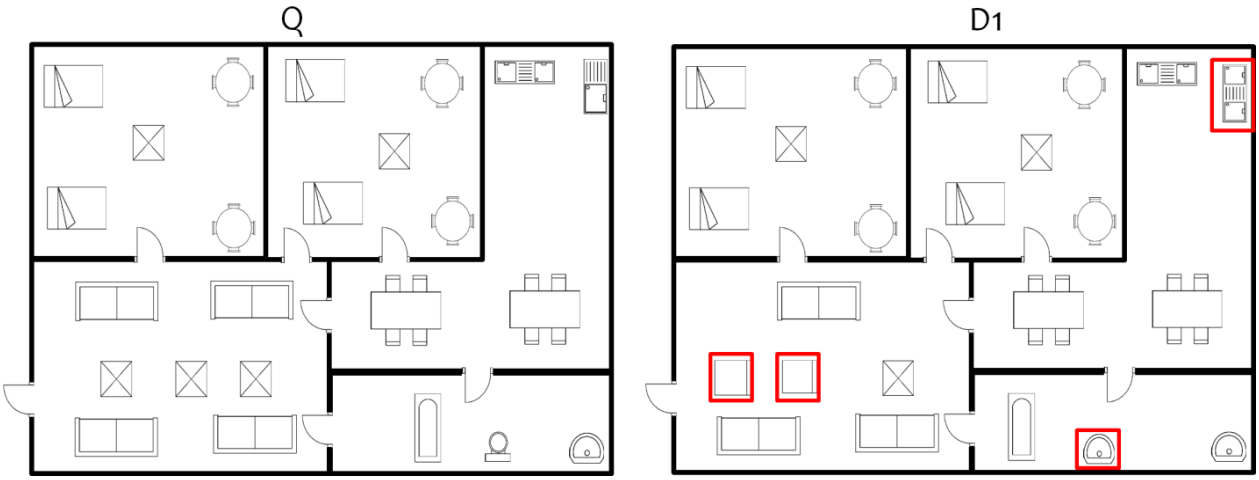


Figure 5.1. : Q depicts the query floor plan and D1 depicts a resultant retrieved floor plan from the database of floor plans. The objects enclosed in red boxes in D1 are furniture components with similar shape as the components in Q, but different types.

An example in support of the above argument is given in Fig. 5.1. Here, the shape of the square table and the small sofa (marked in red in D1) are quite similar, therefore, while retrieval through the deep framework, query floor plan Q and database floor plan D1 are deemed similar and D1 is ranked higher in the retrieved results, in spite of their dissimilarity. Moreover, the layout sub-categories in the ROBIN dataset are mainly proposed to be same in the global layout shape, which justifies the higher mean average precision value using DANIEL framework, but does not result into efficient qualitative results.

The other trait where the deep framework lacks is that there is no provision of a weighted combination of the features, which does not let the user give preferences to individual characteristics, according to the requirement. Choices cannot be set using the proposed deep framework, as the layers extract and combine the features implicitly without the provision of manipulating the weights by the user.

The hand-crafted approach, with non-deep features proposed in Chapter 3 was the baseline, and first of its kind framework for retrieval in floor plans. However, it had limitations like (i) the matching was done in two phase, i.e. layout matching followed by decor matching, and thus not a unified/ combined approach, (ii) a few key features were overlooked that define a buyer's requirement, e.g., room dimensions were not considered. In this Chapter, a framework for "fine-grained" retrieval of floor plan images from the repository is proposed.

The key contributions of this work are:

1. A novel end-to-end framework for extracting high level semantic features like the area and room-wise decor arrangement for the task of fine-grained retrieval

2. A technique to perform feature fusion to aggregate high-level semantic features to retrieve floor plans.

3. Extensive experimentation on a large dataset to demonstrate robustness and scalability. The technique proposed in this Chapter can find application in online property sale/rent scenarios where the buyer has preferred features related to the room semantics.

The Chapter is organized as follows: Sec. 5.1 gives a brief overview of the proposed framework. The fine-grained feature extraction process is presented in Sec. 5.2. The matching algorithm is described in Sec. 5.3. Details of the experimental findings are discussed in Sec. 5.4, followed by critical observations in Sec. 5.5. Section 5.6 summarizes this Chapter and leads the thesis towards the final approach.

5.1 BRIEF OVERVIEW

Figure 5.2 depicts the complete framework of the proposed technique. The entire framework has two phases: (i) feature extraction (solid line) and (ii) retrieval (dotted line). Structural analysis of a floor plan is performed. For this purpose, the layouts are segmented into rooms by identifying the walls separating all the rooms. Then, in a floor plan, the high-level semantic features like adjacencies between rooms, carpet area of rooms [Bannister, 2010] and also the number and type of furniture inside each room, termed as the Furniture Composition Record are extracted. During the retrieval phase, these features are extracted from a query floor plan and a weighted sum is computed of all the features extracted to generate a matching score for a particular layout as compared to another layout in the repository. By default, all the weights while matching are set to 1 to favor an equal sum of all the high-level semantic features. This score then serves as the parameter for fine-grained retrieval of similar floor plans. Additionally, any feature can be given a higher weight as compared to the other features during the matching process which further, aids in preferring a feature more during retrieval. Among the publicly available floor plan datasets, the *ROBIN* [Sharma *et al.*, 2017] dataset is chosen to validate the technique shown in this Chapter.

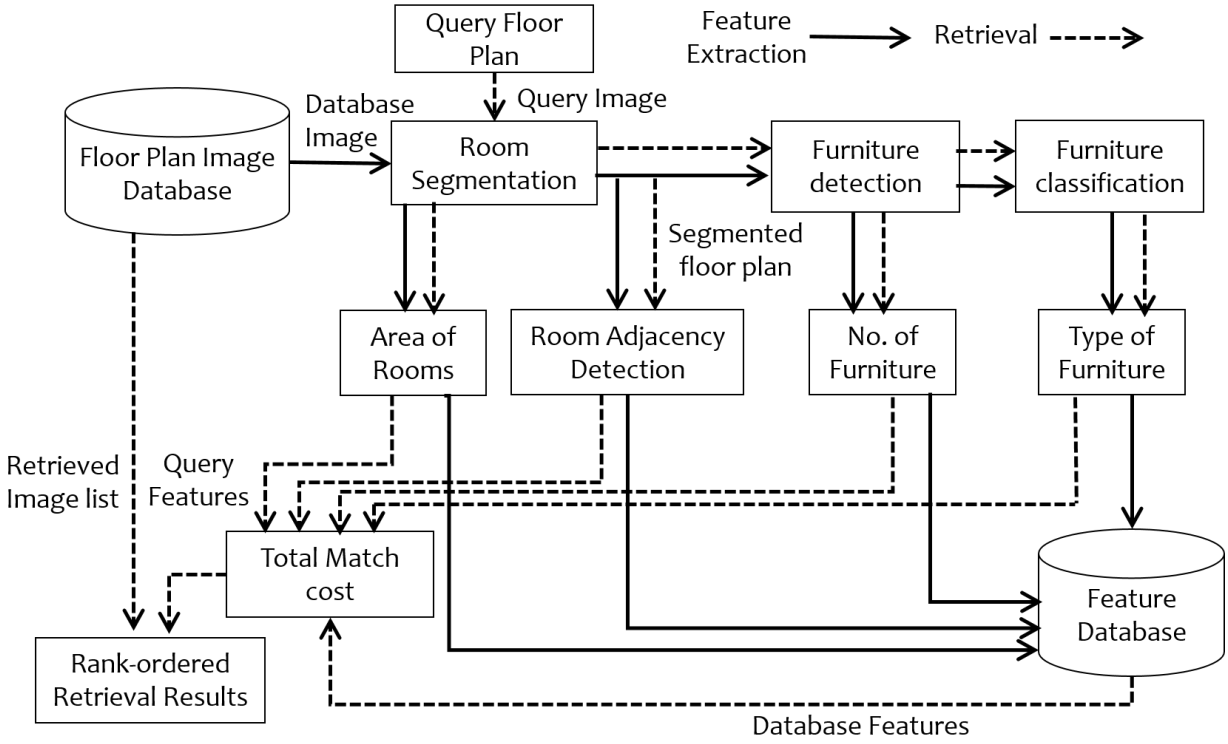


Figure 5.2. : Basic framework depicting extraction of high-level semantic features and weighted fusion for retrieval of floor plans.

5.2 HIGH LEVEL SEMANTIC FEATURES

As a preprocessing step, a morphological closing operation is carried out to segregate the layout boundary from furniture present inside the layout. The doors inside the plan are present in various orientations. They are detected and the gaps at their locations are closed to construct definite boundaries for the rooms. This recognizes the walls in the layout to represent the room structures and thus, delineates the rooms within the layout from each other. Next, connected component analysis using 8-neighborhood criteria is performed to label each room in the layout separately. The rooms in the layout are numbered to get a consistent ordering by their distance from a reference point situated at the top-left-most end of the floor plan image. A diagram showing the same is presented in Fig. 5.4.

Euclidean distance between the reference point taken to be as the top-leftmost point of the floor plan canvas and the centroid of each room is calculated. The closest room to the reference point, i.e., with the minimum distance to the reference point is labeled 1, and other rooms are labeled accordingly. Such an ordering makes the system rotation sensitive. Figure 5.3 (b) and (d) depicts the resultant room layout segmentation and labeling given a floor plan. Such an ordering helps in establishing a better correspondence between rooms while feature extraction and matching. The three features that are being extracted namely, Room Adjacency String, Carpet Area Ratio and Furniture Composition Record are described below.

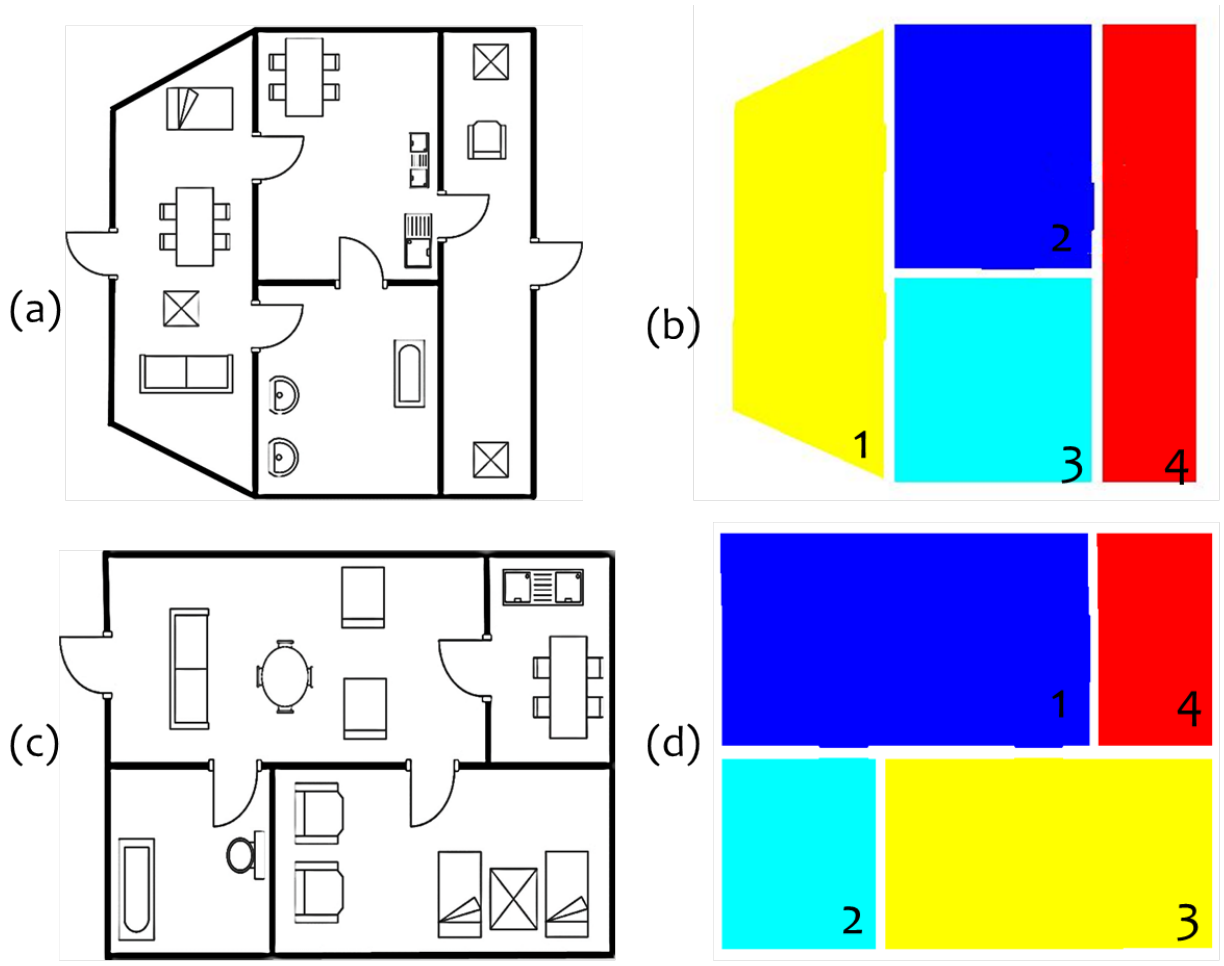


Figure 5.3. : Room level segmentation results on a pair of floor plan images from ROBIN dataset.

5.2.1 Feature 1: Room Adjacency String (RAS)

Room segmenting and labeling is followed by determining the adjacencies between the rooms to mark which rooms share boundaries in a layout using the Room Adjacency String (RAS) feature. This feature can be a preference if buyers are keen to choose the arrangement of rooms in a layout, e.g., *the master bedroom should be near the kids' bedroom.* For this purpose, an adjacency matrix is constructed from the segmented and order-wise labeled layout. For k^{th} floor plan image in the database its topological graph is denoted as $G_k = (V_k, E_k)$, where V_k is the set of vertices also denoting the number of rooms in the layout and E_k is the edge-set as depicted in Fig.5.5 (b). The adjacency matrix, for the graph G_k is denoted as Adj_k which is a $|V_k| \times |V_k|$ matrix, where, $|\cdot|$ denotes cardinality of the set V_k. For each row of the adjacency matrix corresponding to each room, a unique binary string is computed, which represents the adjacencies for that particular room. E.g. the room adjacency of room 2 in the segmented layout in Fig. 5.5 (Example 1) (a) is highlighted in Fig. 5.5 (Example 1) (c). RAS of 1011 depicts Room 2 is adjacent with Rooms 1, 3 and 4.

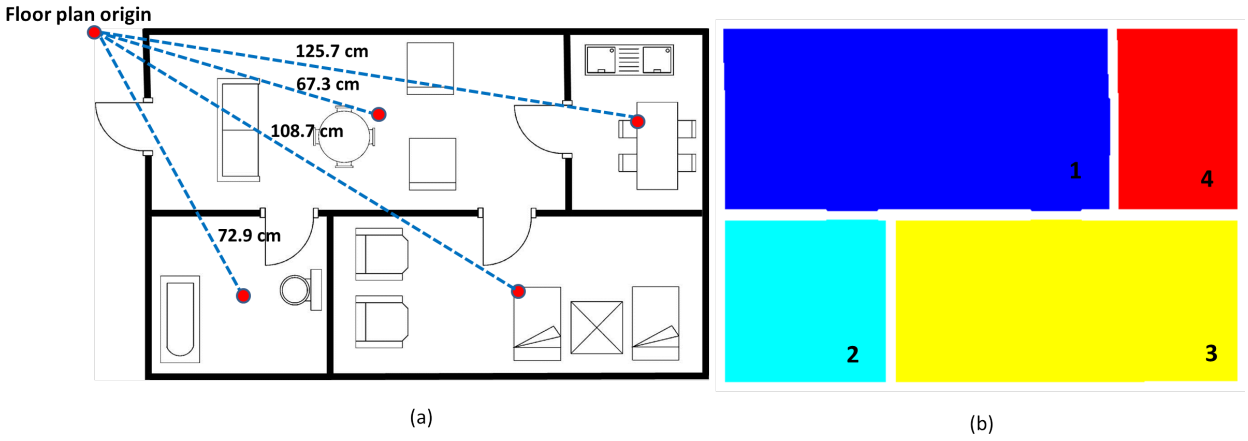


Figure 5.4. : Figure depicting the ordered labeling of rooms after segmentation. (a) Distance is calculated from a leftmost origin point up to all room centroids. Room with the lowest distance from the origin is ordered number 1 and the other rooms are labeled in similar fashion. (b) Subsequent room ordering results.

5.2.2 Feature 2: Carpet Area Ratio (CAR)

CAR is a unique feature that represents the ratios of carpet area enclosed within the walls i.e. the actual area to lay the carpet. This area does not include the thickness of the inner walls. The carpet area (Ar) of a room (L^k) is calculated using the formula for calculation of area of a polygon as specified in [Braden, 1986]:

$$Ar(L^k) = \left| \frac{(x_1y_2 - y_1x_2) + (x_2y_3 - y_2x_3) + \cdots + (x_ny_1 - y_nx_1)}{2} \right| \tag{5.1}$$

where, the pair $(x_i, y_i); \forall i = 1, \cdots, n$ denotes the n vertices of a room, taken in clockwise order, and the operator $|.|$ denotes the absolute value of the function. The motivation behind extracting this feature is the fact that usually users buying layouts have preferences in terms of the size of each room in the floor plan.

CAR ($A(\cdot)$) serves as a unique feature while a user is looking for layouts. The ordered rooms are labeled and the area between their boundaries and their respective ratios is stored which serves as the CAR. The area of a particular room (L^k) is then divided by the total area of the layout (L) to give an idea of the size of the particular room corresponding to the whole layout (refer Eq. 5.2). $Ar(L)$ is also calculated using Eq. 5.1, where (x_i, y_i) are the vertices of the polygon obtained by tracing the global layout of L.

$$A(L^k) = \frac{Ar(L^k)}{Ar(L)} \tag{5.2}$$

Figure 5.6 (b) and (d) show the areas corresponding to the different rooms in the layout shown in Fig. 5.6 (a) and (c) in the form of a pie-chart. Here, the inner entries are the actual areas of the rooms and the outer entries, represent the CAR feature of a room.

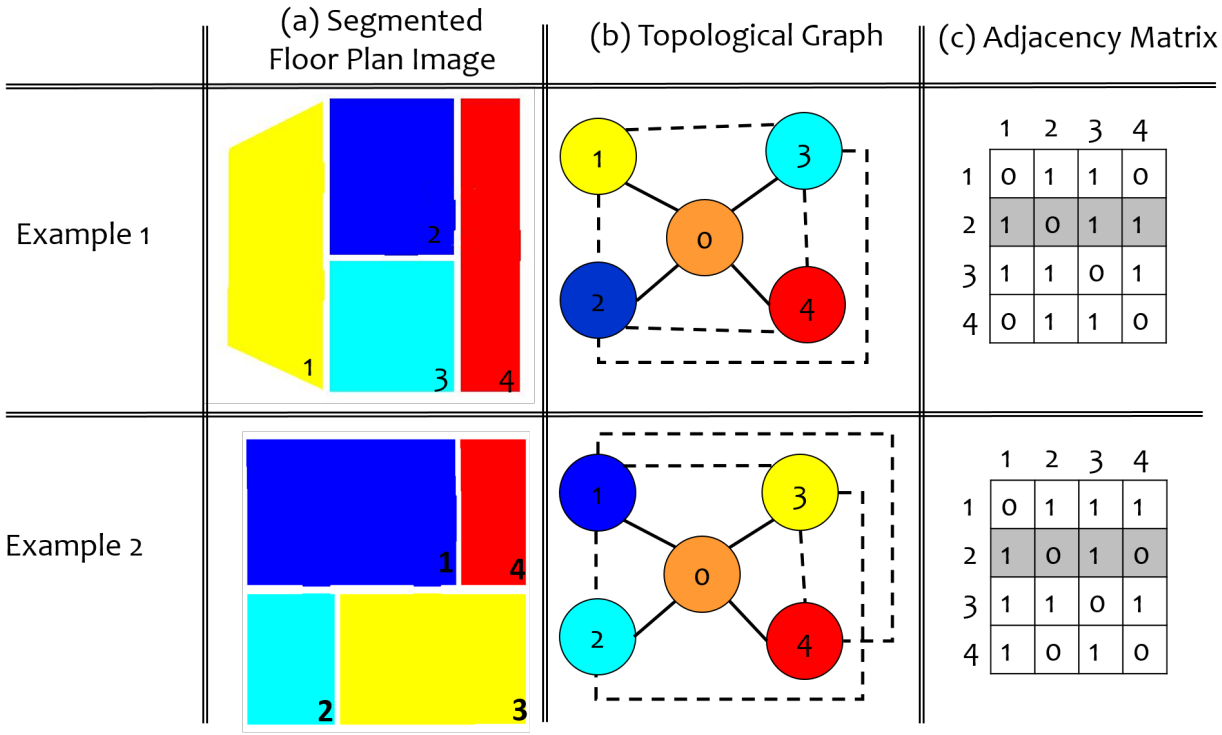


Figure 5.5. : Example of Room Adjacency String (RAS) feature: (a) Segmented rooms (b) Topological Graph (c) Adjacency matrix (Adjacency string corresponding to Room 2 is highlighted).

5.2.3 Feature 3: Furniture Composition Record (FCR)

The third feature that is computed is the Furniture Composition Record (FCR) that comprises of number and type of various furniture inside a particular room. A furniture is localized inside a room according to their position in the layout, by detecting the blobs inside the floor plan image. Further, their count (C) is stored corresponding to each room. This helps to compare the rooms according to the consumers priority of being overcrowded with decor or not. Next, the various furniture are classified by performing a connected component analysis on the detected blobs and taking the area of individual components inside the different furniture symbols. Then a ratio of the largest 3 area components is taken, and a unique signature is obtained for each furniture which can be further used for categorizing them as shown in Fig. 5.7. Algorithm 3 describes each step of the furniture categorization process in detail where the input is a room from a floor plan image without the walls. The set $\mathscr{T}$ is generated as the output of the furniture categorization step. In this step, each blob in a room is categorized and their signature is compared with the already stored furniture template signatures and assigned a type T. The union of the type of all the furniture inside a room then constitute the set $\mathscr{T}$. This set finally contains the type of furniture inside each room.

Furniture type gives a unique signature to a room. For example, a room with a bed, table, armchair etc. would primarily be a bedroom as shown in Fig. 5.8. So, while comparing a pair of layouts, it is worth to compare features of the similar type of rooms, instead of all possible rooms. Hence, categorizing a room based on type of furniture can prove helpful while matching. The feature extraction accuracies of all the 3 features is listed in Tab. 5.1. Accuracy of a particular feature is calculated by taking ground-truth information for each floor plan of the dataset in terms

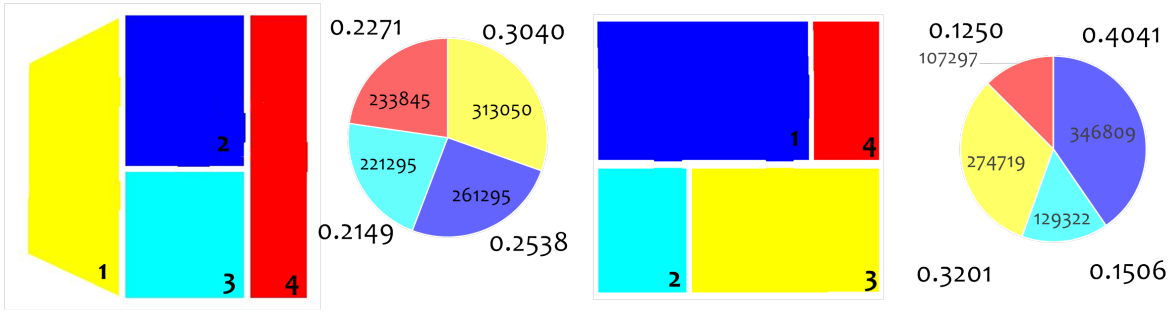


Figure 5.6. : Carpet Area Ratio (CAR) feature: (a), (c) Segmented, ordered layout, while (b), (d) CAR feature corresponding to Example 1 and 2 respectively.

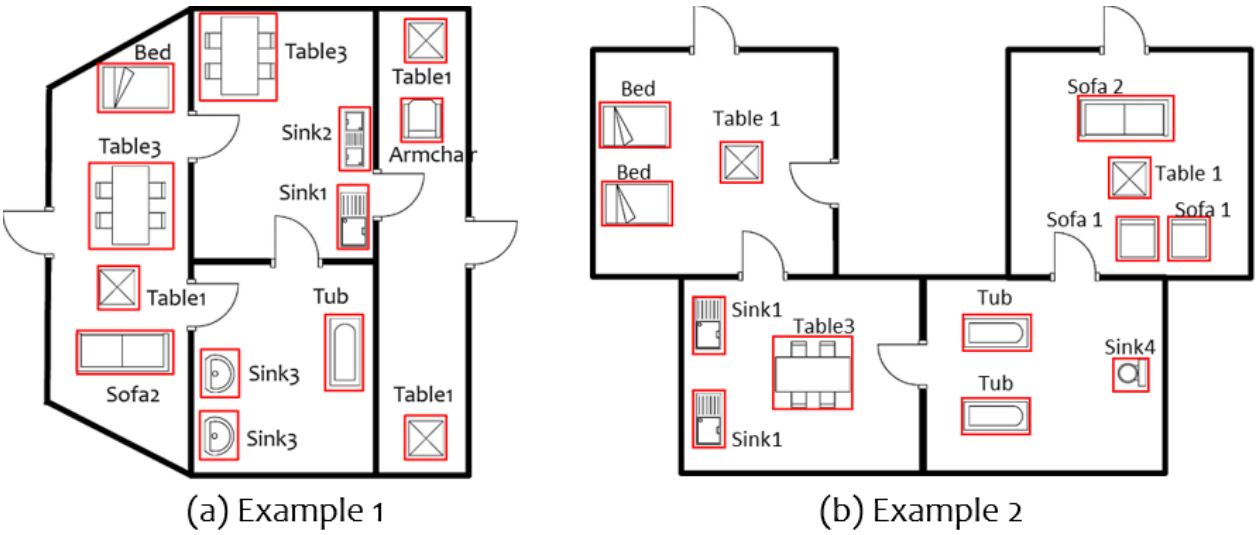


Figure 5.7. : Furniture detection and categorization in each room.

of the adjacencies, areas and furniture composition that each room in a floor plan has and further, taking a ratio of the features extracted from floor plans while running the algorithm and the ground-truth results. It was observed that each feature is extracted to a perfection using the steps mentioned above, except the furniture categorization feature, which performs to an accuracy of 89%. This accuracy while feature extraction helps in identifying the user's requirement efficiently while feature fusion and retrieval.

5.3 FINE-GRAINED MATCHING AND RETRIEVAL

Matching of a pair of floor plans using the extracted features is essential for retrieving similar layouts to the query layout. In the following sections the procedure to calculate matching score corresponding to various high level semantic features is described. Let, L_i^k and L_j^k represent

Table 5.1. : Percentage Accuracy of the extracted high level semantic features

Feature	Extraction Accuracy
Room Segmentation/ Room Adjacency	100%
Carpet Area	100%
Number of Furniture	100%
Type of Furniture	89%

k^{th} room in two different layouts (L), where i and j range from 1 to n, i.e. the total number of floor plans in the dataset, while k ranges from 1 to total number of rooms (V) in a layout. If there is a difference in the number of rooms in the layout then the minimum number of rooms between the layouts is taken for comparison. Algorithm 4 describes the matching procedure in a step-by-step fashion.

5.3.1 RAS Matching Score

Edit distance, represented as $e(\cdot,\cdot)$ [Levenshtein, 1966] is used, as the measure to differentiate between a pair of adjacency strings corresponding to same numbered rooms in two different layouts as described in Sec.5.2.1. As an example room 2 for layout 1 in Fig. 5.5 (Example 1) (b) has RAS 1011 and room 2 for layout 2 in Fig. 5.5 (Example 2) (b) has RAS 1010. In this case, the two adjacency strings differ only at the fourth bit whereas, rest bits are similar. Therefore, edit distance, i.e. the operations required to convert one string to another results to 1. Similarly, in case of room 4 for both the layouts the RAS are 0110 and 1010 respectively, differing in the first

Algorithm 3 Furniture detection and categorization

Input: Room image without outer walls (I), Furniture Template Signatures (F)
Output: Furniture Count (C), Furniture Type Set ($\mathscr{T}$)

C=0, $\mathscr{T}$=$\{\phi\}$
$\mathscr{B}$=Morphological fill operation(I)
$C = |\mathscr{B}|$ ▷ $|\cdot|$: *Cardinality*
for $j = 1$ to C **do**
  $\mathscr{C} = CC(\mathscr{B}_j)$ ▷ CC()=Connected Component
  for $k = 1$ to $|\mathscr{C}|$ **do**
    $\mathscr{A}_k = Area(c_k)$, where $c_k \in \mathscr{C}$
  end for
  $A = Sort_{desc}(\mathscr{A})$
  $S(\mathscr{B}_j) = \{(A_1/A_3), (A_2/A_3), 1)\}$ ▷ $S(\cdot)$: Signature
  for $k = 1$ to $|F|$ **do**
    if $S(\mathscr{B}_j) == F_k$ **then**
      $T(\mathscr{B}_j) = T(F_k)$ ▷ $T(\cdot)$: Type of Furniture
    end if
  end for
  $\mathscr{T} = \{\mathscr{T} \cup (T(\mathscr{B}_j)\}$
end for

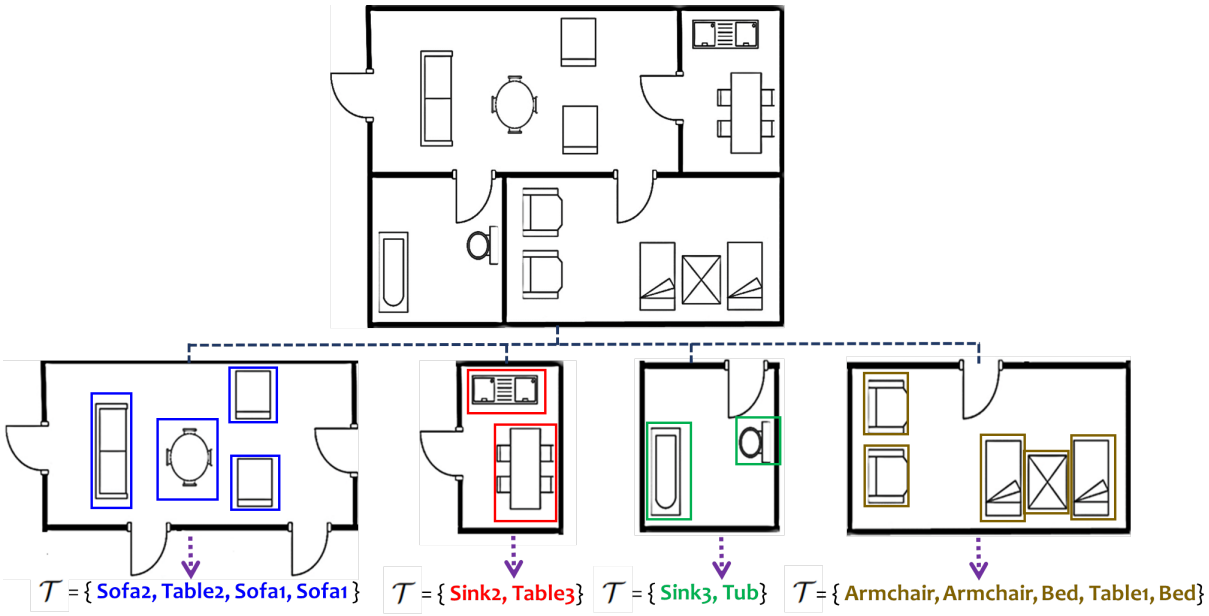


Figure 5.8. : Furniture detection and categorization in each room. Illustration underneath each room gives the array $\mathcal{T}$ containing the type of furniture in each room.

and second bit, resulting in an edit distance of 2. The RAS matching score (ρ) is computed as:

$$\rho(i,j,k) = \frac{e(L_i^k, L_j^k)}{max(V_i, V_j)} \tag{5.3}$$

Here the matrix ρ corresponds to the difference in room adjacencies between all similar numbered rooms in two layouts. Thus, higher ρ value suggests dissimilarities in adjacencies between two rooms. In a similar manner adjacency string matching of the rest of the rooms of a layout as compared to another layout is done. For normalizing this particular score in the range [0,1], it is divided by the factor equal to the maximum number of rooms between layout i and j, denoted as V_i and V_j. The idea behind normalizing in this manner is, that the maximum edit distance in the adjacency string can be obtained only if one of the strings is completely dissimilar from another, and also has a different length as compared to another. In that case the maximum operations taken to convert one string to the other would be equal to the maximum length between both the strings.

5.3.2 CAR Matching Score

The next feature for matching between two layouts is Carpet Area Ratio, $A(\cdot)$ of a particular room in both the layouts as described in Sec. 5.2.2. The CAR matching score (ψ) is computed as:

$$\psi(i,j,k) = A(L_i^k) - A(L_j^k) \tag{5.4}$$

Here the matrix ψ corresponds to the difference in Carpet Area Ratios between all similar numbered rooms in two layouts. The feature area ratio is already normalized in the range [0,1].

Algorithm 4 Fine grained retrieval using high level semantics

Input: Query Image (q), Database Image (d)
Output: Matching Score ($M(q,d)$)

for all d=1 to n **do**
  for all k=1 to min(V_q, V_d) **do**
    $\rho(q,d,k) = e(L_q^k, L_d^k)$ ▷ $e(\cdot)$: edit distance
    $\psi(q,d,k) = A(L_q^k) - A(L_d^k)$ ▷ $A(\cdot)$: Area
    $\phi(q,d,k) = C(L_q^k) - C(L_d^k)$ ▷ $C(\cdot)$:furn-count
    $\theta(q,d,k) = S(\mathscr{T}(L_q^k), \mathscr{T}(L_q^k))$ ▷ $\mathscr{T}(\cdot)$: furn-type
    $Normalize(\rho, \psi, \phi, \theta)$
  end for
  $M(q,d) = \sum_k(\rho(q,d,k)) + \sum_k(\psi(q,d,k)) + \sum_k(\phi(q,d,k)) + \sum_k(\theta(q,d,k))$
  $M(q,d) = M(q,d)/4$
Return $M(q,d)$
end for

Therefore, no post normalization required. A low value of ψ suggests high similarity in area feature of two rooms.

5.3.3 Furniture Composition Record Matching Score

The room decor component has two sub-components, namely the number of furnitures inside two same numbered rooms and type of furnitures inside the rooms.

Number of furnitures

Number of furnitures, $C(\cdot)$ inside a particular room is computed using the technique discussed in Sec. 5.2.3 and corresponding number of furnitures score, ϕ, is calculated as:

$$\phi(i,j,k) = \frac{C(L_i^k) - C(L_j^k)}{max(C(L_i^k), C(L_i^k))} \tag{5.5}$$

Here the matrix ϕ corresponds to the difference in number of furnitures between all similar numbered rooms in two layouts. This helps us compare the rooms according to the consumers priority of being overcrowded with decor or not. The difference of the number of furnitures corresponding to ϕ should be low in both the layouts if they are similar.

Type of furnitures

This particular feature categorizes type of room, for example a bedroom, bathroom or a kitchen based on its furniture arrangement. It helps us distinguish between two same numbered rooms, with different furniture types. Firstly, the classification of the furnitures in each room is done and their classified labels are stored in an array with rows corresponding to each room number and the subsequent column corresponding to the labels of the furniture in that particular room. Motivation behind matching this feature is that a buyer might specify that the third room in the

layout should be a bedroom and hence, while retrieval one should retrieve floor plans with buyer's preferred placement of room in the complete floor plan. Thus, floor plan having a kitchen as the third room should be retrieved later as compared to a floor plan having the third room as the bedroom. In the below mentioned equation, the matrix θ corresponds to the difference in type of furnitures between all similar numbered rooms in two layouts. Using this, the furniture label strings are compared for similar numbered rooms in two layouts and the penalty is incremented if label strings do not match, i.e. if dissimilar furniture is found between both rooms. The more similar the two rooms are in their furniture types the lower would be the value of θ. The type of furniture in a particular layout is represented as $T(\cdot)$, string comparison is represented as $S(\cdot,\cdot)$ and the type of furniture score is represented as θ, and is computed as follows.

$$\theta(i,j,k) = \frac{S(\mathscr{T}(L_i^k), \mathscr{T}(L_j^k)}{max(C(L_i^k), C(L_i^k))} \tag{5.6}$$

5.3.4 Cumulative Match Score (CMS)

After computing the various feature matching scores for individual rooms in a query layout, CMS between a pair of layouts is obtained by adding the feature scores corresponding to all the common rooms present in both the layouts. CMS for a feature (ρ, ψ, ϕ, θ in our case) F is denoted as F^+ and is computed as:

$$F^+(i,j) = \frac{\sum_{k=1}^{min(V_i,V_j)} F(i,j,k)}{min(V_i,V_j)} \tag{5.7}$$

Using this particular equation the corresponding cumulative feature scores ρ^+, ψ^+, ϕ^+ and θ^+ are obtained. Thus, finally, a matching score for a complete layout is obtained by taking a sum of individual features for all the rooms.

5.3.5 Total Match Score

Total Match Score, M for a pair (query and database) of layouts is calculated using the weighted sum of four extracted feature scores namely RAS, CAR score, number and type of furniture score (FCR). Weights are assigned to each feature component to investigate the effect of individual feature during retrieval. This allows user to give more preference to a certain feature while retrieval. In such a case, the matching score has additional individual weight coefficients η_1, η_2, η_3 and η_4 all in the range $[0,1]$, where the default value of the coefficients is 1. In this case, the Match Score M is obtained, as shown in the equation below:

$$M(i,j) = 1 - \frac{\rho^+(i,j) + \psi^+(i,j) + \phi^+(i,j) + \theta^+(i,j)}{4} \tag{5.8}$$

It is to be noted that this score has individual components normalized in the range [0,1]. Also, the Total match score is divided by a factor 4 to be finally normalized in the range [0,1].

As discussed before, the weight coefficients can be incremented and decremented to set preferences to some features while diluting the effect of others, by tuning the values of η_1, η_2, η_3

and η_4 in the equation mentioned below :

$$M(i,j) = 1 - \frac{\eta_1 \rho^+(i,j) + \eta_2 \psi^+(i,j) + \eta_3 \phi^+(i,j) + \eta_4 \theta^+(i,j)}{4} \tag{5.9}$$

The next section describes the qualitative and quantitative assessment of the proposed framework.

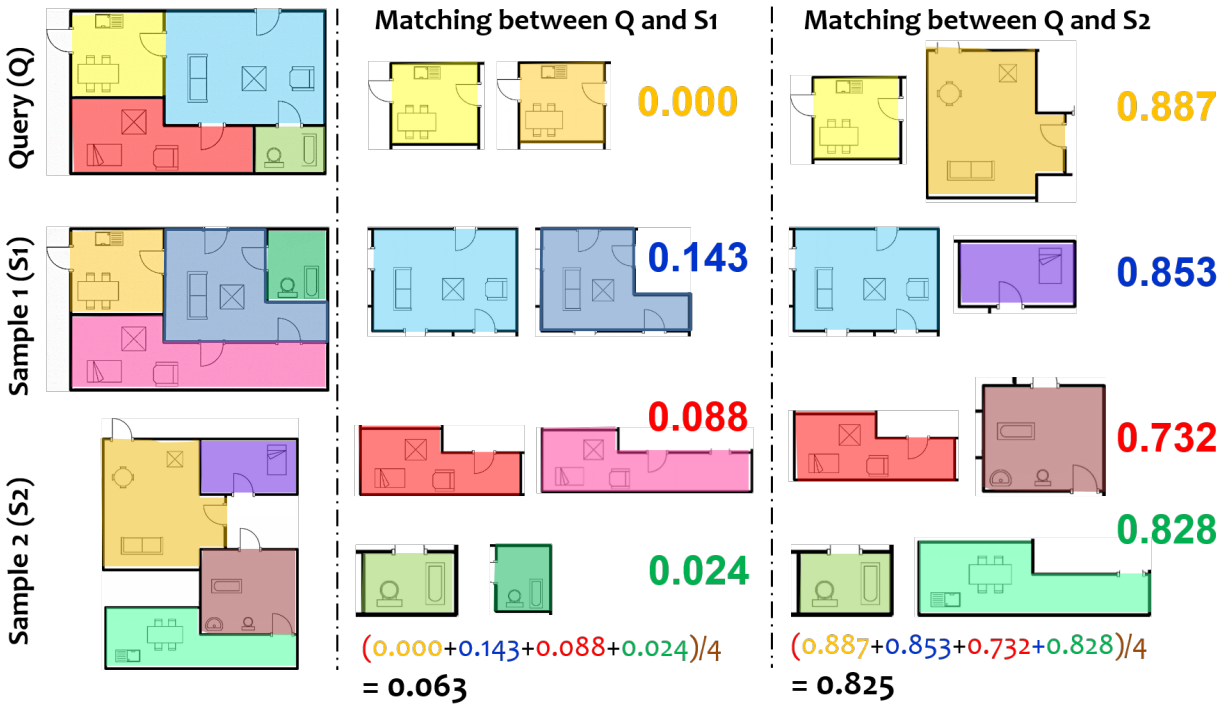


Figure 5.9. : Distance score corresponding to each room of the query layout (Q) and the database layouts (S_1 and S_2). Attributed to the difference in features, lower the score, more similar the floor plans are. Hance, finally the score is subtracted from 1 to make the final greater Matching score to represent the floor plans that match more. Color codes are used to highlight the room correspondence.

5.4 EXPERIMENTS AND RESULTS

The experiments are performed on the *ROBIN* dataset that contains three categories of layouts each with 17 subcategories containing a total of 510 floor plans. The dataset has 11 different categories of furniture inside each layout. The experiments were performed on a machine with Intel Xeon I7 processor with 16 cores.

Details of the matching process are shown in Fig. 5.9. There is a high degree of similarity (in terms of FCR) between the individual rooms of the query floor plan (Q) and that of sample floor plan S1. However, minor differences lie in the area feature (CAR) that can be observed by looking at the second room's ("blue" color) area in both the layouts. The two layouts also differ in their adjacency feature, which can be observed by looking at the fourth room (highlighted in "green"). In the case of Q, the fourth room is adjacent to the third and the second room. On the other hand,

Figure 5.10. : Retrieval results for four different query architectural floor plans taken from the *ROBIN* dataset. Note that the floor plans highlighted in red are incorrectly retrieved as global layout shape differs from query.

in the case of S1, it is adjacent only to the third room. However, database sample floor plan S2, as can be visually seen, differs in many respects as compared to Q. The total scores, as per Eq. 5.9, also justify these differences. For example in the fourth room (color coded in "green") in both Q and S2, all the features, namely the FCR, CAR as well adjacencies differ resulting in a large cost of 0.828. Such observations between Q and database floor plan S2, justify high dissimilarity score attributed to the different global shapes and thus, different CAR, RAS and FCR features of both the layouts. Hence, S2 is ranked later than S1 during retrieval.

5.4.1 Qualitative results

First of all the architectural layouts are segmented into rooms followed by detecting the various types of furniture present in each room. Thus, a semantic analysis of a floor plan is

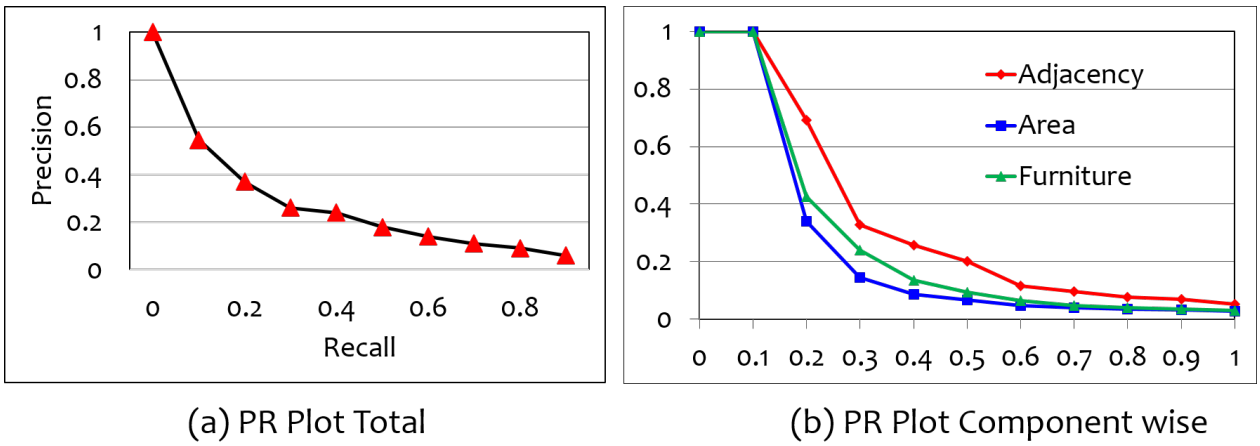
obtained which can be further used for processing. The fine-grained matching techniques proposed in this Chapter helped in the rank ordering of the matched layouts based on the differences in the adjacencies between corresponding rooms, the area of the rooms, number, and type of furniture inside the rooms. The rank-ordered retrieval results for six such queries are shown in Fig. 5.10. The first rank ordered result retrieved from the layout database is the query image itself. This is due to the fact that the match score between the retrieved layout and the query layout is maximum. The subsequent results differ in the scores in the area, adjacencies and decor features inside each room.

As is evident from Fig. 5.10(d) the rank one result is the query layout as mentioned above. Justification of the rank 2 retrieved result can be seen by observing its differences/ similarities with the query floor plan. For example, note that the adjacencies of the first room in the query layout are with room 2 and 4, that is exactly similar to the adjacencies shared by room 1 in rank 2 retrieved result. Thus, on the adjacency parameter, both layouts match completely. Although the first room is slightly bigger area-wise in the query, the furniture type and number are approximately similar in both the layouts. Hence, the match score between the query layout and the retrieved rank 2 layout is high which attributes to the result. An interesting observation is that the rank 3 result retrieves layout from a different sub-category of the dataset as it differs in the global layout shape. It is to be noted that such a retrieval is due to the fact of various common features between the query and rank 3 result which can be analyzed similarly as shown in Fig. 5.9.

5.4.2 Quantitative results

The performance of the system is quantified using a Precision and Recall (PR) metric (see Fig. 5.11). The precision values are averaged over all the query images for a particular recall value. Given a query, retrieved layouts should belong to the same sub-category of layouts as the query, i.e., it should maintain the global shape and also keeping in mind the preference set by the property seeker during requirement driven querying. Thus, if the global shape of the layouts differ then that is considered as an incorrectly retrieved result.

Figure 5.11 (a) refers to the overall PR plot according to the match score (see Eq. 5.8), i.e. giving equal weights for FCR, CAR and RAS scores and retrieving the results. As shown in



(a) PR Plot Total (b) PR Plot Component wise

Figure 5.11. : Precision and Recall (PR) plot corresponding to a) Cumulative sum of all the features with equal weights b) Preferring one feature more than the others while retrieval.

Fig. 5.11 (a), given a query sample the rank 1 result is the query floor plan itself. This leads to the highest precision value of 1 during the initial recall. With further retrieval, the average precision value decreases due to some incorrectly retrieved results as compared to the query layout. As discussed in Sec. 5.3.5, while retrieval a user can have a preferred feature. For example, a user might be fixated over preferring area feature over the others while retrieving. The performance of the approach mentioned in this Chapter is analyzed by simultaneously increasing the weight of a feature, while decreasing weights of other features to study the impact of individual high level semantic features during retrieval, thus obtaining PR plot as in Fig. 5.11 (b), corresponding to preferring different features during retrieval. In Fig. 5.11 (b), "red" line corresponds to the PR plot given high weight to adjacency feature during retrieval. Similarly, if the user wants the furniture type and number to be very similar to the query layout then weight of the cost of types of furniture during matching can be increased. The PR plot in Fig. 5.11 (b) with the "green" line depicts the quantitative retrieval results using such an approach. Similarly, the "blue" line in the same plot depicts the results obtained when the area parameter is given more weight. The area under the PR plot is the highest when the adjacency feature ("red" line) is given the highest weight as compared to other features. Because layouts, which are falling in the same category of a global shape generally have similar room adjacencies. Even though their areas and furniture arrangement might differ a little. Thus, the retrieval accuracy is higher when adjacency parameter is favored as is evident from its PR plot.

5.5 DISCUSSION

The proposed approach, efficiently performs all the processing steps given in Fig. 5.2. As observed, the system generally succeeds in retrieving the samples from the same sub-category as the query sample, only faltering at some occasions, where, our proposed FCR and CAR features overpower the RAS feature. For run time analysis of this approach, the time taken (averaged over 510 samples) in seconds at every step was computed. Segmentation into rooms was a computationally expensive step owing to a number of intermediate processing steps. The average time taken by the segmentation step was 22.543 secs. Furniture detection and Furniture classification took 3.968 secs and 5.815 secs respectively. The four feature scores proposed by us namely CAR, RAS, No. of furnitures and Type of Furnitures take 4.253 secs, 3.453 secs, 3.234 secs and 5.736 secs respectively. It was noted that these feature scores took more time in layouts with 5-rooms, due to higher number of doors and inner components owing to the increased rooms. Calculating the cumulative matching score between two layouts and thereby performing the retrieval costed 12.257 secs at an average.

Due to a lack of availability of the implementation of such a composite framework of fine grained retrieval of floor plans in the literature, the results couldn't be compared with other retrieval based approaches. The current approach is motivated by the idea proposed in Chapter 3 in the area of floor plan retrieval [Sharma *et al.*, 2016a]. The key difference between the work discussed in this Chapter and that proposed earlier in this thesis are: (i) previous work does not establish room-wise uniform ordering and correspondence, (ii) the earlier approach took only adjacency and decor arrangement in the whole layout as the parameters during retrieval. In this Chapter, there is an inclusion of high level semantic features like ordering of rooms, area and furniture arrangement in a particular room to aid better fine grained retrieval of floor plans in the *ROBIN* dataset. For making a fair comparison, the results of the baseline method [Sharma *et al.*, 2016a] on *ROBIN* dataset are reproduced and the mean average precision values (MAP) are compared with the newly proposed method in this Chapter. As shown in Tab. 5.2, the approach proposed in this chapter has led to an increase in the mean average precision value as compared to the previous approaches.

The proposed method also has the edge over the approach proposed in Chapter 4 [Sharma

Table 5.2. : Comparison of techniques on ROBIN and SESYD Dataset

Approach	MAP (ROBIN)	MAP (SESYD)
Baseline [Sharma *et al.*, 2016a]	0.2479	1.0
OASIS with RLH [Chechik *et al.*, 2009]	0.0956	1.0
OASIS with BOW [Chechik *et al.*, 2009]	0.1892	0.99
Proposed	**0.3047**	1.0

et al., 2017], where a Convolutional Neural Network based deep framework for retrieval in floor plans is proposed. This deep framework although quantitatively performs better than the method proposed in this Chapter. However, it has a few drawbacks. For example, there is no provision of giving preferences to individual features during retrieval as the layers extract and combine the features implicitly without the provision of manipulating the weight by the user. On the other hand, the provision of a weighted combination of the features proposed in this Chapter lets the user give preferences to individual features, according to his requirement (justified by Fig. 5.11 (b)), thus making it more relevant from a user's point of view. Moreover, the CNN framework is a simple network that mainly distinguishes by the shape of an object or a layout. It would not perform well while doing fine-grain categorization of furniture components, as well as if two layouts from different classes have almost similar shapes.

As the proposed technique deals with similarity in floor plan images thus, the results are also compared with the Online Algorithm for Large Scale Image Similarity Learning (OASIS) [Chechik *et al.*, 2009] coupled with the Run Length Histogram technique (RLH) [de las Heras *et al.*, 2013] and Bag of Words technique (BOW) [Lazebnik *et al.*, 2006b]. These algorithms were chosen for comparison as they are widely popular in object retrieval in document images. It was observed that the proposed feature matching yielded a much better average precision value as compared to OASIS approach (See Tab. 5.2) because OASIS technique focuses on the difference of similarity values between relevant and irrelevant image pairs while ignoring the similarity value between highly similar images. As already discussed, reproducing the approaches mentioned in Tab. 5.2 on the SESYD public dataset yielded a perfect MAP value of 1 due to the low interclass similarity in the samples.

In [Ahmed *et al.*, 2014], the authors proposed a sketch based floor plan retrieval framework, where query layouts are represented as sketches using only primitive geometric shapes such as rectangles. In contrast, in this work, the room decor represented by complex geometric primitives is considered. Furthermore, room decor labeled fine-grained analysis was missing in the work by [Ahmed *et al.*, 2014]. This framework is rotation sensitive to allow relative position of rooms as a feature. Customers may follow Vastu tips and prefer "*master bedroom should be towards the south facing the wall, so that sleeping person's head is facing south*". Making the retrieval rotation invariant will fail to serve this purpose.

5.6 SUMMARY

The proposed framework for fine-grained retrieval of floor plans uses high-level semantic features like area, number, type of furniture and also adjacencies between the rooms. The inclusion of the high-level features during retrieval is a novel idea that can find application in scenarios where the user has specific choices in terms of the fully furnished floor plans. Also, weighted feature fusion

helps in weighing features when a user is querying, and the user is fixated over certain features to be present during retrieval mandatorily.

Till this point in the thesis, only query by example in the form of image has been used. However, the primary concern in this case is that the user must have an image of a floor plan available for querying. In a real scenario, a user may not have a digitized floor plan image to search for similar floor plans. A more intuitive mode of querying is to scribble the requirement as a sketch and query accordingly. Therefore, in the next Chapter, a sketch based input query is discussed, where, the user can draw the layout and the internal furniture components as well. In the current world of smart, hand-held devices this provides an easy mode of search for the end users. Thus, the next Chapter deals with changing the mode of query from images to sketch, to ease querying and make the application more viable.

6

Domain invariant feature transform for interactive floor plan retrieval

Content-based image retrieval (CBIR) emerged as an attempt to deal not only with the absence or insufficiency of annotations for most of the images, but also to support alternative retrieval approaches, relying on visual perception, which is more appropriate in many scenarios. Within CBIR, sketch-based image retrieval (SBIR) aims at retrieving images that are similar to a sketch made by the user (typically a simple set of drawing lines). Thus, SBIR is particularly adapted in situations where a user has a perceptual image of what he is searching. In this scenario, a sketch image is useful specially when the image dataset is not annotated and the user has no similar example image to use as a query input. The challenges associated with SBIR are:

1. Finding a relevant visual content representation associated with a similarity measure that allows effective comparison with a query that is not a picture, but rather a drawing made by a user who sometimes is not very skillful.

2. Making retrieval scalable to large image datasets by building an appropriate index structure which is able to exploit the content representation and similarity measure in a better way.

3. Sketches are highly abstract and carry a huge level of variation from person-to-person.

4. Sketches don't provide much of visual cues like colour as present in images.

5. The inherent difference between two domains (sketch and image) also makes the problem more challenging.

In Chapter 3, 4 and 5, the proposed methodologies revolve around extracting semantic and structural features from floor plan images. Exploring both hand-crafted and deep learning features has been done in the previous chapters to extract and represent meaningful content in images and finally retrieve similar floor plan images from the database. The important thing to be noted here is that the mode of the query until this chapter is through an image. In this Chapter, the query mode is interactive, i.e., it can either be an image based query or a sketch based query. This Chapter is organized as follows: Sec. 6.1 gives a brief overview of the approaches proposed in this Chapter, Sec. 6.2 gives an insight into the first approach using Cyclic GAN that has been proposed in this Chapter and Sec. 6.3 details out the second approach using autoencoders and Cyclic GANs for the task of retrieval. Section 6.4 gives a qualitative and quantitative analysis of both approaches. Finally, Sec. 6.5 concludes the Chapter.

6.1 BRIEF OVERVIEW

Sketching using pen and paper is more intuitive and easy to represent ideas as compared to other modalities. With the popularity of digital hand-held devices, users are keen on using sketches to exchange ideas. Users can sketch the floor plans on a tablet in the same fashion as they would on

paper. With the increase in the usage of touchscreens and hand-held devices, sketch-based querying for finding a dream home can be a convenient proposition. Sketch-based floor plan retrieval can benefit the architects and the users while searching for floor plans in the early design phases of a building. However, sketch-based processing and retrieval come with their challenges. In this Chapter, techniques for retrieval of floor plans given an interactive query mode are proposed. Two approaches to facilitate sketch based retrieval of floor plans are presented in this Chapter. Figure 6.1 shows the flow charts of the two approaches.

1. **Approach 1:** An initial attempt aims at using a Cyclic Generative Adversarial Network (Cyclic GAN) for domain mapping between floor plan sketches and floor plan images along with Convolutional Neural Network (CNN) to extract features from the mapped query and retrieval set.

2. **Approach 2:** Although approach 1 performs well for sketch-based retrieval of floor plans, there is still a scope of improvement in terms of the precision values during retrieval. Also, there is no provision of image-based retrieval in the proposed approach 1. This serves as the motivation for an efficient unified framework to meet the requirements of both image to image as well as the sketch to image retrieval. Therefore, to improve the results of the retrieval mode and to propose a composite solution to multimodal retrieval, a unified framework for retrieval of similar floor plan images given query mode as floor plan sketches or images is proposed. This approach uses the conjunction of CNN for query classification, Cyclic GAN for sample generation and Autoencoder for domain mapping and proves to be efficient than approach 1 in terms of retrieval.

In the next section the methodology associated with Approach 1, taking into account Cyclic GANs and CNN for feature extraction and domain mapping is discussed in detail.

6.2 APPROACH 1: CYCLIC GAN FOR DOMAIN ADAPTATION AND SKETCH BASED FLOOR PLAN RETRIEVAL

In this section, a brief overview of the first approach for similar floor plan retrieval using sketch modality is discussed. Figure 6.2 depicts the complete framework of the proposed technique. If the query is a sketch, the steps carried out are as follows. The entire framework is divided into three main phases. They are: (1) domain mapping through Cycle Generative Adversarial Networks (Cyclic GANs); (2) feature representation through Convolutional Neural Networks and (3) matching and retrieval. Given an input sketch S, the goal is to retrieve similar looking floor plan images (I) from the ROBIN dataset [Sharma *et al.*, 2017] corresponding to the query sketch. The Cyclic GAN is trained using images from ROBIN and sketches from S-ROBIN to learn domain representation. As direct retrieval of floor plan images given a sketch is not straightforward (refer the low performance of CNN1 in Fig. 6.14. (b)). Therefore, first the ROBIN dataset images are converted into their sketch counterparts (called GAN generated sketch dataset (GS-ROBIN)) using a Cyclic GAN trained to map the sketch and image domain. Thereafter, the post-processing of the generated sketches (GS) is done to get their enhanced versions. Then a CNN is trained over the generated sketches to learn the deep feature representation for the floor plan retrieval task. The learned deep representation helps to extract deep features from the GS-ROBIN database. The domain mapping is necessary because of large variation between the quality of images/sketches generated using Cyclic GAN and original images/sketches from the ROBIN/S-ROBIN dataset, as shown in Fig. 6.3. During the retrieval stage, the framework extracts the deep features from the query sketch also using the same deep representation. The extracted deep features are then matched with the database features. A similarity score for a particular layout is thus calculated

for all the query samples. To know which individual layer of the deep framework performs best, similarity using all the individual deep features/layers is calculated, and at the same time, the effect of CNN trained on the ROBIN dataset, the S-ROBIN dataset, and the GS-ROBIN dataset is studied. Details of the same are discussed in Sec. 6.4.5.

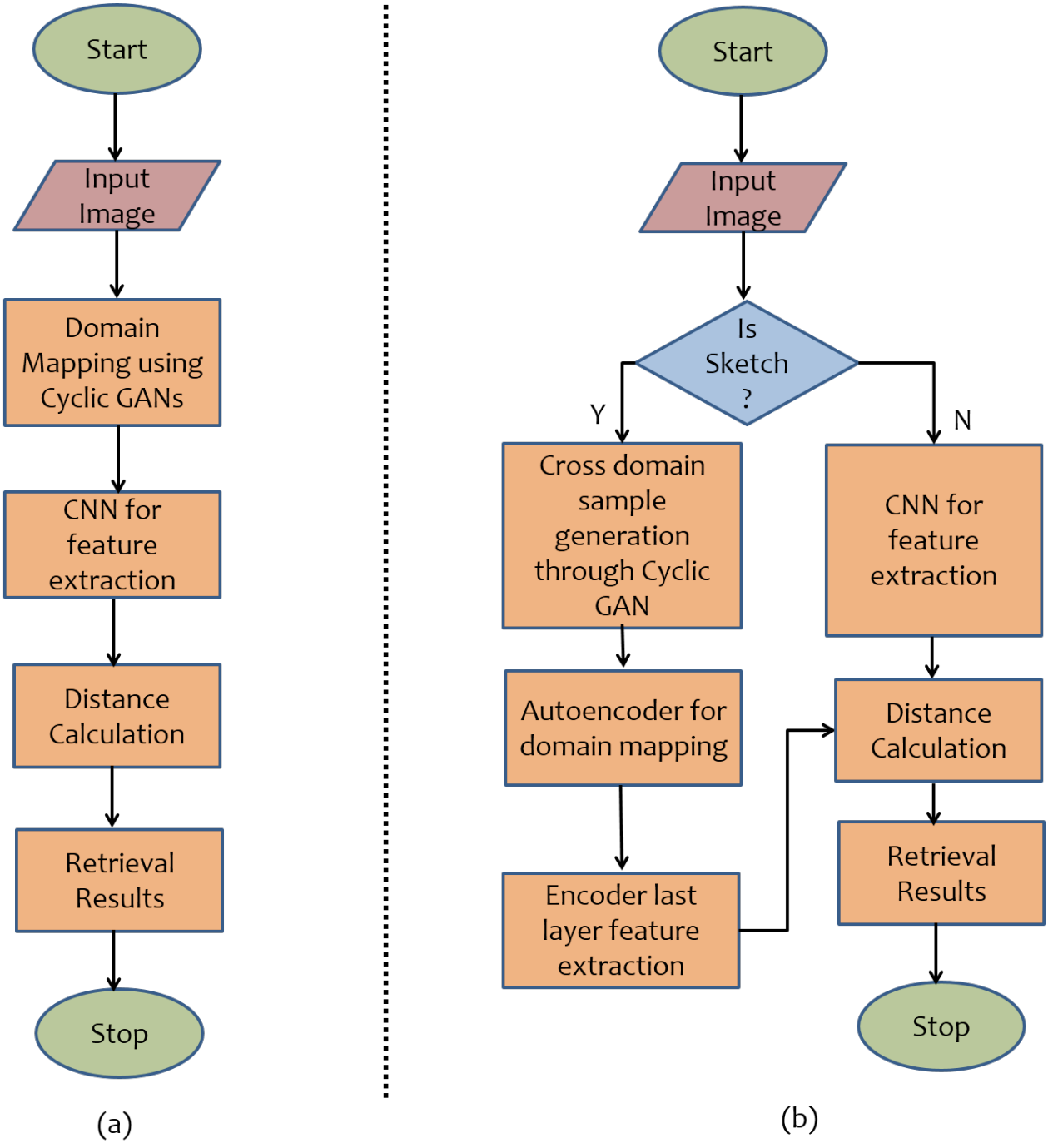


Figure 6.1. : Flowchart comparing the different sub-components in two approaches for sketch based retrieval. (a) Approach 1: Basic framework using cyclic GAN for domain mapping and sketch based retrieval (b) Approach 2: An improved unified framework using cyclic GAN and autoencoders in conjunction for floor plan retrieval.

6.2.1 Domain Mapping through Cyclic GAN

In this section, the use of Cyclic GAN for facilitating the "domain shift" between sketch and image floor plan domain is illustrated. In this case, the purpose is to choose a floor plan image domain (I) and then transform it into an image of sketched floor plan domain (S), and vice versa. Since one-to-one mapping does not exist between the sketch and the image floor plans in the training set, the problem becomes more challenging Zhu *et al.* [2017]. Having this relaxation of the non-existence of one-to-one mapping makes this formulation quite powerful. The need for a paired image in the target-domain is eliminated by the two-step transformation that involves mapping of source domain image to the target and back to the original image. Mapping the image to the target domain is improved by competing for the generator with the discriminator in the Cyclic GAN model. For training the Cyclic GAN model, a training set consisting of 70% samples from the ROBIN and S-ROBIN datasets is used.

Figure 6.4 depicts the overall objective of the proposed model for domain adaptation. Let, the image domain be denoted as I, and the sketch domain be denoted as S. An i^{th} floor plan image sample is denoted as f_I^i, where $i = 1,2,\ldots,m$. Here m is the number of floor plan image samples taken from ROBIN. On the other hand, the j^{th} floor plan sketch sample from S-ROBIN is denoted as f_S^j, where $j = 1,2,\ldots,n$. In this case, n is the number of sketches in S-ROBIN. To convert a floor plan sketch to its corresponding image version, a mapping function $G(.)$ is introduced. The task of $G(.)$ is to take a floor plan sketch from S-ROBIN and convert it to an image, which closely resembles a floor plan image ($\tilde{f}_I^j$) from ROBIN. This process is denoted in Fig. 6.4. Similarly, to convert a floor plan image from ROBIN to that of a sketch (belonging to S-ROBIN), another mapping function $F(.)$ is introduced. This function performs the same task as that of $G(.)$, but in the reverse order. The hypothesis is that the generated images, i.e., outputs of $G(.)$ and $F(.)$ will eventually match the target distributions. To compare between the actual and the desired distributions, two adversarial discriminators D_S and D_I have been introduced. The task of D_I is to check how good the mapping is from a given sketch to an image, whereas D_S aims to achieve the reverse. During the training phase, examples from both S-ROBIN and ROBIN dataset are given to the network to learn the mapping functions $G(.)$ and $F(.)$.

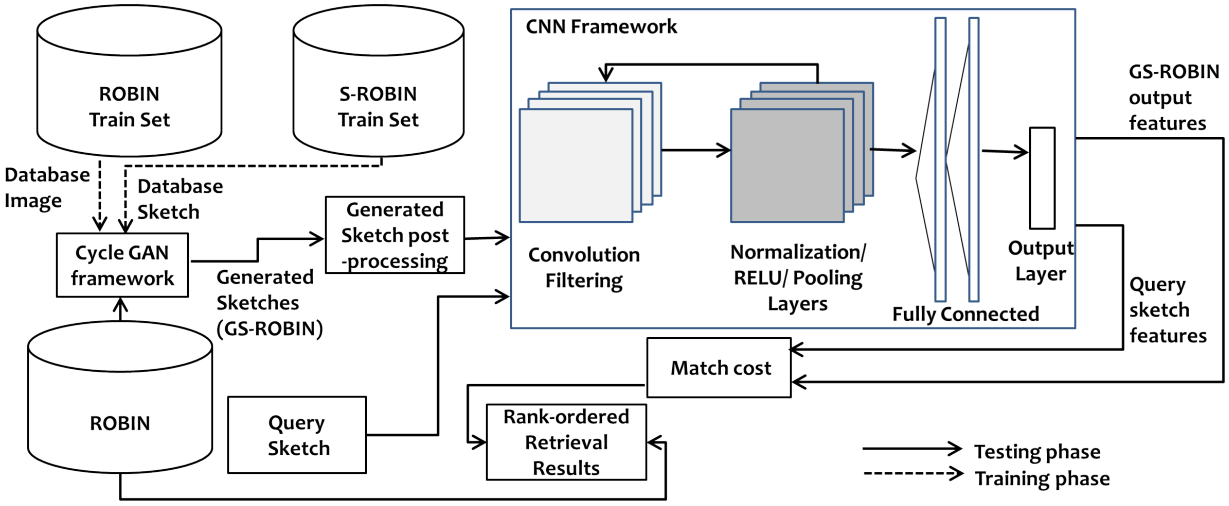


Figure 6.2. : Framework for sketch based retrieval of floor plans. Domain adaptation is facilitated by a Cyclic GAN.

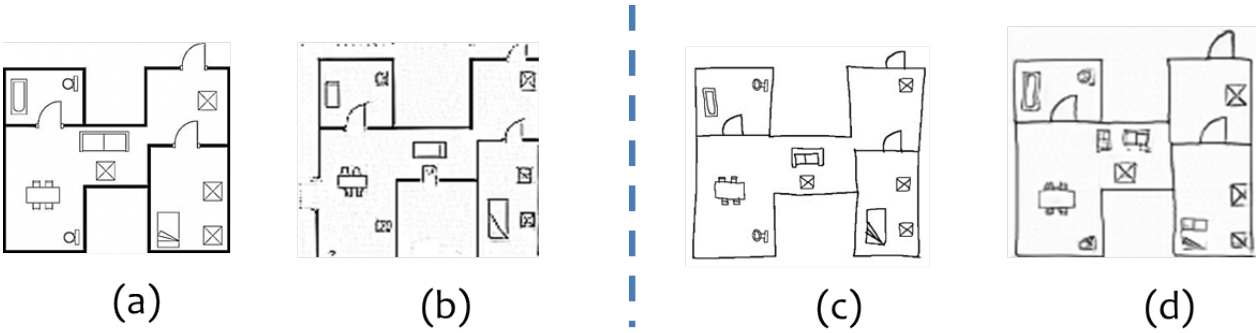


Figure 6.3. : Comparing the original sketches/images with outputs generated from Cyclic GAN. (a) Original floor image from ROBIN dataset. (b) Generated floor plan image using Cyclic GAN. (c) Original floor plan sketch from S-ROBIN dataset. (d) Generated floor plan sketch using Cyclic GAN. Kindly note the quality of generated sketches is better than that of generated images.

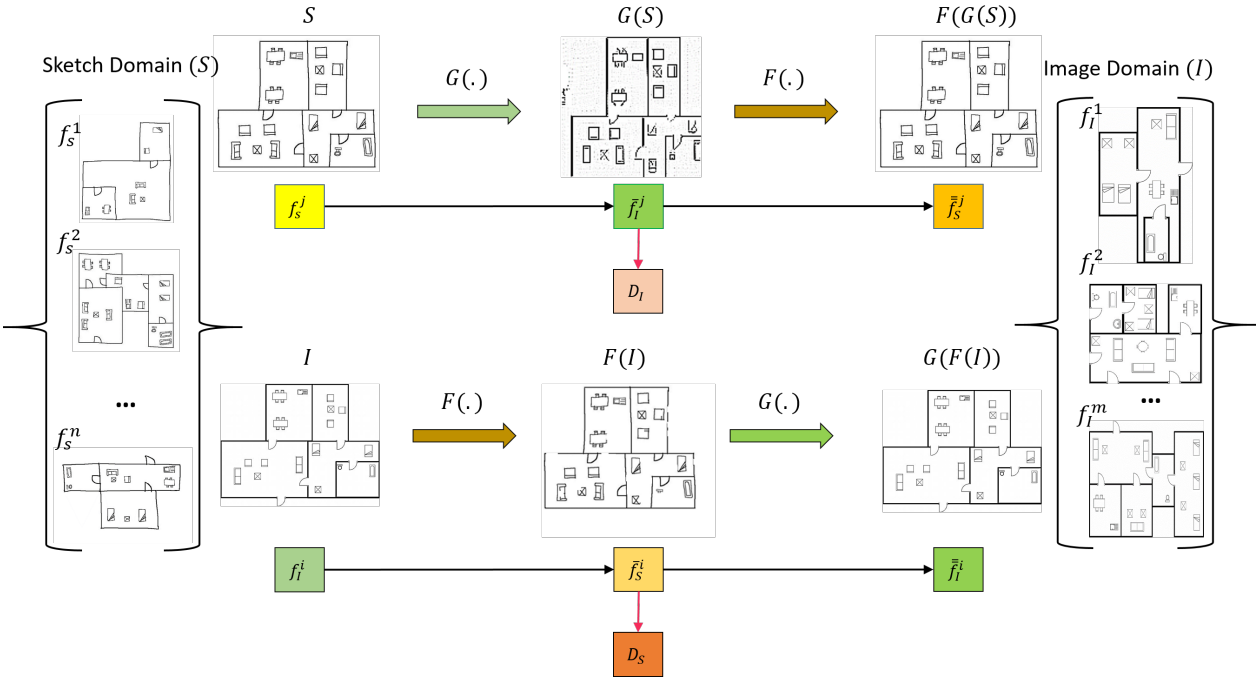


Figure 6.4. : Cyclic GAN framework, corresponding to the proposed approach with converting the sketches from the S-ROBIN dataset into GAN generated images (constituting GI-ROBIN dataset) closely similar in features to the to-be retrieved images and images from the ROBIN dataset to sketches (constituting GS-ROBIN dataset).

Let the probability distribution of a sample $f_S \in S$ be denoted as $f_S \rightarrow p(f_S)$, and for a sample $f_I \in I$ be denoted as $f_I \rightarrow p(f_I)$. Then the adversarial loss function for mapping a floor plan sketch to an image ($S2I_A(.)$) can be written as:

$$S2I_A(G, D_I, S, I) = E_{f_I \rightarrow p(f_I)}[logD_I(f_I)] + E_{f_S \rightarrow p(f_S)}[log(1 - D_I(F(f_S)))] \tag{6.1}$$

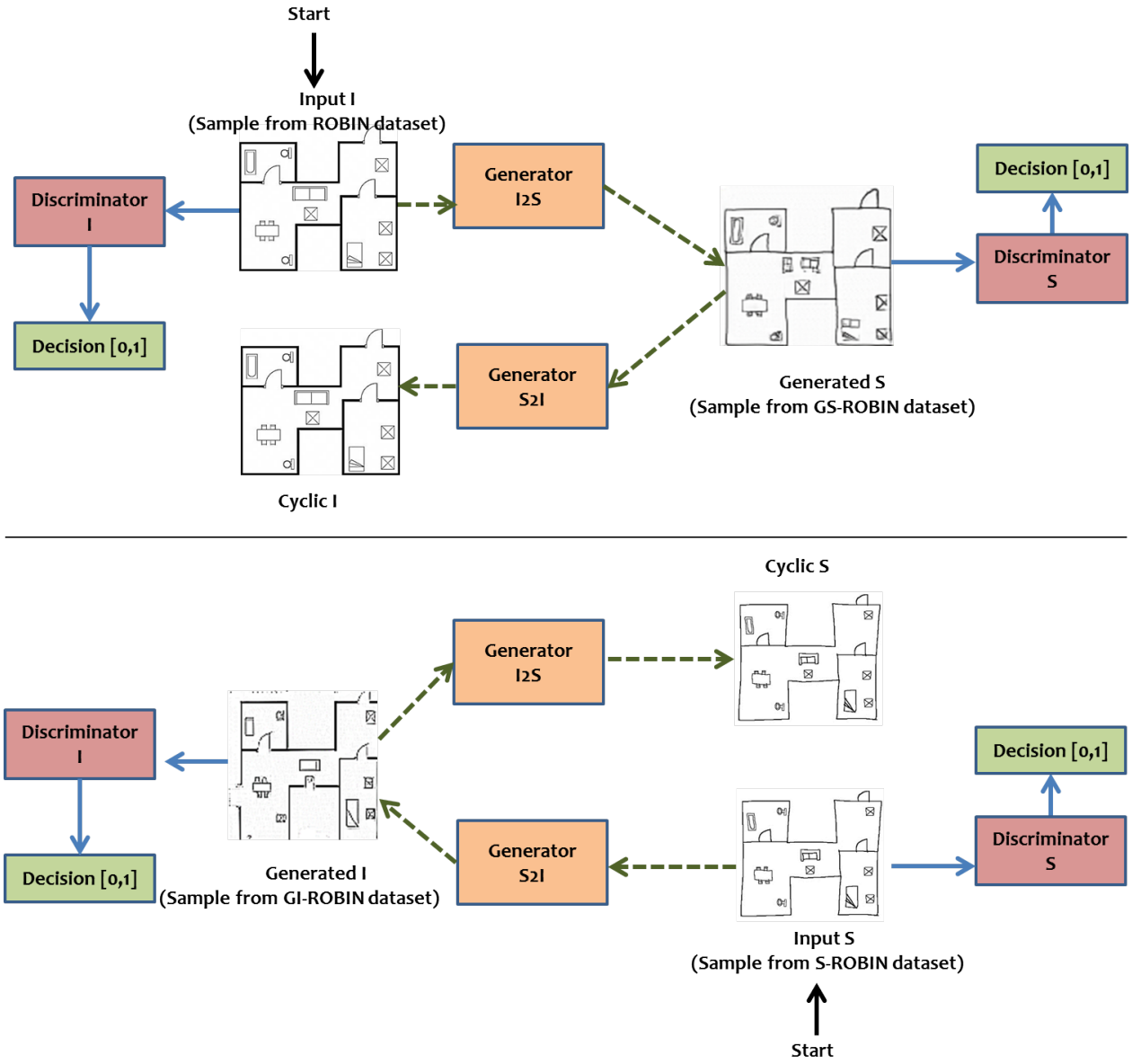


Figure 6.5. : Network architecture for the Cyclic GAN framework, representing the inputs from ROBIN and S-ROBIN dataset. The outputs being generated constitute the GS-ROBIN dataset.

In Eq. 6.1, two operations are taking place. Here the discriminator term D_I, tries to maximize the objective function by distinguishing between the generated floor plan images and the given sketches. Whereas, the generator function $G(.)$ tries to minimize the objective and synthesize image samples as close as possible to the sketch. Similarly, the adversarial loss function for the reverse mapping process, i.e. image to sketch $I2S_A(.)$ is formulated as:

$$I2S_A(F, D_S, I, S) = E_{f_S \to p_{(f_S)}}[logD_S(f_S)] + E_{f_I \to p_{(f_I)}}[log(1 - D_S(F(f_I)))] \tag{6.2}$$

As shown in Fig. 6.4, the generated target domain image using $G(.)$, and $F(.)$, is close to that of the source domain. The learned mapping functions are able to map to any random permutation

of images. However, it doesn't ensure that for a given floor plan image f_I^i, a corresponding f_S^i is obtained. In other words, given an image sample $f_s \in S$, if the generator function $G(.)$ is able to generate f_I^G (here the sub-script I stands for the image domain, and the super-script G denotes that it is a generated image), then the mapping function $F(.)$ must be able to take f_I^G and bring back f_S. So the sketch to image generation cycle completes when the proposed network is able to re-generate the input samples, from the synthesized (or GAN generated) samples, and thus termed as "cycle-consistent". Similar consistency is also desired for image to sketch domain translation. The cycle consistency loss ($CCL(.)$) function between the two mapping functions is formulated as:

$$CCL(G,F) = E_{f_S \to p(f_S)}[||F(G(f_S)) - f_S||_1] + E_{f_I \to p(f_I)}[||G(F(f_I)) - f_I||_1] \tag{6.3}$$

The final objective function ($X(.)$), that combines the two adversarial loss functions and the cycle consistency loss function is given by :

$$X(G,F,D_I,D_S) = S2I_A(G,D_I,S,I) + I2S_A(F,D_S,I,S) + \alpha CCL(G,F) \tag{6.4}$$

Here the hyper-parameter α controls the relative significance of the pair of loss functions, i.e. the adversarial loss and cyclic loss. Using this objective function $X(.)$, the optimum value for the two mapping functions $G(.)$ and $F(.)$ (by minimizing over G and F, while maximizing D_I and D_S) is obtained. The value of α was empirically determined to be 10, which yields the best retrieval accuracy for all the experiments performed for this problem. Another diagram to depict the Cyclic GAN framework is shown in Fig. 6.5.

Figure 6.6 depicts the schematic structure of the generator and discriminator network. There are two parts to the network structure, the generator network and the discriminator network. There are three components to the generator network, namely, an encoder, a transformer, and a decoder. The encoder is a convolutional neural network that extracts the features from a given floor plan dataset. The second building block is a transformer network. This layer has been constructed using 6 ResNet blocks to transform from I domain to S domain (refer Fig. 6.4). In Fig. 6.6, the internal structure of an individual ResNet block is shown. Each block is a neural network layer

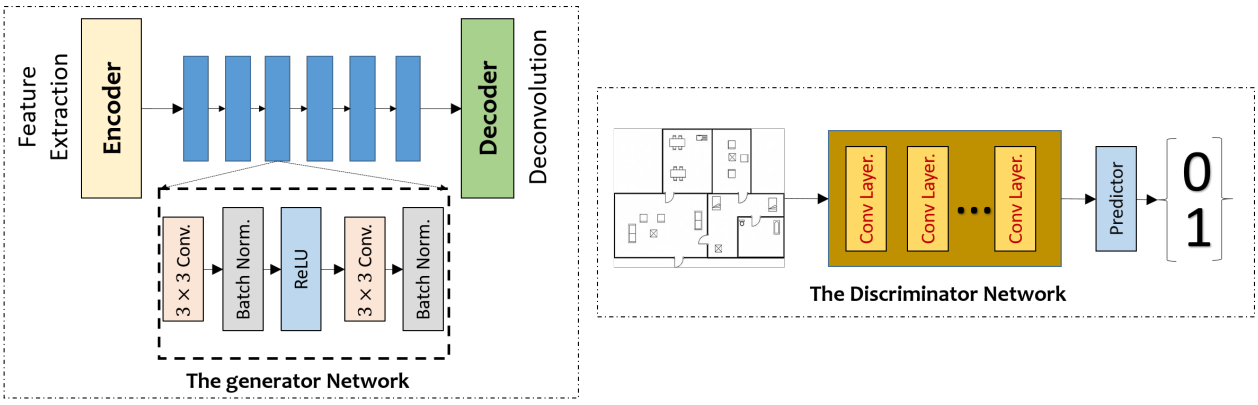


Figure 6.6. : Schematic structure of the Cyclic GAN network used in the proposed framework for image floor plan to synthesized sketched floor plan mapping, and vice-versa.

which consists of two convolution layers where a residue of input is added to the output. This is done to ensure properties of previous layer inputs are available for later layers as well, so that their output does not deviate much from original input resulting into abrupt results. The third component is a decoder network. The decoder block recreates the low level features from the feature vector by applying a deconvolution layer. The next part of the network is a discriminator network. The input to the discriminator network is an image. The output of the discriminator network is a prediction value 0 or 1, that denotes whether the generated image is original or the output from the generator. The discriminator is implemented as a series of convolutional neural networks. After this training and testing phase, a set of generated images corresponding to the sketches, as well as, a set of sketches corresponding to the ROBIN dataset are obtained. Thus, GI-ROBIN dataset generated from the S-ROBIN dataset, and GS-ROBIN dataset generated from the ROBIN dataset, are obtained which are further used for retrieval purposes. To improve the quality of the generated images and sketches, they are sharpened as a post-processing step.

After this training and testing phase, a set of generated images corresponding to the sketches, as well as, a set of sketches corresponding to the ROBIN dataset are obtained. It was observed that if the obvious path, of converting the query sketches into corresponding image counterparts and then retrieving from the ROBIN dataset was taken, the retrieval performance was quite weak giving an average precision value as low as 0.1631 and 0.1219, upon being trained on ROBIN as well as S-ROBIN dataset respectively (refer Fig. 6.14. (a)). Therefore, the less intuitive path of converting the ROBIN dataset into sketches and then performing retrieval was taken to yield better results.

6.2.2 Feature Representation through CNN

CNNs have proved to be very effective for feature extraction and classification purposes. In the recent past, research gaining insight into CNN models includes exploring new non-linear activation functions, new training techniques, optimal network configurations and making them system time efficient and reducing overfitting.

In this work as proposed in Chapter 4, a combination of convolutional, pooling, normalization and fully connected layers, similar to described in the work AlexNet [Krizhevsky *et al.*, 2012a], is taken to obtain an effective feature representation from floor plans. Changes are made in the model as compared to AlexNet by not using relighting data augmentation during training; and by changing the order of pooling and normalization layers [Jia *et al.*, 2014]. For learning a representation of CNN (deep) layers, training of the deep model was done on ROBIN, S-ROBIN and GS-ROBIN dataset. It was observed that the deep learning framework trained on GS-ROBIN dataset performs the best during retrieval (see Fig. 6.14. (b)). The effectiveness of each layer was also investigated by taking the activations of all the layers (convolution, pooling, norm, and fully connected). The major motivation behind using lower convolution neural network layers is to understand the effectiveness of low-level features as compared to higher-level feature representation for floor plan retrieval tasks. Since one only needs to compute the feed-forward network based on the matrix multiplication for one time, the whole scheme is highly time efficient. Therefore, this is how the features from both the sketched query samples as well as the ROBIN generated sketches (GS-ROBIN dataset) are generated. In matching and retrieval stage, query samples are matched with database samples to effectively retrieve the rank-ordered samples from the database.

6.2.3 Matching and Retrieval

As the retrieval is performed on the generated sketches from the ROBIN dataset, therefore, let $(x_\phi^{GS}, y_\phi^{GS})$, $\phi = 1,2...,m$, represent the generated sketches, where, m is the total number of samples in the GS-ROBIN dataset, x_ϕ^{GS} is the observed variable and y_ϕ^{GS} is the corresponding class label in the retrieval database. Let $f(x_\phi^{GS}) : \{f_1(x_\phi^{GS}), f_2(x_\phi^{GS}), .., f_H(x_\phi^{GS})\}$ be the set of hidden layer features extracted from x_ϕ^{GS} GS-ROBIN dataset sample, where $f_p(x_\phi^{GS})$ represents p^{th} hidden layer feature extracted from x_ϕ^{GS} sample in database and H represents the total number of hidden layers in deep feature hierarchy. For experimentation purpose, the features have been extracted from 12 hidden layers of CNN model. For each sample in the database, features from each hidden layer $(f_p(x_\phi^{GS}))$ are extracted and stored in the feature database as discussed in Sec. 6.2.2. In a similar manner, features are extracted from the query sketches as well. Firstly, the performance of individual hidden layer for proposed floor plan retrieval task is observed. Let $f_s(q) : \{f_{s1}(q), f_{s2}(q), .., f_{sH}(q)\}$ be the the set of hidden layer features extracted from a query sketch (q). Then, the Matching Score (M) between query sketch (q) and a sample from the database (x_ϕ^{GS}) using p^{th} hidden layer feature is calculated (and used for ranking), as:

$$M_{approach_1} = \sum_{i=1}^{|f_p(x_\phi^{GS})|} |f_p^i(x_\phi^{GS}) - f_{sp}^i(q)|_2 \tag{6.5}$$

Secondly, the best hidden layer representation for each retrieval sample is obtained by combining the matching score outputs of each hidden layer using a *min* operation. The Matching Score (M) between a query sketch (q) and a generated sketch from the ROBIN dataset (x_ϕ^{GS}) using all the hidden layer features is calculated (and used for ranking), as:

$$M_{approach_1} = argmin_p \sum_{i=1}^{|f_p(x_\phi^{GS})|} |f_p^i(x_\phi^{GS}) - f_{sp}^i(q)|_2, \quad p \in (1,...,H) \tag{6.6}$$

Using Eq. 6.5 and 6.6, given a query sketch the similar generated sketches from the GS-ROBIN dataset are obtained. It is to be noted, that as these sketches have been generated from ROBIN dataset, therefore, transitively they correspond to the images they are generated from. Thus, through this indirect relation, it is shown that the images belonging to the ROBIN dataset [Sharma *et al.*, 2017] are obtained as the retrieved result given the query sketch.

In the next section, Approach 2, catering to sketch based retrieval is discussed in detail. This approach additionally proposes a unified framework for dealing with both image and sketch based queries and finally performaing efficient matching and retrieval of similar floor plans given the dual query mode.

6.3 APPROACH 2: UNIFIED FRAMEWORK FOR RETRIEVAL USING AUTOENCODER AND CYCLIC GAN

Figure 6.7 depicts the complete framework of the proposed technique. Firstly, a query is classified as a floor plan sketch or image using a Convolutional Neural Network (CNN). If

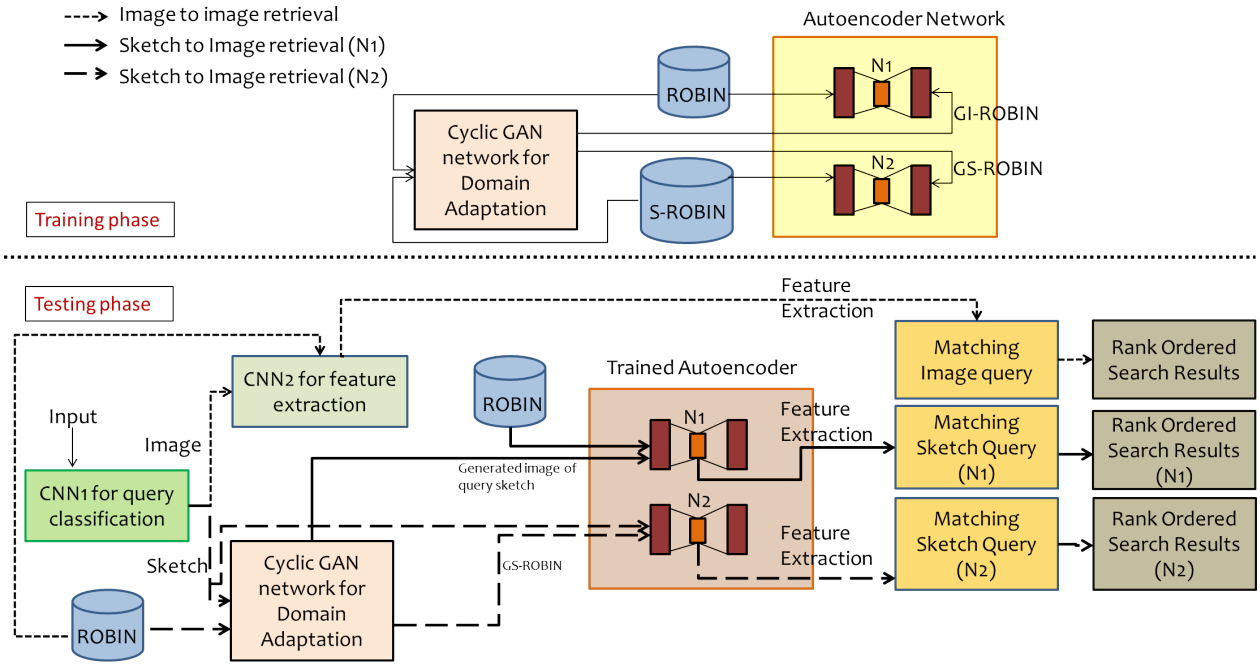


Figure 6.7. : Unified framework for multimodal retrieval.

the query is a floor plan image, the query and the database images are passed through a CNN for feature extraction, and retrieval is carried out using a distance metric to define similarity as proposed in Chapter 4. However, if the query is a sketch, the steps are taken as follows. The entire framework is divided into three main phases. They are: (1) cross-domain sample generation through Cyclic Generative Adversarial Networks (Cyclic GANs); (2) domain mapping through autoencoder and Cyclic GANs, and (3) matching and retrieval. Given an input sketch S, the goal is to retrieve similar looking floor plan images (I) from the ROBIN dataset [Sharma *et al.*, 2017] corresponding to the query sketch. The Cyclic GAN is trained using images from ROBIN and sketches from S-ROBIN to learn domain representation. Direct retrieval of floor plan images, given a sketch, is not straightforward (refer to the low performance of DANIEL in Fig. 6.18) proposition. Therefore, firstly the ROBIN dataset images are converted into their sketch counterparts (called GAN generated sketch dataset (GS-ROBIN)) and the S-ROBIN dataset into their image counterparts (called GAN generated image dataset (GI-ROBIN)) using a Cyclic GAN trained to map the sketch and image domain (as shown in Fig. 6.4).

Further, to map both the floor plan sketches/ retrieval set floor plan images and the cyclic GAN generated outputs, to a common space, experiments are performed by training two autoencoders, taking input as images from ROBIN (network N1) and sketches from S-ROBIN (network N2). The basic autoencoder network is tweaked a little, by keeping the target as GAN generated images (GI-ROBIN) in case of N1 and GAN generated sketches (GS-ROBIN) in the case of N2. This autoencoder learning helps in domain mapping of (a) Images from ROBIN and Generated Images from S-ROBIN (i.e., GI-ROBIN) using autoencoder N1, (b) Sketches from S-ROBIN and Generated Sketches from ROBIN (i.e., GS-ROBIN) using autoencoder N2. Further, the compressed features of both query sketch and floor plan images in the database are extracted from the last layer of the encoder of the trained autoencoder network and used for matching and retrieval purposes. In the next section, details about the various modules of the unified framework are discussed.

6.3.1 CNN for query classification

In this module of the framework a Convolutional Neural Network (CNN), trained on both the floor plan sketch and image datasets, ROBIN and S-ROBIN, is used to classify whether a query is a sketch or an image. A combination of convolutional, pooling, normalization, ReLU and fully connected layers (see Fig. 6.8) along with dropout regularization technique, similar to described in the work AlexNet [Krizhevsky *et al.*, 2012a], is used to obtain an effective feature representation from both floor plan images and sketches.

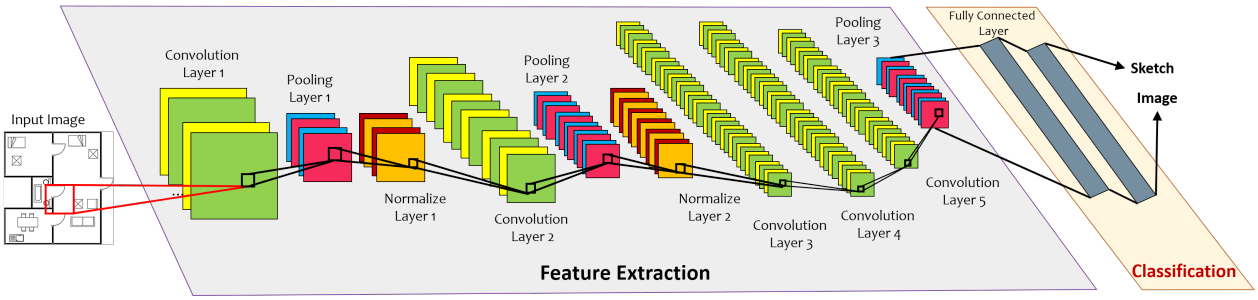


Figure 6.8. : Layerwise depiction of the Convolutional Neural Network (CNN) for query classification

As proposed in Chapter 4, to create a distinction from the original AlexNet framework, the model is not trained with the relighting data-augmentation; and pooling is done before normalization [Jia *et al.*, 2014]. To obtain the feature representation from both the floor plan sketches and images, they are fed into the input layer of the learned CNN model. The sketched floor plans are labeled as 1 and the floor plan images are labeled as 0. The model takes the activations of all the layers viz. a viz. convolution, pooling, normalization, and fully connected to extract the low-level features from the sketch and image floor plan datasets. Since the feed-forward network performs matrix multiplication only once, it makes the whole scheme highly time efficient. The datasets are divided into training and testing sets, which are comprised of 90% and 10% samples respectively and the network is appropriately trained. Finally, on giving the test set to the trained network, the probability of each sample from the mixed bag of S-ROBIN and ROBIN is evaluated to determine which class (0 or 1) they lie into and finally, each test sample is classified into a floor plan image or a sketch. If it is a floor plan image, retrieval proceeds as in [Sharma *et al.*, 2017]. However, if it is a floor plan sketch then domain adaptation as proposed in the subsequent subsections is done.

The next task is cross-domain sample generation using Cyclic GANs, which is done in a similar manner as discussed in Sec. 6.2.1. After the training and testing phase of the Cyclic GAN, a set of generated images corresponding to the sketches, as well as, a set of sketches corresponding to the ROBIN dataset is obtained. Thus, after this step, two sets, GI-ROBIN dataset generated from the S-ROBIN-dataset, and GS-ROBIN dataset generated from the ROBIN dataset, are obtained which are used for retrieval purposes.

6.3.2 Autoencoder for domain mapping

If the query is classified as a sketch, it is first converted into its image representation using Cyclic GAN as discussed in Sec. 6.2.1. As observed in Fig. 6.3, function $G(S)$ generates an image corresponding to the floor plan sketch S that looks approximately similar to how a corresponding image would be in the ROBIN dataset. However, the approximation is still a little far from the exact actual image representation (Fig. 6.3 (b)). Therefore, if direct matching between a generated image from the query sketch and samples of the ROBIN dataset (retrieval set) is performed, the average

precision values obtained were as low as 0.2, which although, is still better than matching sketches and images directly (average precision in this case was obtained to be 0.04) but requires significant improvement. Therefore, to overcome these shortcomings, the introduction of autoencoders is proposed. This helps in mapping the sketch query and sketches generated from the floor plan images, or in another case the image generated from query sketch and the retrieval set floor plan images onto a common space for matching purposes. An autoencoder neural network is an unsupervised learning algorithm that applies backpropagation and sets the target values to be equal to the inputs, i.e., it uses $y^i = x^i$. The autoencoders here are not being used to generate output image similar to the input image at the output layer, rather the use of autoencoders is proposed to learn a common subspace representation of two similar looking domains. Therefore, in the proposed case, additional output image (generated using Cyclic GAN) corresponding to the input image is used for training the autoencoders (Fig. 6.9). This additional output image helps to learn intermediate representation of both input and output domains. This intermediate representation acts as a common subspace representation in our case. The novelty in this case lies in the fact that, unlike traditional autoencoders, which try to generate an output similar to the input fed to them by computing the loss between the input and the target output and updating the weights, the training process of the proposed autoencoder is slightly modified. In the proposed case, a new way of learning representation is performed, where, predefined outputs are used along with the decoder generated output for autoencoder training. The aim is to minimize the difference between the decoder generated output and provided output (generated through Cyclic GAN). The autoencoder is trained in two ways, firstly, the autoencoder labeled as N1 is trained keeping the input as images from ROBIN dataset (i $\in$ ROBIN), whereas, the additional target output as images generated from sketches through the Cyclic GAN network (i' $\in$ GI-ROBIN). This helps in learning the intermediate representation till the loss between the decoder generated output and the sketch generated image domain converges to a small value. For this task, an autoencoder is used as shown in Fig. 6.9.

An autoencoder consists of two parts, the encoder and the decoder, which can be represented as $\mathscr{E}$ and $\mathscr{D}$ respectively. Where, $\mathscr{E} : i \rightarrow z$ and $\mathscr{D} : z \rightarrow \hat{i}$, and z is the compressed representation of the input. Note that during training the target has been set as the images generated form sketches of the S-ROBIN dataset, so that the autoencoder learns to map both domains.

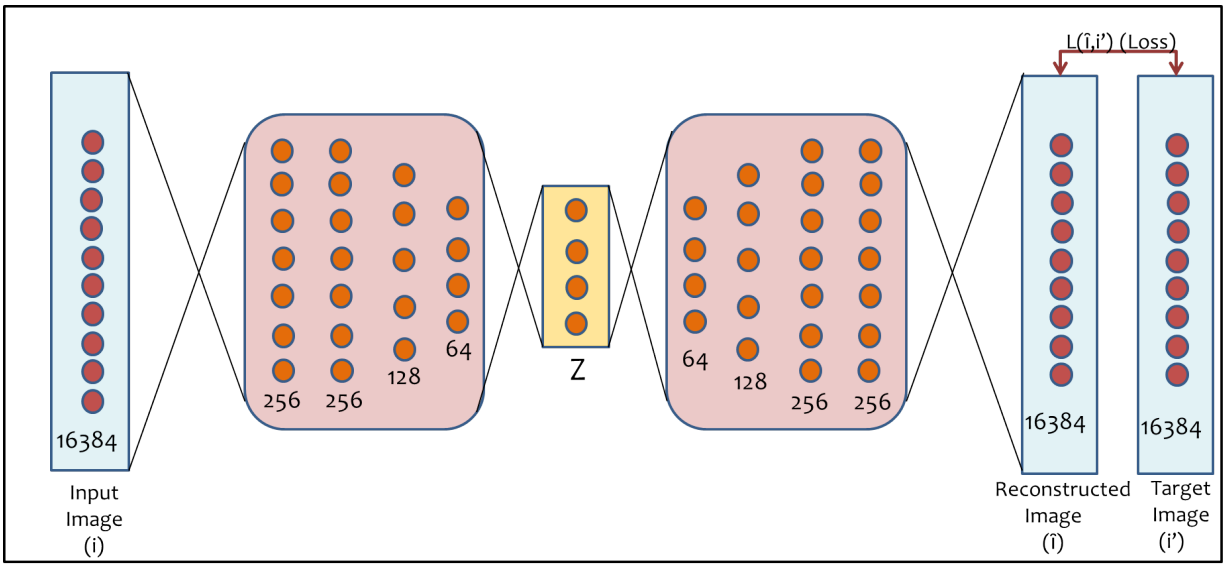


Figure 6.9. : Layerwise depiction of the autoencoder network.

Further, the squared loss function to minimize the reconstruction error to train the network is defined as follows:

$$\mathscr{L}(\hat{i}, i') = ||\hat{i} - i'||^2 \tag{6.7}$$

Note that, $\hat{i}$ is the reconstructed image from the trained autoencoder corresponding to the input. The distinction of our autoencoder network from a simple autoencoder is that the loss is calculated between the GAN generated floor plan image and the reconstructed image. This helps in efficient mapping of both sketch and image domains by taking aid from the Cyclic GAN network, for conversion of sketch query to image or retrieval set images to sketch. The parameters for the autoencoder network are listed in Tab. 6.3.

After the network is fully trained, features from the last layer of the encoder are extracted for both the query sketch generated image that has been obtained using a Cyclic GAN and images from ROBIN dataset that serve as the retrieval set. This helps to obtain a compressed feature vector for both, which is further used for matching and retrieval. This feature extraction helps in mapping both domain of images (I) and domain of sketches (S) to a common domain called (C) where, now the matching and retrieval process takes place.

On similar lines, the autoencoder network, labeled N2, was also trained keeping the input as sketches from the S-ROBIN dataset and the target as generated sketches (GS-ROBIN dataset) obtained by the Cyclic GAN network. It was interesting to note, that due to the quality of the generated sketches from the Cyclic GAN being better, the autoencoder learned a better representation and mapping of both the domains using the N2 network. As per our knowledge, this is a novel way to learn a common embedding space between the sketch and image domain, using a modification in the autoencoder framework by not keeping the input and the expected output to be the same.

6.3.3 Matching and Retrieval

During testing, if the query is a floor plan image then matching and retrieval proceeds as proposed in Sec. 4.3. However, if the query encountered in the framework is classified as a sketch, then two paths can be taken for retrieval purposes (solid-line and dashed line) as shown in Fig. 6.7. The query sketch can be converted into its image representation using the Cyclic GAN. Compressed vector can be obtained from the final layer of the encoder, by passing query sketch generated image and the ROBIN dataset through the autoencoder network N1. The features thus obtained can be used for matching and retrieval purposes. On similar lines, the other path that can be adopted is if the query is a sketch, then the query sketch, along with the sketch counterparts of the ROBIN dataset (GS-ROBIN dataset) can be passed through N2 autoencoder network, where again, compressed vector obtained from the last layer of the trained encoder can be used as a feature vector for both query sketch and the retrieval set. To illustrate the matching process for one such path, let $f(x_i) : \{f_1(x_i), f_2(x_i), .., f_n(x_i)\}$ be the feature vector extracted for x_i image where $i \in$ ROBIN dataset, and n represents the size of this feature vector, which in our case is 64. For each sample in the database, features are extracted and stored in the feature database. In a similar manner, features are extracted from the query sketch generated image. Then, the Matching Score ($M_{Approach_2}$) between query sketch generated image(gi) and a sample from the database (x_i) is calculated (and used for ranking), using three different distance metrics:

Euclidean Distance Metric

$$M_{Approach_2} = \{ \sum_{a=1}^{|f(x_i)|} \{f_a(gi) - f_a(x_i)\}^2 \}^{\frac{1}{2}} \tag{6.8}$$

Manhattan Distance

$$M_{Approach_2} = \sum_{a=1}^{|f(x_i)|} |f_a(gi) - f_a(x_i)|_2 \tag{6.9}$$

Cosine Distance

$$M_{Approach_2} = \sum_{a=1}^{|f(x_i)|} \frac{\sum_{a=1}^{|f(x_i)|} \{f_a(gi).f_a(x_i)\}}{\{\sum_{a=1}^{|f(x_i)|} \{f_a(gi)\}^2\}^{\frac{1}{2}}.\{\sum_{a=1}^{|f(x_i)|} \{f_a(x_i)\}^2\}^{\frac{1}{2}}} \tag{6.10}$$

The distance is calculated using 10-fold cross-validation where, the query sketch samples consist of 70% of the sketched floor plans from S-ROBIN dataset and the retrieval set comprises of all the 510 floor plans in the ROBIN dataset. Based on this distance the database floor plan images are rank ordered. Shorter the distance, more similar a pair of floor plan is. The rank ordered samples are shown to the user. Next, the details of the experiments performed, implementation details, and the results obtained from them are discussed.

6.4 EXPERIMENTS AND RESULTS

6.4.1 Dataset Creation

The experimentation was done on 2 datasets : (1) ROBIN dataset[Sharma *et al.*, 2017] and (2) S-ROBIN dataset. For the framework all the floor plan images of ROBIN to sketches were replicated. Since, there is no publicly available sketch floor plan dataset for the DAR community, a new dataset was created and named as S-ROBIN as discussed in Annexure A. S-ROBIN dataset has 510 real world floor plan sketches divided into 3 broad categories and 17 sub-categories differing in the overall global outer shape of the layouts. It is to be noted that class-belongingness of each sketched layout is determined in the dataset by its outer shape, for example all the hand-drawn floor plans in one sub-category will have the same L-shaped outer layout. To capture these hand drawn sketched floor plans, the volunteers were asked to draw the floor plans on a digital platform using a Wacom Tablet. The resolution ranges from 591×517 pixels to 1483×884. Since working in the off-line mode is considered, the hand drawn floor plans are stored as an image and used for experimentation. All the images were saved in a gray scale format.

6.4.2 Implementation Details

For training the Cyclic GAN, the datasets ROBIN and S-ROBIN were divided into training (70%) and testing (30%) sets. Further, the Cyclic GAN with the help of the losses discussed in Sec. 6.2.1, learns the mapping between both domains. During testing, due to the cyclic nature of the network, both the generated sketches from [Sharma *et al.*, 2017] and the corresponding images generated from the S-ROBIN dataset are obtained. It was observed that visually the sketches generated from ROBIN were of better quality. To make it clear to the reader, let query sketch

be S, the ROBIN dataset images to be retrieved be $\{I\}$ and the generated sketches from ROBIN ($\{I\}$) be $\{GS\}$. Then as S and $\{GS\}$ are in the same domain, similar sketches in $\{GS\}$ are retrieved smoothly through the framework. Further, it is to be noted, these generated sketches $\{GS\}$ exactly correspond to their image counterparts from $\{I\}$. Hence, it is safe to say that through the framework relationship between S and $\{GS\}$ has been determined. Also, relationship between $\{GS\}$ and $\{I\}$ pre-exists. Therefore, transitively, relationship between S and $\{I\}$ has thus been established. It is this indirect relationship that helps to justify that ROBIN dataset images similar to the query sketch can be shown as the retrieval result obtained through the framework.

Approach 1:

For feature representation in Approach 1, three CNNs were trained using the ROBIN dataset, S-ROBIN dataset and the GS-ROBIN dataset, respectively. All the three datasets are again divided into training set with 70% samples and test set with 30% samples. As a retrieval database, the entire GS-ROBIN dataset is used which is generated from the ROBIN dataset and thus, transitively represents it. Each floor plan image has been resized to a size of 128×128 for both matching and feature representation task. For deep model configuration, the architecture described in [Krizhevsky *et al.*, 2012a] is followed.

Approach 2:

After the classification task, if the query is a floor plan image, feature extraction and retrieval proceeds as in [Sharma *et al.*, 2017]. If the query is a sketched floor plan then the Cyclic GAN network comes into picture to convert the sketched floor plans into their image counterparts, which are then used for training the autoencoder, to bridge the difference between the sketched and the image domain.

6.4.3 Parameter Setting

Convolutional Neural Network:

The parameters for the CNN network for both feature extraction in Approach 1 and query classification in Approach 2, are similar as mentioned in Chapter 4. Table 6.1 lists out these parameters again.

Cyclic GAN:

Setting appropriate values for the hyper parameter plays a pivotal role for the success of the Cyclic GAN and the deep learning implementation. Table 6.2 lists out the parameters and their corresponding values used while implementing the framework. The effect of higher number of iterations to generate good quality image is depicted in Fig. 6.10. It is to be noted that the input is given in the form of an image and number of filters in the 1^{st} layer of the generator determine the output features of the convolution layer. This is then passed to the next convolution layer, which subsequently leads to extraction of progressively higher level features aiding in domain mapping. It can be observed that given an original floor plan image our proposed network is able to generate a sketched version of it with very high accuracy. As the number of iterations increases the quality improves. It was empirically found that after the optimum value of number of iterations is achieved, the quality of the image doesn't improve significantly, and hence to have a trade-off between time and quality, an appropriate value for the number of iterations is chosen, which in our case was chosen to be 60000 iterations.

Autoencoder framework:

Layers	1	2	3	4	5	6
Type	conv1+ maxpool1+ norm1	conv2+ maxpool2+ norm2	conv3	conv4	conv5+ maxpool3	full (fc6,fc7)
Channels	96	256	384	384	256	4096
Filter Size	11*11	5*5	3*3	3*3	3*3	-
Convolution Stride	4*4	1*1	1*1	1*1	1*1	-
Pooling Size	3*3	3*3	-	-	3*3	-
Pooling Stride	2*2	2*2	-	-	2*2	-
Padding Size	-	1*1	1*1	1*1	1*1	-

Table 6.1. : Network Parameters for the proposed framework

Table 6.2. : Parameter Values used while implementing Cyclic GAN

Parameter	**Value**
# filters in the 1^{st} layer of the generator	64
# filters in the 1^{st} layer of the discriminator	64
batch size	1
pool size	50
img width and img height	128

The autoencoder used in Approach 2, plays a pivotal part in mapping both the sketched and image floor plans onto a common space for efficient retrieval. The images/sketches are reshaped to a size 128×128 and vectorized into a 16384×1 vector, before being fed to the autoencoder network.

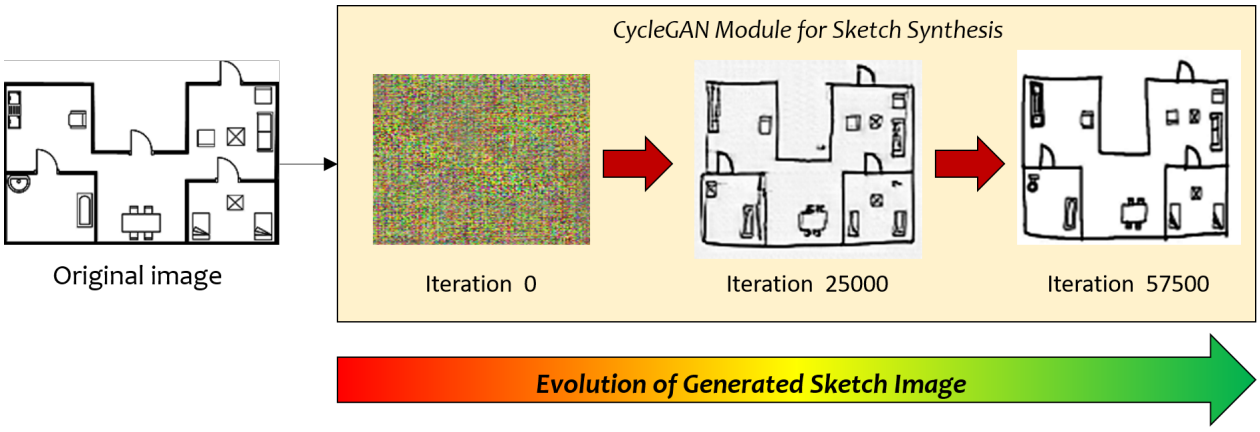


Figure 6.10. : A diagram depicting stages of sketch synthesis from an image through our Cyclic GAN framework. Note how, the discriminator and generator improve with increasing the number of training iterations.

100,000 iterations were undertaken for the autoencoder to train properly and 70% of the samples from the ROBIN and S-ROBIN datasets were used as training set for training both CNN, Cyclic GAN and the autoencoder. Further, the details of the parameters involved in the autoencoder network are listed in Tab. 6.3.

Table 6.3. : Parameter Values for Encoder Network

Component	Parameter	Value
Encoder	No. filters in the 1^{st} layer	256
	No. filters in the 2^{nd} layer	256
	No. filters in the 3^{rd} layer	128
	No. filters in the 4^{th} layer	64
Decoder	No. filters in the 1^{st} layer	64
	No. filters in the 2^{nd} layer	128
	No. filters in the 3^{rd} layer	256
	No. filters in the 4^{th} layer	256

6.4.4 Qualitative Results

Approach 1:

Approach 1 is able to successfully retrieve a rank order set of floor plan images from the ROBIN dataset given a query in the form of sketch, using the proposed domain mapping framework consisting of Cyclic GAN and CNN. The top 5 rank ordered retrieval results for three different hand-drawn queries, are shown in Fig. 6.11 (a),(b) and (c). The samples which are erroneously retrieved, i.e. not from the same class as that of the query class, are highlighted with a "red" bounding box. For all the results shown in Fig. 6.11, the features extracted from the first normalization layer of the CNN stack have been used, owing to the fact that it yields the best performance. The first rank ordered result retrieved from the ROBIN dataset, is the image exactly matching the query sketch. This is due to maximum matching score between the retrieved layout and the query layout. The subsequent results differ in the matching scores and are thus, ranked lower. In case of Fig. 6.11 (a) and (b), all the top 5 retrieved samples belong to the query class. Global layouts of all the floor plans are identical and so is the number of rooms in the floor plans. It is to be noted that there is a difference in the shape and position of the individual rooms, as well as, the furniture components for the rank ordered samples. There are 4 rooms in each floor plan (refer Fig. 6.11 (a)), however this framework is able to rank order the samples correctly by ordering the most similar ones in terms of the features present inside the floor plans.

Figure 6.11(c) depicts one of the cases where this algorithm does not perform perfectly. As it can be noted, for a given query sketch, the 5^{th} ranked result is erroneously retrieved (highlighted by "red") by this technique. Justification of the Rank 5 erroneously retrieved result can be understood through qualitative analysis. Observe that the rooms in both the layouts share very similar adjacency. The bottom part of the floor plan is almost identical. Moreover, the size and type of furniture present in the individual rooms is quite similar in case of the query and the Rank 5 sample. Hence, similarity score between the query floor plan and the retrieved Rank 5 layout is high.

Approach 2:

The framework proposed in this approach can successfully retrieve a rank order set of floor plan images from the ROBIN dataset given a query in the form of floor plan sketch/image. It is

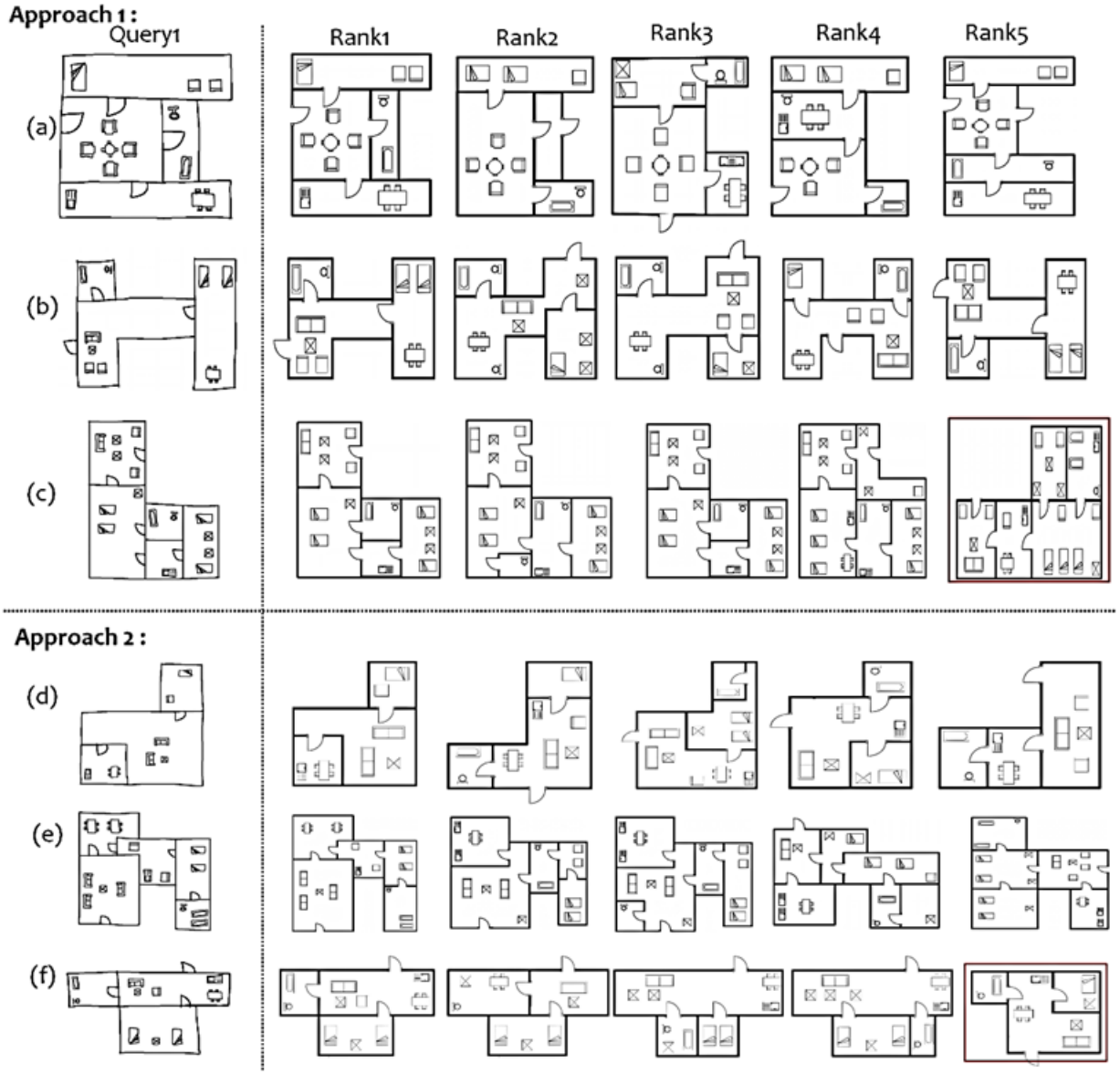


Figure 6.11. : Top five rank ordered retrieval results for the two proposed approaches for three different hand-drawn query floor plans.

to be noted that the retrieval set is always the ROBIN dataset. Experimentation was done taking random queries from both S-ROBIN and ROBIN dataset. The top 5 rank ordered retrieval results for three different hand-drawn queries are shown in Fig. 6.11 (d), (e) and (f). The samples which are erroneously retrieved, i.e., not from the same class as that of the query class, are highlighted by a bounding box. As shown in Fig. 6.11, it can be observed that the conjunction of Cyclic GAN and autoencoder for mapping as proposed in the framework, is efficiently able to capture the similarity between the query sketch and the floor plan images. In the case of Fig. 6.11 (d) and (e), all the top 5 retrieved samples belong to the query class. Global layouts of the all the floor plans are identical

and so is the number of rooms in the floor plans. It is to be noted that there is a difference in the shape and position of the individual rooms, as well as, the furniture components for the rank ordered samples. There are three rooms in each floor plan (refer Fig. 6.11 (d)). However, this framework can rank order the samples correctly by ordering the most similar ones in terms of the features present inside the floor plans.

Figure 6.11(f) depicts one of the cases where the algorithm does not perform perfectly. As it can be noted, for a given query sketch, the 5^{th} ranked result is erroneously retrieved (highlighted by "red") by this technique. Justification of the Rank 5 erroneously retrieved result can be understood through qualitative analysis. Observe that the rooms in both the layouts share very similar adjacency. The similarity lies in specific parts such as the 1^{st}, 2^{nd}, and 3^{rd} rooms have the same furniture components as the query floor plan, although the overall layout differs.

Similarly, qualitative analysis of the framework for the query being an image was also done as shown in Fig. 6.12. For all the results shown in Fig. 6.12, the features extracted from the first normalization layer of the CNN stack are used, as it yields the best performance. The first rank ordered result retrieved from the ROBIN dataset, is the image exactly matching the query image. The reason is the maximum matching score between the retrieved layout and the query layout. The subsequent results differ in the matching scores and are thus, ranked lower.

Both the proposed frameworks are also able to perform well for queries belonging to unseen classes. Figure 6.13 (1) and (2) depict two such scenarios where the sketch query does not match any of the samples in the ROBIN dataset. Still, this framework is capable of retrieving the closest

Figure 6.12. : Top five rank ordered retrieval result of proposed framework for three different query floor plan images using Approach 2 proposed in the Chapter.

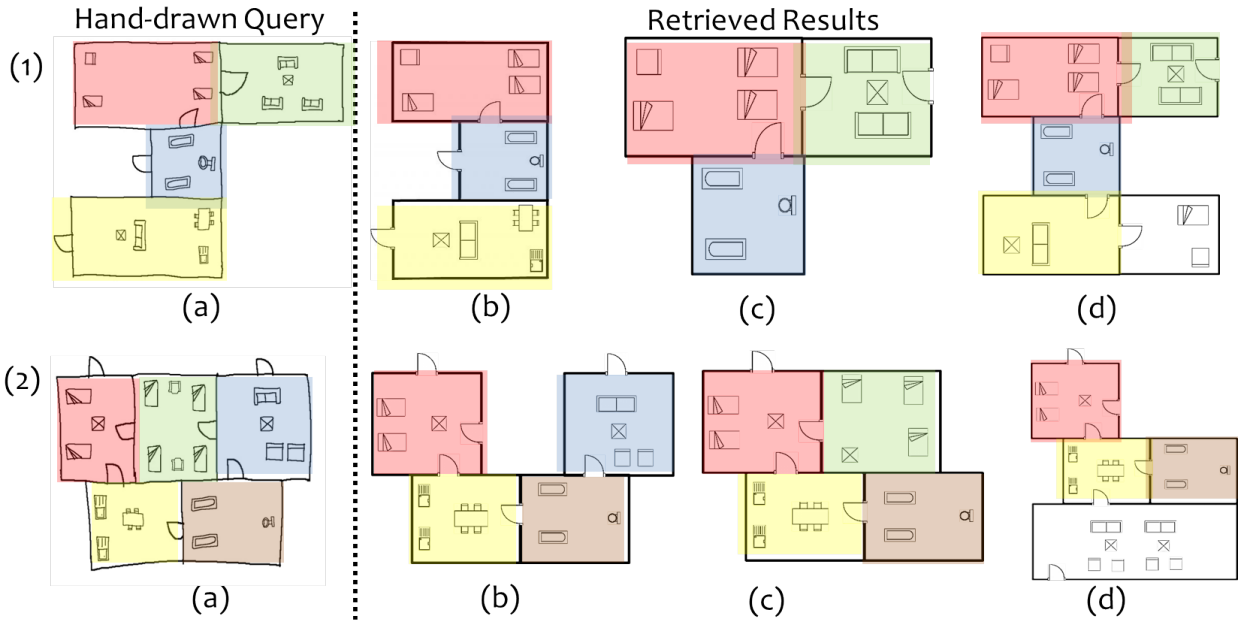


Figure 6.13. : Two examples showing results for querying the system for partial matches. (a) Query Sketch, (b),(c),(d) Retrieved results. Colour Coding establishes room correspondence between the query and the results.

similar results, in terms of room furniture and room placement. E.g., note the similarity between the rooms highlighted in "red", "blue" and "yellow" of the query sketch Fig. 6.13 (1) (a) and the retrieved result of Fig. 6.13 (1) (b). The decor components and room placement are quite similar. The "green" part of Fig. 6.13 (1) (a) is missing in Fig. 6.13 (1) (b). Similar observations can be made for the other results also. Although the shape of the layout is a subset of the shape of the query sketch and does not match completely, this framework is able to correctly retrieve and rank-order the results. Thus, this can help to suggest varied finished designs to architects/property buyers if they have some abstract ideas in mind.

6.4.5 Quantitative Results

The performance of both the approaches is quantified using the Precision (P) and Recall (R) metric. The precision values have been averaged over all the queries for the particular recall values. Given a query sketch, retrieved layouts should belong to the same sub-category of layouts as the query, keeping in mind the preference set by the property seeker during querying. Here, the global outer shape of the layout is the criteria for class belongingness of a given floor plan.

Approach 1:

As discussed, direct retrieval of floor plan images given a sketch is not straightforward (refer to the low performance of the CNN framework in Fig. 6.18 using DANIEL proposed in Chapter 4). Moreover, converting query sketches into corresponding image counterparts and then retrieving from the ROBIN dataset yielded weak performance (refer CNN A, B (trained on ROBIN and S-ROBIN respectively) in Fig.6.14 (a)). Therefore, sketch counterparts of ROBIN dataset generated through CyclicGAN are used for retrieval. The experimentation was carried out with CNNs trained on ROBIN (CNN_1), S-ROBIN (CNN_2) and the GS-ROBIN dataset (CNN_3), for feature

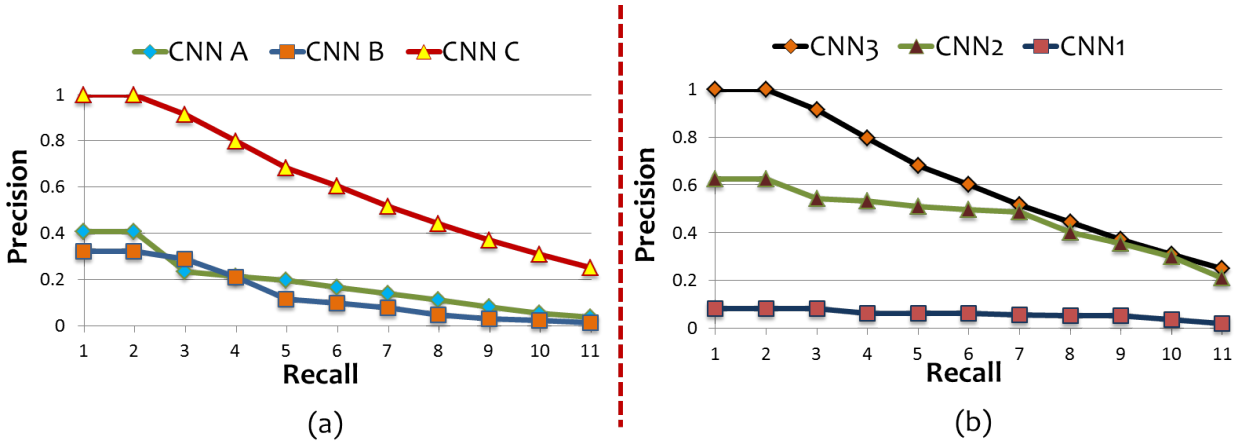


Figure 6.14. : (a) Effect of converting query sketch into image through Cyclic GAN and performing retrieval (CNN A (trained on ROBIN) and CNN B (trained on S-ROBIN)) as opposed to converting the retrieval database into sketches (CNN C). (b) Effect of training CNNs on ROBIN (CNN_1), S-ROBIN (CNN_2) and GS-ROBIN dataset (CNN_3)

extraction purpose, and interesting quantitative results were obtained showing that CNN trained on the GS-ROBIN dataset performs the best in terms of retrieval of similar floor plans (refer Fig.6.14 (b)). To understand the effectiveness of the proposed CNN layers, the PR and Mean Average Precision (MAP) of floor plan retrieval task was evaluated using each layer separately (as shown in Fig. 6.15). It was analyzed that normalized and pooling layers are quite powerful as compared to convolutional layer for floor plan retrieval task. It was observed that the normalized layer 1 yielded the best average precision value of 0.63 while retrieval. An analysis was also performed for the proposed min based approach (refer Eq. 6.6) during retrieval task and it was observed that the precision values through the min-based approach were obtained to be 0.525.

Approach 2:

Using the unified framework proposed in Approach 2, the mean average precision (MAP) value obtained if the query is an image is 0.56 as reported in Chapter 4.

However, if the query is a sketched floor plan, as discussed before, direct retrieval of floor plan images given a sketch is not straightforward (refer the low performance of CNN framework, DANIEL in Fig. 6.18). The experimentation was done with training the autoencoder network using sketches as well as images as described in Sec. 6.3.2. The PR plot obtained for retrieval using network N1 and N2 is shown in Fig. 6.16.

As shown in Fig. 6.11, given a query sketch the rank 1 result is the corresponding query floor plan image itself. This leads to the highest precision value of 1 during the initial recall. With further retrieval the average precision value decreases due to some incorrectly retrieved results belonging to other categories of layouts as compared to the query layout. It is to be noted that the area under the PR curve (refer Fig. 6.16) is greater for network N2, trained on sketch datasets due to the fact that the generated sketches from Cyclic GAN network are qualitatively better. The average precision value is obtained to be 0.642 using N2 network for domain mapping and 0.621 using the N1 network. Also, Tab. 6.4 lists out how each distance metric shown in Sec. 6.3.3 performs while matching and retrieval.

Figure 6.17, compares both the approaches and their analyzed sub-components. In Approach

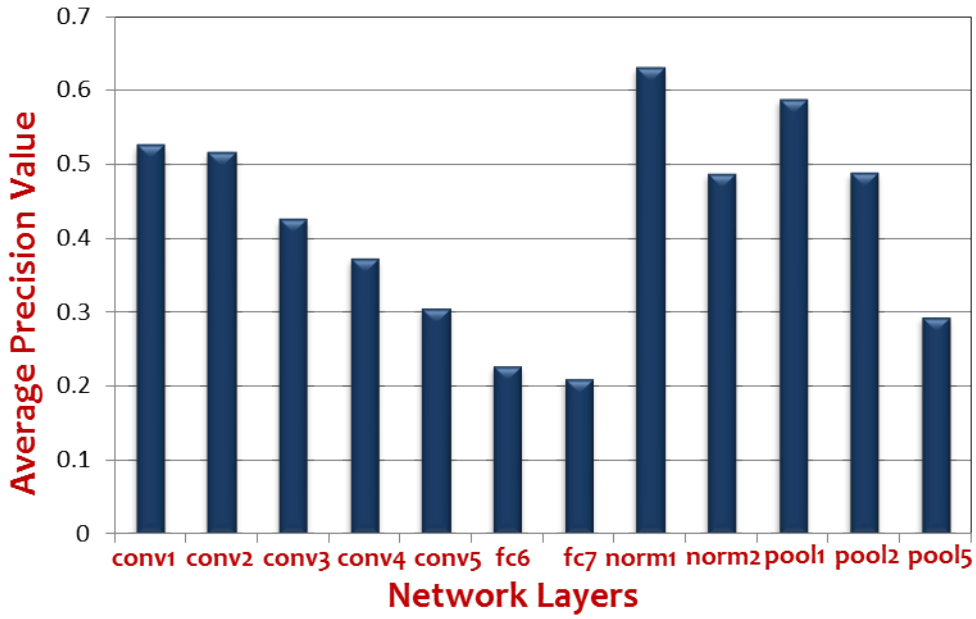


Figure 6.15. : Precision and Recall (PR) plot, comparing result of the framework, using all the hidden layers CNN features.

Table 6.4. : Mean Average Precision values for the proposed framework considering different distance metrics.

Distance Metric	Average Precision Value
Euclidean Distance	0.642
Cosine Distance	0.636
Manhattan Distance	0.625

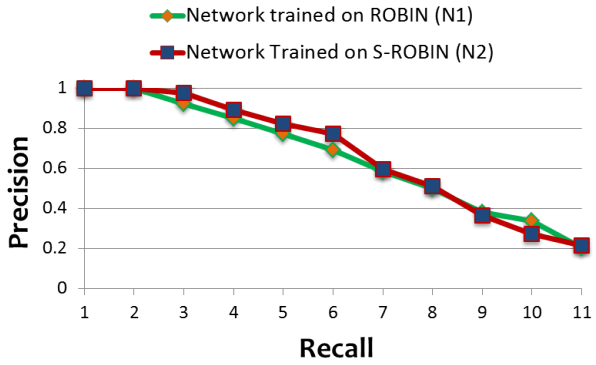


Figure 6.16. : Precision and Recall (PR) plot, comparing result of this framework on autoencoder trained with ROBIN dataset (N1) and S-ROBIN dataset (N2).

1, retrieval results are demonstrated using (1) the min-based approach taking the minimum of distances obtained by retrieval from all layers of the CNN framework and then calculating the Precision and Recall. (2) The best precision obtained from each layer of the CNN framework. This layer was obtained to be normalization one layer, which out-performed all other layers of the CNN framework in terms of retrieval. In Approach 2, N1 autoencoder is trained using images from ROBIN dataset as input and the target images as sketch generated images, (GI-ROBIN dataset) obtained through Cyclic GAN network. N2 autoencoder network, on the other hand, is trained using sketches from S-ROBIN dataset as input and the target images as the image generated sketches, GS-ROBIN dataset obtained through Cyclic GAN network.

It can be observed looking at the area under the curve in Fig. 6.17, that Approach 2, using N2 autoencoder network outperformed all the other approaches while retrieval and gave a MAP value of 0.642 during retrieval.

Comparative studies with state-of-the-art (SOA) techniques were carried out to show the effectiveness of the proposed approaches. Figure 6.18 depicts the PR plot comparing the result of best of the two approaches (Approach 1 using norm1 layer and Approach 2 using N2 network), with the other SOA techniques. There is a significant increase in the performance while using the two proposed approaches as compared to other features or techniques keeping sketch as query mode during retrieval.

Kindly note, the under-performance of the basic deep learning framework [Sharma *et al.*, 2017] (average precision value : 0.04393) using straight-forward feature extraction from CNNs.

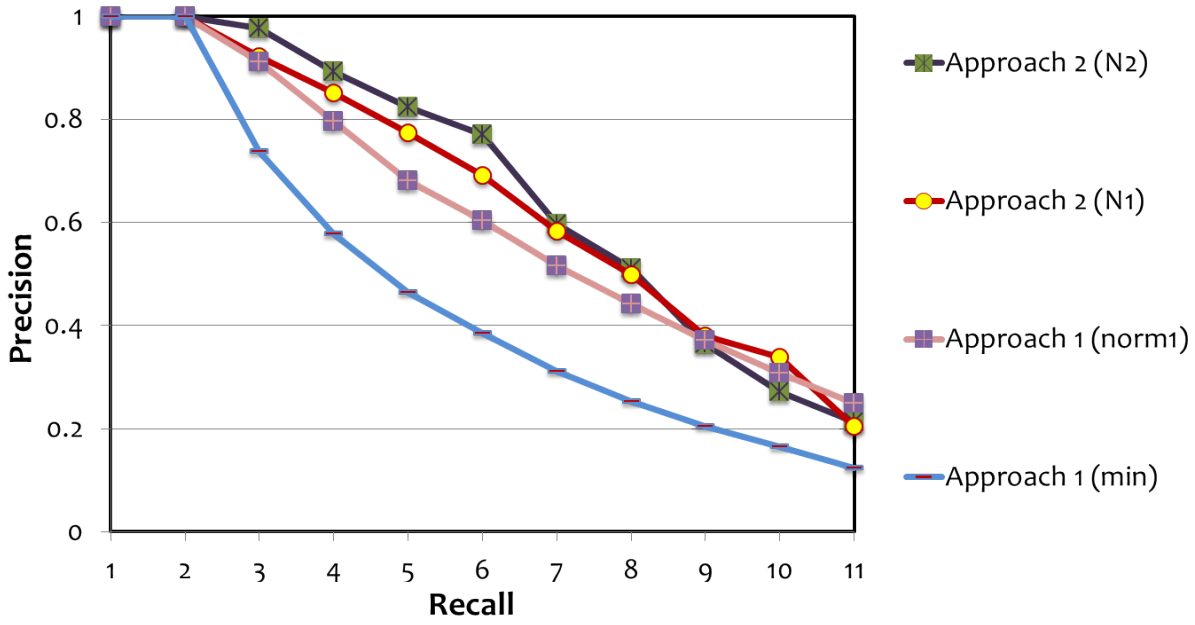


Figure 6.17. : PR plot, comparing the two approaches. Approach 1 extracts features and performs retrieval taking minimum of the precision values obtained by each layer (Approach 1 (min)) and taking the best layer (normalization 1 layer) giving the highest MAP value (Approach 1 (norm1)). Whereas, Approach 2 has two autoencoder networks, N1 trained on ROBIN and GI-ROBIN and N2 trained on S-ROBIN and GS-ROBIN.

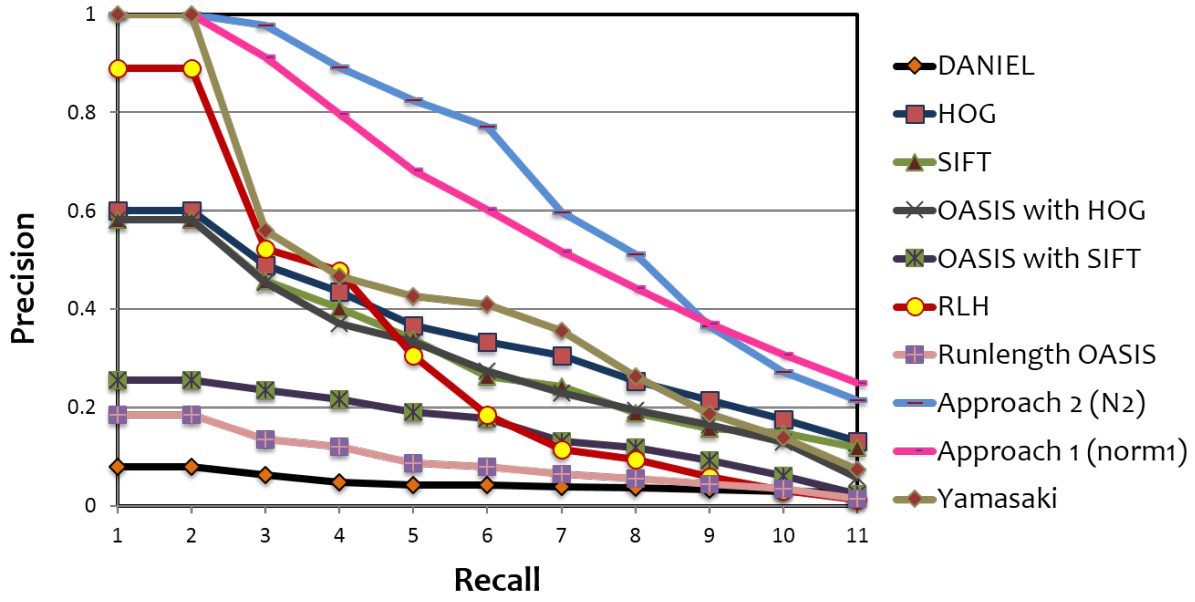


Figure 6.18. : PR plot, comparing the best of the proposed two approaches (Approach 1 (norm1)) and (Approach 2 (N2)) with SOA techniques.

The proposed approach was also compared with the very recent technique mentioned in [Yamasaki *et al.*, 2018] that segments the floor plans using CNN and matches them using Maximum Common Subgraph. It was observed that the proposed technique performs considerably well as compared to [Yamasaki *et al.*, 2018] (improvement in average precision value by 0.254), because the matching criteria in Yamasaki *et al.* [2018] only pertains to matching adjacencies and is also not well scaled for domain adaptation from sketch to image. Also, SIFT, HOG, Runlength Histogram and OASIS approaches perform weakly as they are not able to capture the abstract and sparse nature of sketches and are unable to map between the features of images and sketches. Thus, the comparative results justify the importance of the introduction of both the domain-mapping approaches given a sketch based query and image-based retrieval samples.

6.5 SUMMARY

In this chapter, two frameworks are proposed to handle multimodal retrieval in floor plans. Firstly, a deep learning framework using GAN model is proposed for sketch-based retrieval of building floor plan images. Sketch-based queries make the framework more convenient for the end user to access. Moreover, sketch based queries help in capturing the user's intent in the best possible manner. Due to the limitation on the available images in the database, it is not always necessary to find a perfect mapping of what the user desires to query in the database. Hence, drawing one's ideas on a device which can capture the basic structure of a floor plan and return similar images can prove to be beneficial as an application. The framework is capable of achieving an average precision value of 0.63 upon experimenting with 510 real-world floor plan images and their sketch replicas. Also, the framework is capable of partial matching of sketched floor plans. These abstract floor plans match in certain features, if not all, to the floor plan images in the database. Even if the user has some basic idea of what he desires his floor plans to be then the user can use this

framework for retrieval of similar floor plans.

The second framework applies domain adaptation techniques that help to map both sketches and images to a single framework and are capable of performing better retrieval. There was a need to develop a unified framework capable of both image to image and sketch to image retrieval to serve as a composite tool for multimodal retrieval in floor plans. It was observed that autoencoders in conjunction with Cyclic GANs performed better during retrieval of architectural floor plans giving an average precision value of 0.642 which is improvement by a value 0.1 over the network containing only Cyclic GANs for domain adaptation.

Annexure A
Floor plan datasets ROBIN and S-ROBIN

Dataset plays a pivotal role in any image processing problem. Availability of a public dataset sometimes makes the life of a researcher easy. However, this is true for only generic problems, and not so for problems which are of some specific nature. Floor plan image analysis is such a problem where a large collection of samples is hardly available. For this reason two public datasets have already been proposed by researchers. They are: (i) SESYD [Delalandre *et al.*, 2010b] and (ii) CVC-FP [de las Heras *et al.*, 2015]. The former has 10 classes of floor plans, where there are 100 samples/class. The latter has 122 scanned floor plan documents divided into 4 categories based on the origin and style. The SESYD dataset is inappropriate for application to a retrieval task due to very less difference between a pair in a class of floor plan images. Also, in the CVC-FP dataset, the samples are insufficient in number for the task of floor plan retrieval.

To train and evaluate the contributions presented in this thesis, two floor plan databases, named Repository Of BuildIng floor plaNs (ROBIN) and Sketched-Repository Of BuildIng floor plaNs (S-ROBIN) are created. S-ROBIN are the sketch replicas of the ROBIN dataset containing floor plan images. These datasets have significantly aided in the content analysis and retrieval of similar floor plans as proposed in this thesis. The primary motivation while creating the ROBIN dataset was to have: i) sufficient number of sample to execute the experiments, and ii) variation (inter and intra class) among the samples. While searching for a dream house, the customer generally has the follwing sequence of requirements in his/her mind; i) the number of rooms inside the house, ii) the global shape of the house, iii) what is the pattern in which the rooms are connected to each other, and iv) if a user is looking for a well furnished home. then which furniture should be placed in the house and in which arrangement. In the following subsections, each of the points mentioned above are elaborated in detail.

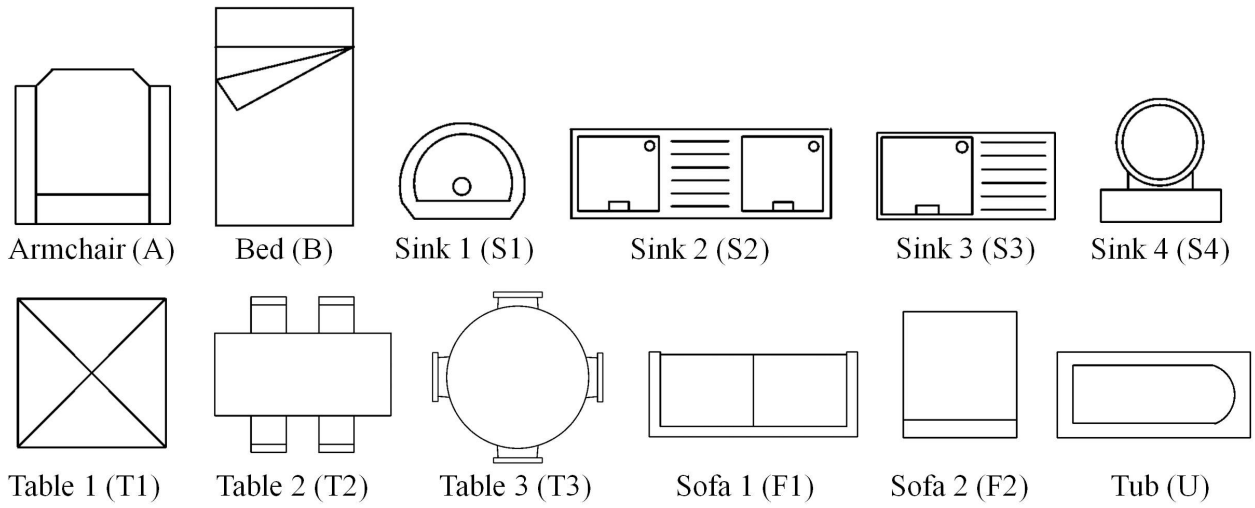


Figure A.1. : Symbol set used as furniture in ROBIN and S-ROBIN dataset.

A.1 NUMBER OF ROOMS

In ROBIN there are three categories of floor plans, depicting the three broad classes of houses. There are houses with 3, 4 and 5 rooms. The purpose and the subsequent label of each room varies depending on the various decors present in them. To be consistent with the existing floor plan datasets, the furniture symbols used in ROBIN dataset are similar to the ones used in SESYD dataset [Delalandre *et al.*, 2010a] as shown in Fig. A.1.

A.2 GLOBAL SHAPE OF THE HOUSE

The second important criteria kept in mind while designing the dataset was the global or overall shape of the layout. To design the floor plans the rooms were arranged either in the horizontal or the vertical direction with respect to each other. While designing, due consideration was given to the fact that, there is no arbitrary shaped floor plan image and also that curvilinear structures in the layout walls are avoided. Moreover, to increase the variability among the samples, difference in arrangement of rooms across 3, 4 and 5 room layout categories was also introduced. Figure A.2 (a), (b) and (c) depict the example of variation in global shapes across a layout category.

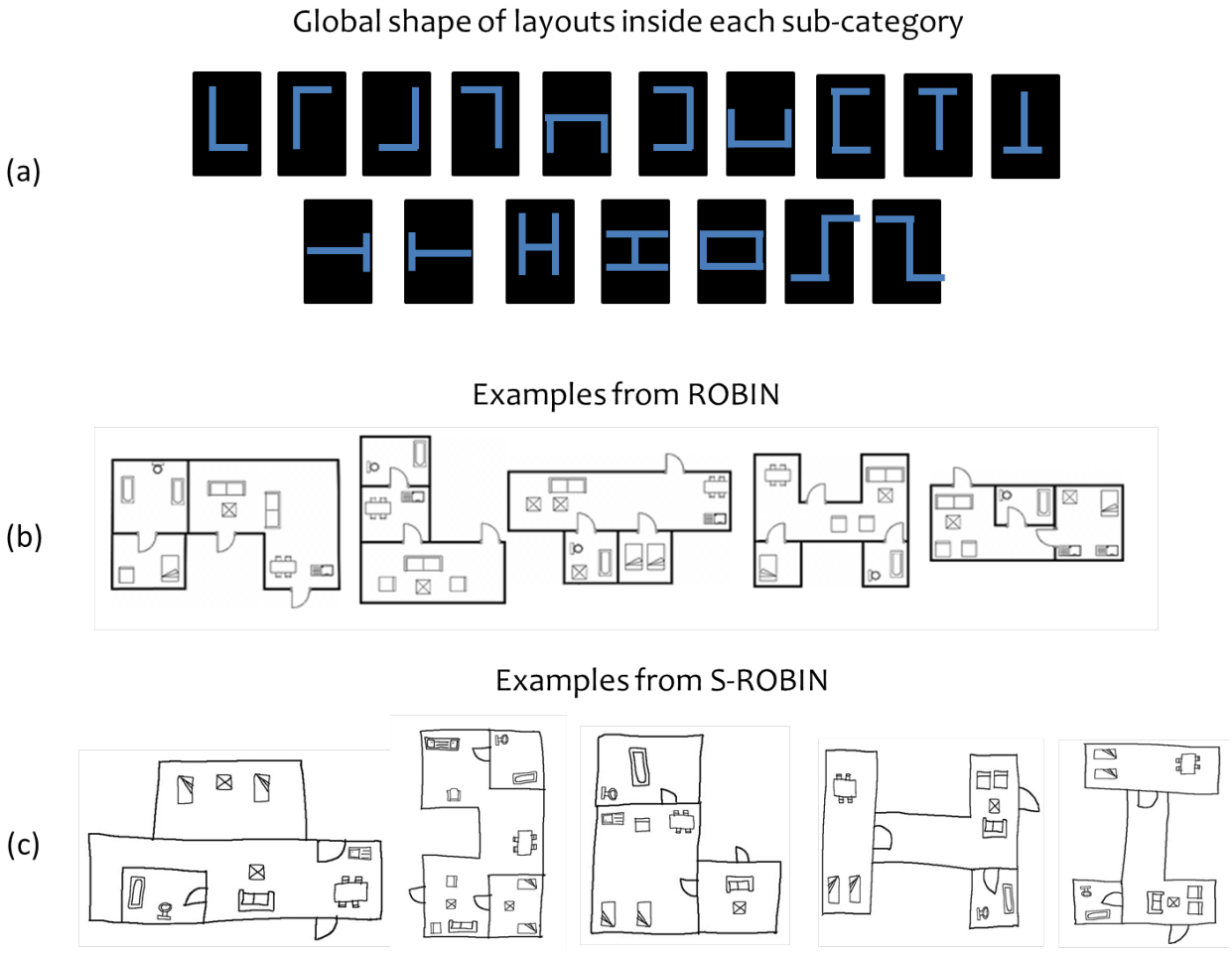


Figure A.2. : 3-Room layouts in the ROBIN and S-ROBIN dataset.

A.3 CONNECTIVITY WITHIN A HOUSE

Doors are placed at the desired places in the floor plans to ensure connectivity between the rooms. Orientation of each door is given special consideration to ensure proper direction of movement within the house. For every floor plan, there is atleast one door lying on the outer wall of the floor plan that opens inside the house and provides entry/exit for the house. Windows are appropriately placed on the walls of the floor plan to ensure proper ventilation and aesthetics. This makes the floor plan more realistic. Figure A.3 (b) can be observed for the arc-type door symbols satisfying entry-exit criteria in the floor plans.

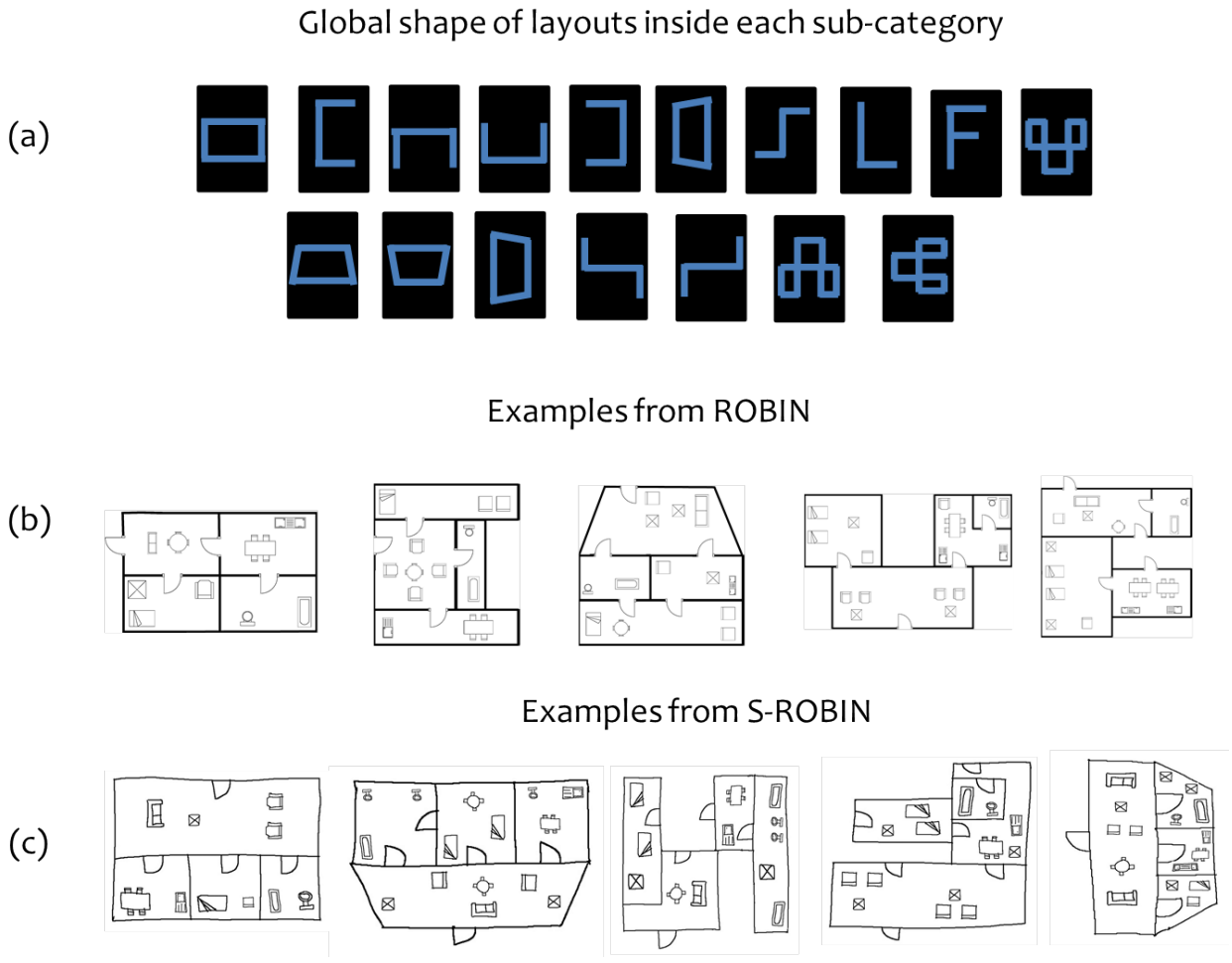


Figure A.3. : 4-Room layouts in the ROBIN and S-ROBIN dataset.

A.4 PLACEMENT OF FURNITURE

Furnitures or decors are inseparable part of a floor plan. To make the floor plans of ROBIN realistic, decor symbols as proposed by Delalandre *et al.* [2010a] were utilized. As shown in Fig. A.1, there are 12 different floor plan symbols. In each floor plan a subset of the entire symbol set was chosen and placed inside every room to give a definition to the room. As the furnitures are indicators of the purpose of the rooms, therefore, while designing, sufficient care has been taken to ensure that a house contains a i) kitchen, ii) bedroom, iv) washroom and a v) lobby. Moreover,

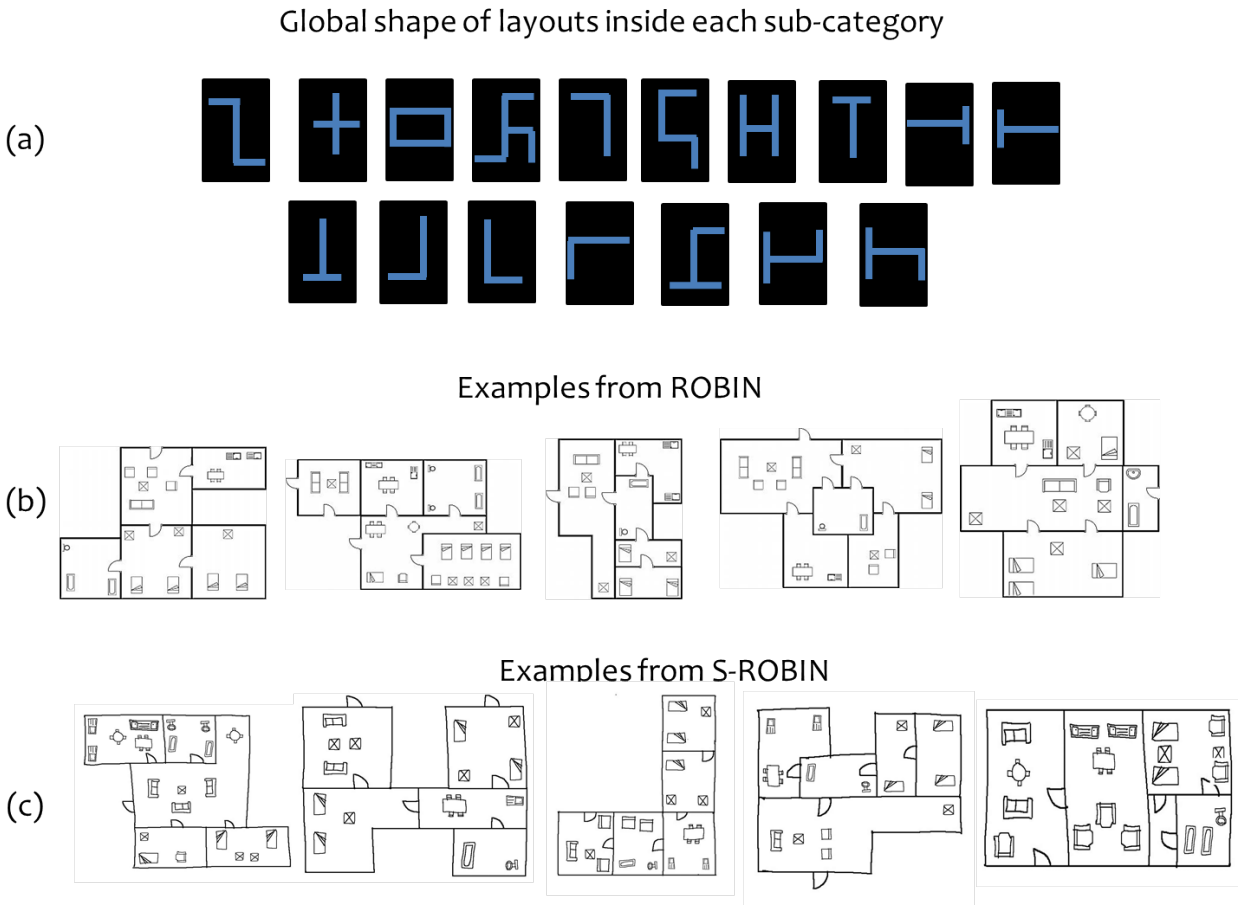


Figure A.4. : 5-Room layouts in the ROBIN and S-ROBIN dataset.

special consideration was given to the fact that the furniture is not placed in an arbitrary manner. For example, furniture should not be placed in a way that it blocks the door. Also, sufficient area should be left vacant after the furniture placement for navigation purposes.

A.5 DATASET CREATION PROCESS

ROBIN dataset was created on similar lines as the SESYD dataset [Delalandre *et al.*, 2010a]. Basic room structure of the floor plan was built using rectangular shape in Microsoft powerpoint. Thickness of each wall was kept uniform to be around 40 pixels. The decor symbols were inserted inside the rooms from the symbol library and door symbols were carefully placed on the room walls. On the other hand, to capture hand drawn sketched floor plans in S-ROBIN, volunteers were asked to draw the floor plans on a digital platform using a Wacom tablet. Wacom tablet is an interactive tool, where the user can draw on a board like tablet using a pen stylus conveniently and the resultant floor plan sketch is captured on the Microsoft paint software by adjusting the thickness of the strokes of the pen stylus. Since, offline mode is being used, the hand drawn floor plans are stored as an image and used for experimentation.

A.6 STATISTICS OF THE DATASET

Both the datasets contain 510 floor plans divided into 3 categories of layouts each with 17 sub categories. The unique characteristic of the datasets is that they are designed keeping in mind the need of a potential buyer. Every prospective buyer wants to have certain amenities and functionalities in their house. Thus, in the dataset, there are three broad categories, which are different from each other in terms of the number and type of rooms present in a floor plan. The broad categories are layouts with 3-rooms, 4-rooms and 5-rooms each which differ in the global shape of layout across their sub-categories. Each broad category is further classified into 17 sub-categories depending upon the global layout shape of the floor plan. Standard notations as mentioned by Delalandre *et al.* [2007], are followed for all the floor plans so that the datasets can be used for other floor plan analysis tasks as well, and not only for retrieval. The dataset has 12 different categories of furnitures inside each layout. These datasets help in better visualization of the floor plans and aid in efficient capturing of various high-level features while fine-grained retrieval. Illustrations of the various sub-categories in the floor plans and how their shapes are modeled are shown in Fig. A.2, Fig. A.3 and Fig. A.4. The size of the floor plans in ROBIN vary from 825×708 to 2916×1845 whereas the resolution in S-ROBIN dataset ranges from 591×517 pixels to 1483×884.

A.7 CONCLUSION

The rationale behind the creation of ROBIN and S-ROBIN dataset was to create a floor plan repository that can be used for research purposes by the Document Analysis and Retrieval (DAR) community. Both ROBIN and S-ROBIN dataset are sufficient for carrying out tasks like, retrieval, semantic segmentation, door to door navigation, etc. The entire dataset along with the annotations can be downloaded from [S-ROBIN, 2018].

Annexure B

GUI for floor plan analysis and retrieval

In this Chapter, a software interface for the task of content-based floor plan retrieval is discussed. The software works in the following manner:

1. The user chooses a sample floor plan image and uploads it to the server.
2. The retrieval engine extracts features from the query sample and matches with all the samples of the database.
3. The retrieved samples are rank ordered and shown to the user.

The software has a Graphical User Interface (GUI) with which the user can interact with the system and retrieve similar floor plans from the database. The software has two operational modes, which are (i) detailed mode, and (ii) direct mode. In the detailed mode, depending on the users' choice the intermediate steps of floor plan retrieval such as, segmentation, graph creation with nodes as segmented rooms and adjacencies as relations between the rooms, and finally retrieval based on the features extracted from the floor plans are shown in this GUI. On the other hand, in the direct mode the intermediate steps are not shown, and only the final result is displayed.

B.1 SYSTEM ARCHITECTURE

The web-based interface, which runs in a client server mode, has been developed to facilitate the framework for semantic retrieval of floor plans. The proposed system runs on Windows 7/XP and requires a few seconds to retrieve the rank-ordered results from the repository of floor plans. Figure B.1 shows the overall architecture for the framework. Developed under a client-server paradigm, the retrieval framework has two types of users to access the application, namely normal user and the administrator. A normal user is allowed to browse the system and perform queries, select whether he/she wants to get results of all the intermediate steps in the framework or only the final retrieval results, place a request to add a new floor plan to the repository, etc. An administrator on the other hand has all the access of the functionalities provided to a normal user, and in addition has the privilege to see the repository updation requests sent by the normal user and add/remove user accounts etc. The types of user access are controlled by a backend user database maintained using Microsoft Access. The retrieval portal is developed using HTML5, JavaScript, CSS. Apache webserver hosts the webpages, while Tomcat Java Server hosts Java servlets for receiving the query provided by the user.

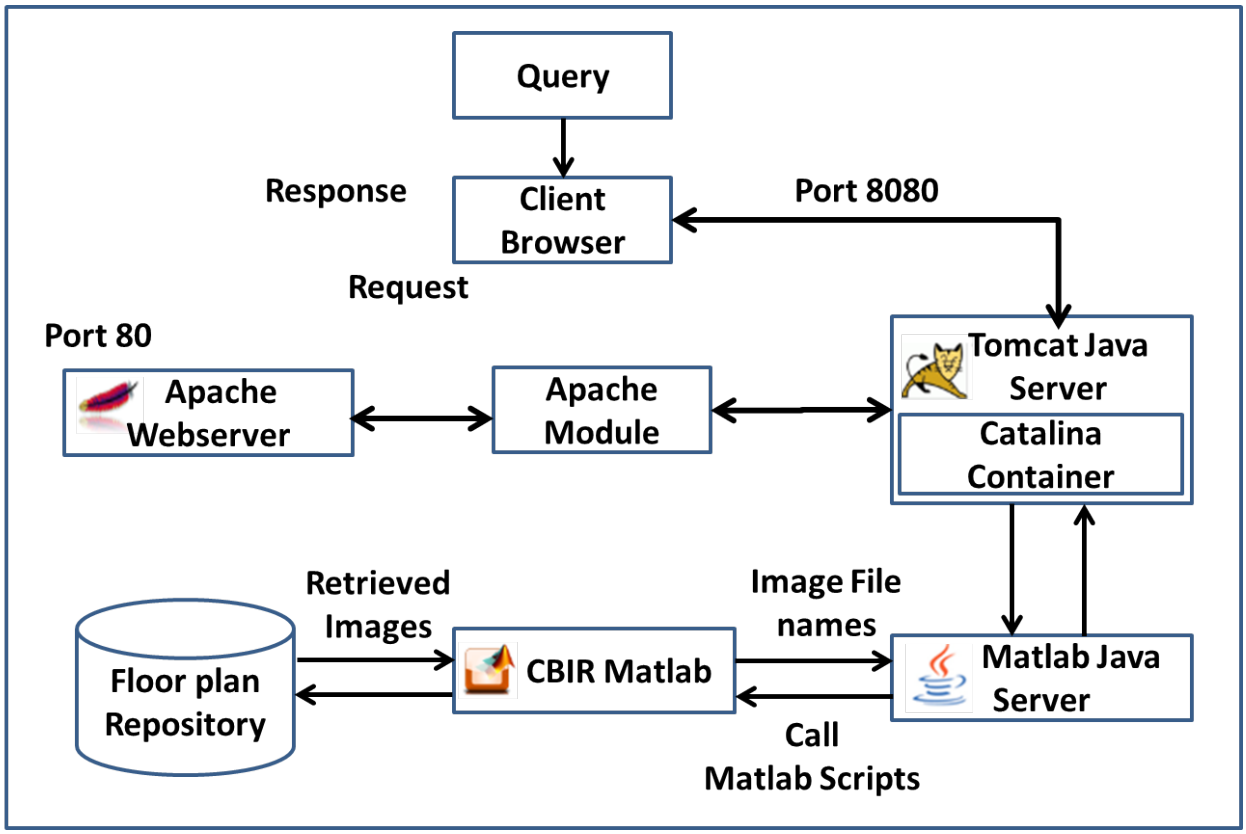


Figure B.1. : Architecture of the developed software for floor plan retrieval.

B.2 SOFTWARE FUNCTIONALITY

Figure B.2 depicts the sequence diagram of the developed software. There are five primary objects. They are: (i) user, (ii) host computer, (iii) Graphical User interface (GUI), (iv) Matlab workspace, and (v) the floor plan database. The user initiates the interaction between the software and the outside world by selecting a query floor plan image from the host computer. Through the GUI the query image is submitted to the Matlab workspace. All the necessary modules are developed in Matlab as .m files. The feature extraction, segmentation, matching and rank ordered list is done in the Matlab workspace. Based on the rank ordered list, floor plans are retrieved from the database and shown to the user. The software also has a functionality to allow the user to choose the number of floor plan samples to be displayed. Depending upon the users' choice (the value of k), the appropriate number of floor plans are shown to the user. In the next section, screen-shots of the key components of the software and their description is presented.

B.3 SCREEN-SHOTS OF THE FLOOR PLAN RETRIEVAL FRAMEWORK

Initially, when the user launches the application he/she comes across the basic screen prompting the user to select a floor plan from the repository to query into the retrieval framework as shown in Fig. B.3. As shown in Fig. B.3, there are three place holders to host the three different type of images. It can be observed that the "Search" button at the bottom of the interface is disabled at the beginning to ensure that the search is not made on null objects.

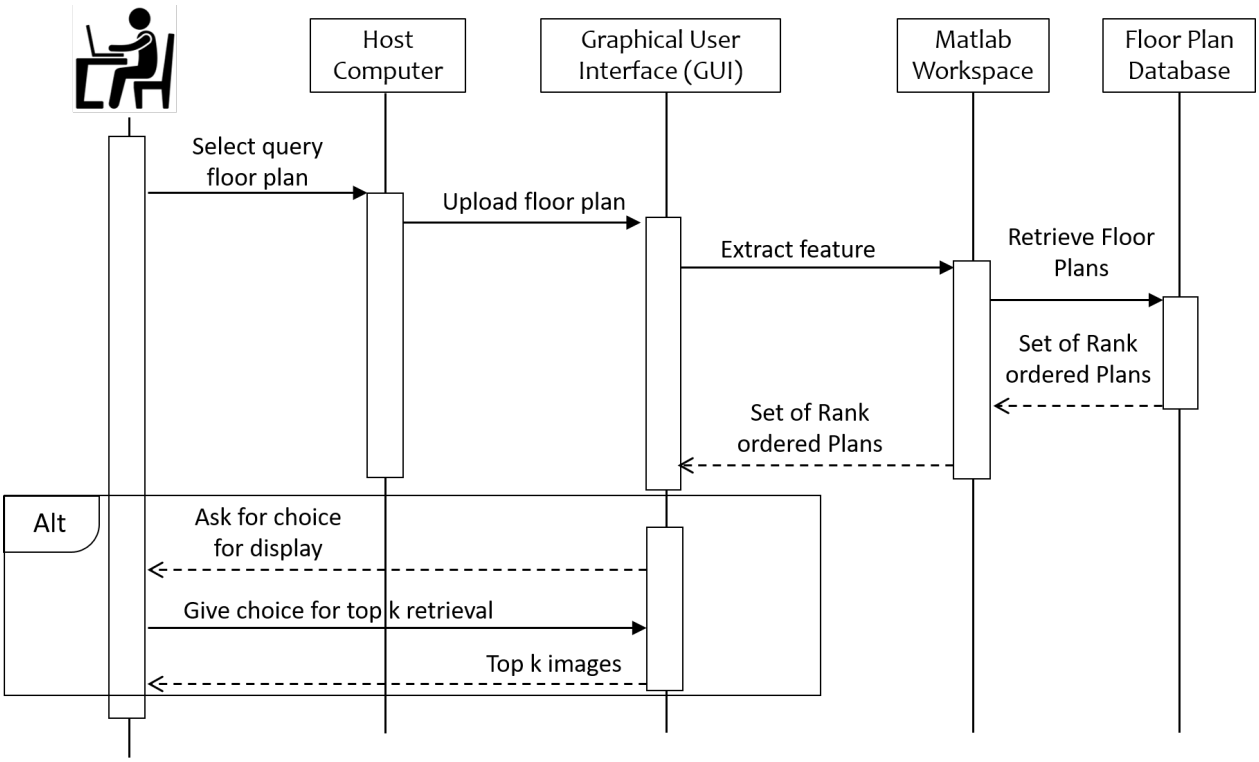


Figure B.2. : Sequence diagram depicting the overall flow of the software.

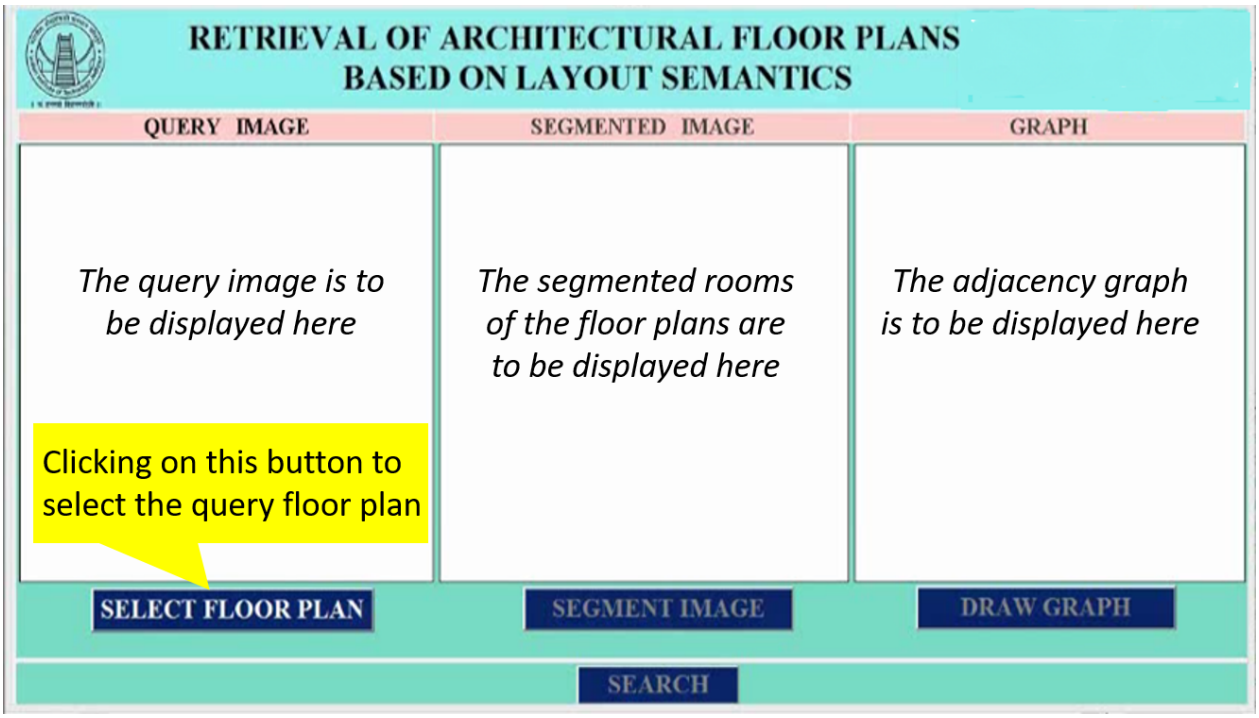


Figure B.3. : Screen prompting the user to select a floor plan of his/her choice.

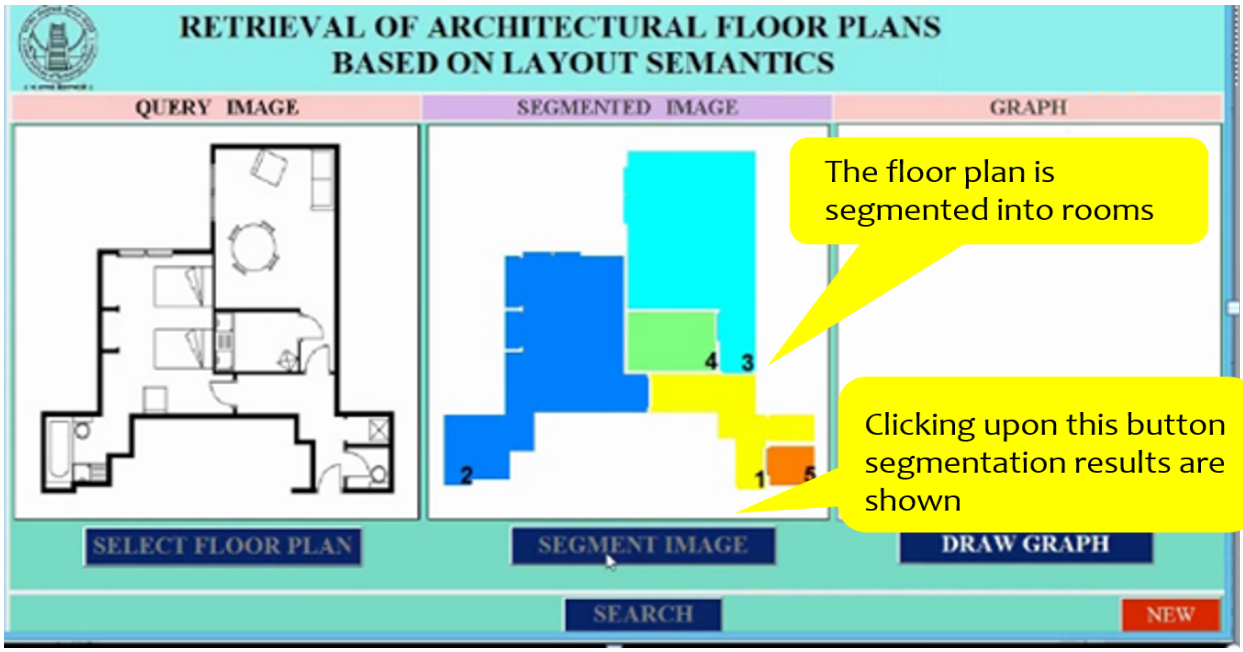


Figure B.4. : Segmentation of the selected plan into labeled rooms

Once the floor plan is chosen and shown in the appropriate place holder the user has the option to choose one of the two modes, i.e. direct or detailed mode. If the user chooses the detailed mode then the intermediate results will be shown in the appropriate place holders. Under the detailed mode, the next task is to segment the floor plan into meaningful entities like rooms and separating it from the decor elements present inside the rooms to further derive the relations between them. Figure B.4 depicts such a scenario, where the intermediate result is shown. Since

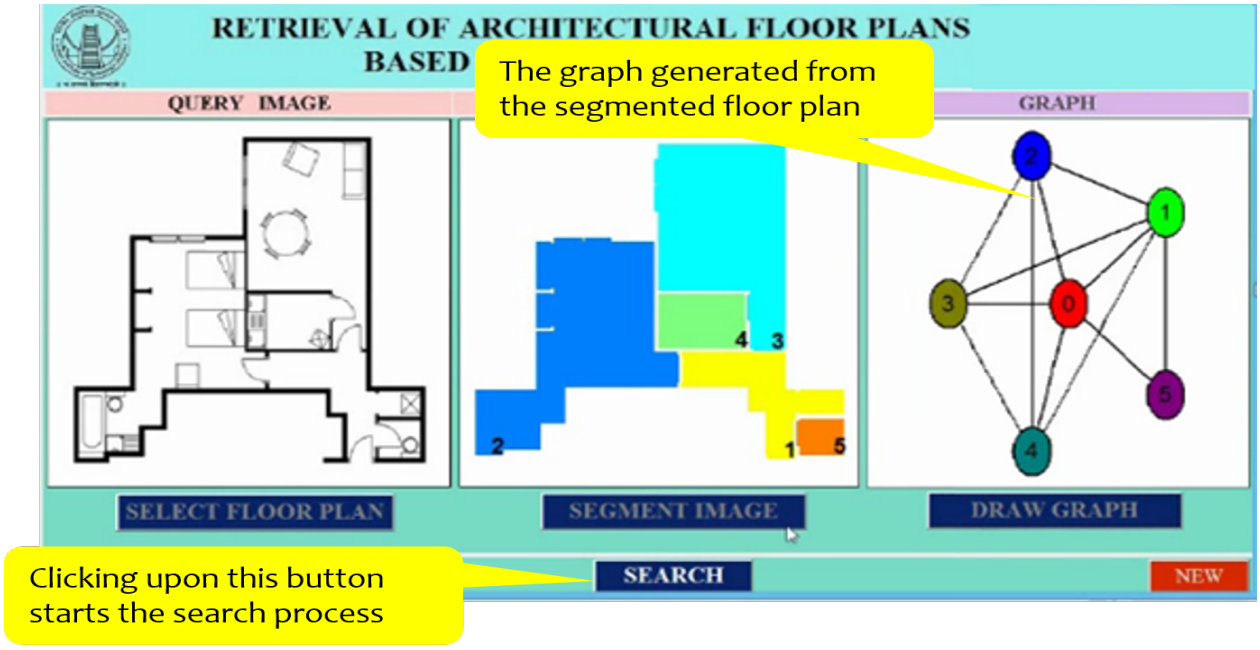


Figure B.5. : Graph embedding of the room level structure of the floor plan.

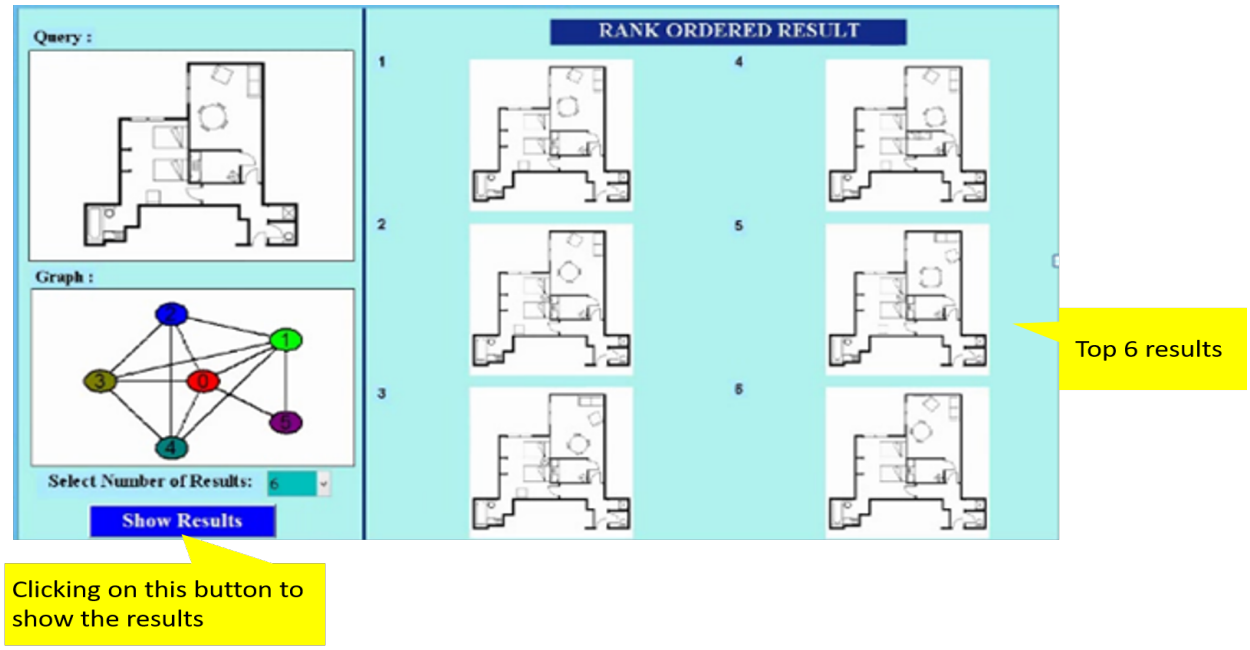


Figure B.6. : Selecting how many results to be displayed while retrieval.

the user has chosen detailed mode, the "Search" button is still disabled as one more step, i.e. the graph creation process is pending.

The next important user experience is shown in Fig. B.5. After the segmented floor plan is obtained with numbered labeling on each room, the next task is to obtain a relationship between these rooms. Therefore, adjacency is used as a parameter to define relationships between them. A graph is created where nodes represent rooms and edges represent which rooms are adjacent to each other. As shown in Fig. B.5, the "Search" button is now activated as the software is ready to do the matching between the query feature and the database features to show the retrieved result.

After the required graph is obtained, spectral analysis of the adjacency matrix is done in the back-end for the query floor plan as well as the database floor plans. The floor plans with similar adjacency are clustered together in the spectral space and thus, the GUI redirects to the second page for retrieval of similar floor plans as shown in Fig. B.6. As shown in Fig. B.6, there are three place holders in the interface. On the left hand side top the query floor plan is shown, while the corresponding graph is shown at the bottom. Just below the graph, there is a drop down list using which the user can choose the value of the number of floor plans to show as result. Upon the selection of number of results to be displayed, the rank ordered retrieval results are displayed as shown in Fig. B.7.

Figure B.7 depicts the final result for top 6 results. As seen in the results the retrieved results are numbered as per their position in the rank ordered list. After the results are displayed, the user can always change the number of results to be displayed. The user needs to choose an appropriate number from the drop-down list and then click the "'Show Results" button. The number of floor plans shown as results are updated accordingly.

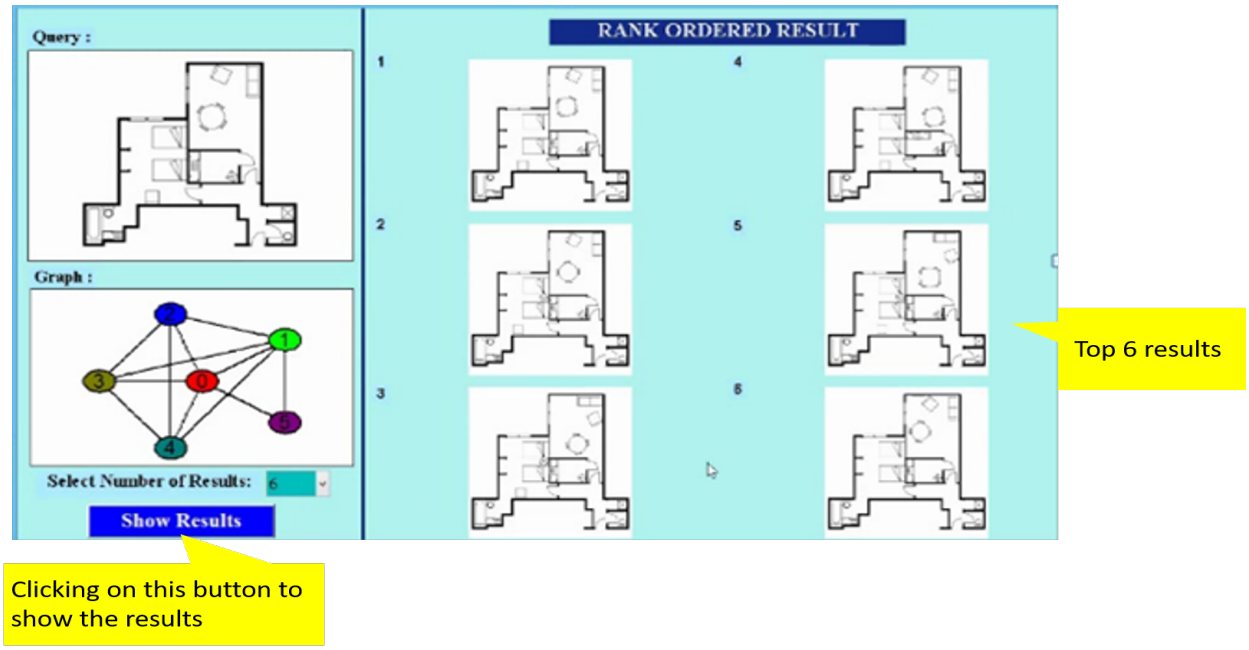


Figure B.7. : User's required floor plans on display as per their similarity from the query floor plan.

B.4 CONCLUSION

This is an initial version of a platform developed for content based floor plan retrieval. At present, the software runs on system with Matlab installed in it or a system that has a Matlab plugin installed. An extension could be to have a mobile version of the same that can be used easily without any specific software prerequisites.

References

Ah-Soon, C. and Tombre, K., "Variations on the analysis of architectural drawings", in "Document Analysis and Recognition, 1997., Proceedings of the Fourth International Conference on", vol. 1, pp. 347–351, IEEE 1997

Ah-Soon, C. and Tombre, K., "Architectural symbol recognition using a network of constraints", *Pattern Recognition Letters*, vol. 22, no. 2, 231–248 2001

Ahmed, S., Liwicki, M., Weber, M., and Dengel, A., "Automatic room detection and room labeling from architectural floor plans", in "Document Analysis Systems (DAS), 2012 10th IAPR International Workshop on", pp. 339–343, IEEE 2012

Ahmed, S., Weber, M., Liwicki, M., Langenhan, C., Dengel, A., and Petzold, F., "Automatic analysis and sketch-based retrieval of architectural floor plans", *Pattern Recognition Letters*, vol. 35, 91–100 2014

Aoki, Y., Shio, A., Arai, H., and Odaka, K., "A prototype system for interpreting hand-sketched floor plans", in "Pattern Recognition, 1996., Proceedings of the 13th International Conference on", vol. 3, pp. 747–751, IEEE 1996

Bannister, A., *Elements of Civil Engineering*, Pearson Education India 2010

Barducci, A. and Marinai, S., "Object recognition in floor plans by graphs of white connected components", in "Pattern Recognition (ICPR), 2012 21st International Conference on", pp. 298–301, IEEE 2012

Bay, H., Tuytelaars, T., and Van Gool, L., "Surf: Speeded up robust features", in "European conference on computer vision", pp. 404–417, Springer 2006

Bengio, Y. *et al.*, "Learning deep architectures for AI", *Foundations and trends® in Machine Learning*, vol. 2, no. 1, 1–127 2009

Braden, B., "The surveyor's area formula", *The College Mathematics Journal*, vol. 17, no. 4, 326–337 1986

Cadène, R., Thome, N., and Cord, M., "Master's Thesis: Deep Learning for Visual Recognition", *arXiv preprint arXiv:1610.05567* 2016

Chechik, G., Shalit, U., Sharma, V., and Bengio, S., "An online algorithm for large scale image similarity learning", in "NIPS", 2009

Chechik, G., Sharma, V., Shalit, U., and Bengio, S., "Large scale online learning of image similarity through ranking", *Journal of Machine Learning Research*, vol. 11, no. Mar, 1109–1135 2010

Chen, W. and Hays, J., "SketchyGAN: Towards Diverse and Realistic Sketch to Image Synthesis", in "Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition", pp. 9416–9425 2018

Cherneff, J., Logcher, R., Connor, J., and Patrikalakis, N., "Knowledge-based interpretation of architectural drawings", *Research in Engineering Design*, vol. 3, no. 4, 195–210 1992

Creswell, A. and Bharath, A. A., "Task specific adversarial cost function", *arXiv preprint arXiv:1609.08661* 2016

da Fonseca, M. J. C. M., *Sketch-based retrieval in large sets of drawings*, Ph.D. thesis, Universidade Técnica de Lisboa 2004

Dalal, N. and Triggs, B., "Histograms of Oriented Gradients for Human Detection", in "CVPR", 2005

de las Heras, L.-P., Fernández, D., Fornés, A., Valveny, E., Sánchez, G., and Lladós, J., "Runlength Histogram Image Signature for Perceptual Retrieval of Architectural Floor Plans", in "International Workshop on Graphics Recognition", pp. 135–146, Springer 2013

de las Heras, L.-P., Ahmed, S., Liwicki, M., Valveny, E., and Sánchez, G., "Statistical segmentation and structural recognition for floor plan interpretation", *IJDAR*, vol. 17, no. 3, 221–237 2014

de las Heras, L.-P., Terrades, O. R., Robles, S., and Sánchez, G., "CVC-FP and SGT: A new database for structural floor plan analysis and its groundtruthing tool", *IJDAR*, vol. 18, no. 1, 15–30 2015

Delalandre, M., Pridmore, T., Valveny, E., Locteau, H., and Trupin, E., "Building synthetic graphical documents for performance evaluation", in "International Workshop on Graphics Recognition", pp. 288–298, Springer 2007

Delalandre, M., Valveny, E., Pridmore, T., and Karatzas, D., "Generation of synthetic documents for performance evaluation of symbol recognition & spotting systems", *IJDAR*, vol. 13, no. 3, 187–207 2010a

Delalandre, M., Valveny, E., Pridmore, T., and Karatzas, D., "Generation of synthetic documents for performance evaluation of symbol recognition & spotting systems", *IJDAR*, vol. 13, no. 3, 187–207 2010b

Doermann, D., Tombre, K. *et al.*, *Handbook of Document Image Processing and Recognition*, Springer 2014

Dosch, P. and Masini, G., "Reconstruction of the 3d structure of a building from the 2d drawings of its floors", in "Document Analysis and Recognition, 1999. ICDAR'99. Proceedings of the Fifth International Conference on", pp. 487–490, IEEE 1999

Dosch, P., Tombre, K., Ah-Soon, C., and Masini, G., "A complete system for the analysis of architectural drawings", *International Journal on Document Analysis and Recognition*, vol. 3, no. 2, 102–116 2000

Dutta, A., Llados, J., and Pal, U., "Symbol spotting in line drawings through graph paths hashing", in "Document Analysis and Recognition (ICDAR), 2011 International Conference on", pp. 982–986, IEEE 2011

Dutta, A., Gibert, J., Lladós, J., Bunke, H., and Pal, U., "Combination of product graph and random walk kernel for symbol spotting in graphical documents", in "Pattern Recognition (ICPR), 2012 21st International Conference on", pp. 1663–1666, IEEE 2012

Dutta, A., Lladós, J., and Pal, U., "A symbol spotting approach in graphical documents by hashing serialized graphs", *Pattern Recognition*, vol. 46, no. 3, 752–768 2013

El Rube, I., Ahmed, M., and Kamel, M., "Wavelet approximation-based affine invariant shape representation functions", *IEEE transactions on pattern analysis and machine intelligence*, vol. 28, no. 2, 323–327 2006

Escalera, S., Fornés, A., Pujol, O., Escudero, A., and Radeva, P., "Circular blurred shape model for symbol spotting in documents", in "Image Processing (ICIP), 2009 16th IEEE International Conference on", pp. 2005–2008, IEEE 2009

Ghifary, M., Kleijn, W. B., Zhang, M., Balduzzi, D., and Li, W., "Deep reconstruction-classification networks for unsupervised domain adaptation", in "European Conference on Computer Vision", pp. 597–613, Springer 2016

Harris, C. and Stephens, M., "A combined corner and edge detector.", in "Alvey vision conference", vol. 15, pp. 10–5244, Citeseer 1988

Hilaire, X. and Tombre, K., "Robust and accurate vectorization of line drawings", *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 28, no. 6, 890–904 2006

Hilbert, M. and López, P., "The world's technological capacity to store, communicate, and compute information", *science*, vol. 332, no. 6025, 60–65 2011

Hörster, E. and Lienhart, R., "Deep networks for image retrieval on large-scale databases", in "International Conference on Multimedia", pp. 643–646, ACM 2008

Ioffe, S. and Szegedy, C., "Batch normalization: Accelerating deep network training by reducing internal covariate shift", *arXiv preprint arXiv:1502.03167* 2015

Jia, Y., Shelhamer, E., Donahue, J., Karayev, S., Long, J., Girshick, R., Guadarrama, S., and Darrell, T., "Caffe: Convolutional Architecture for Fast Feature Embedding", *arXiv preprint*

arXiv:1408.5093 2014

Kasturi, R., O'Gorman, L., and Govindaraju, V., "Document image analysis: A primer", *Sadhana*, vol. 27, no. 1, 3–22 Feb 2002

Kong, X., Valveny, E., Sánchez, G., and Wenyin, L., "Symbol spotting using deformed blurred shape modeling with component indexing and voting scheme", in "Workshop on Graphic Recognition (GREC)", 2011

Krizhevsky, A. and Hinton, G. E., "Using very deep autoencoders for content-based image retrieval.", in "ESANN", 2011

Krizhevsky, A., Sutskever, I., and Hinton, G. E., "Imagenet classification with deep convolutional neural networks", in "ANIPS", 2012a

Krizhevsky, A., Sutskever, I., and Hinton, G. E., "Imagenet classification with deep convolutional neural networks", in "Advances in neural information processing systems", pp. 1097–1105 2012b

Lazebnik, S., Schmid, C., and Ponce, J., "Beyond Bags of Features: Spatial Pyramid Matching for Recognizing Natural Scene Categories", in "CVPR", 2006a

Lazebnik, S., Schmid, C., and Ponce, J., "Beyond bags of features: Spatial pyramid matching for recognizing natural scene categories", in "CVPR", 2006b

Le Bodic, P., Héroux, P., Adam, S., and Lecourtier, Y., "An integer linear program for substitution-tolerant subgraph isomorphism and its use for symbol spotting in technical drawings", *Pattern Recognition*, vol. 45, no. 12, 4214–4224 2012

LeCun, Y. *et al.*, "LeNet-5, convolutional neural networks", *URL: http://yann. lecun. com/exdb/lenet*, p. 20 2015

Levenshtein, V. I., "Binary codes capable of correcting deletions, insertions and reversals", in "Soviet physics doklady", vol. 10, p. 707 1966

Llados, J. and Sanchez, G., "Graph matching versus graph parsing in graphics recognition—A combined approach", *International Journal of Pattern Recognition and Artificial Intelligence*, vol. 18, no. 03, 455–473 2004

Lladós, J., López-Krahe, J., and Martí, E., "A system to understand hand-drawn floor plans using subgraph isomorphism and Hough transform", *Machine Vision and Applications*, vol. 10, no. 3, 150–158 1997

Locteau, H., Adam, S., Trupin, E., Labiche, J., and Héroux, P., "Symbol spotting using full visibility graph representation", in "Workshop on Graphics Recognition", pp. 49–50 2007

Lowe, D. G., "Distinctive image features from scale-invariant keypoints", *International journal of computer vision*, vol. 60, no. 2, 91–110 2004

Lu, T., Yang, H., Yang, R., and Cai, S., "Automatic analysis and integration of architectural drawings", *International Journal of Document Analysis and Recognition (IJDAR)*, vol. 9, no. 1, 31–47 2007

Luo, B., Wilson, R. C., and Hancock, E. R., "Spectral embedding of graphs", *Pattern Recognition*, vol. 36, no. 10, 2213–2230 2003

Luqman, M. M., Brouard, T., Ramel, J.-Y., and Llodos, J., "A content spotting system for line drawing graphic document images", in "Pattern Recognition (ICPR), 2010 20th International Conference on", pp. 3420–3423, IEEE 2010

Macé, S., Locteau, H., Valveny, E., and Tabbone, S., "A system to detect rooms in architectural floor plan images", in "Proceedings of the 9th IAPR International Workshop on Document Analysis Systems", pp. 167–174, ACM 2010

Murase, H. and Nayar, S. K., "Illumination planning for object recognition using parametric eigenspaces", *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 16, no. 12, 1219–1227 1994

Nayef, N. and Breuel, T. M., "Statistical grouping for segmenting symbols parts from line drawings, with application to symbol spotting", in "Document Analysis and Recognition (ICDAR), 2011 International Conference on", pp. 364–368, IEEE 2011

Nguyen, T.-O., Tabbone, S., and Boucher, A., "A symbol spotting approach based on the vector

model and a visual vocabulary", in "Document Analysis and Recognition, 2009. ICDAR'09. 10th International Conference on", pp. 708–712, IEEE 2009

Or, S.-h., Wong, K.-H., Yu, Y.-k., Chang, M. M.-y., and Kong, H., "Highly automatic approach to architectural floorplan image understanding & model generation", *Pattern Recognition*, pp. 25–32 2005

Qureshi, R., Ramel, J., and Cardot, H., "Graphic symbol recognition using flexible matching of attributed relational graphs", in "proceeding of 6th IASTED International Conference on VIIP", pp. 383–388 2006

Qureshi, R. J., Ramel, J.-Y., Barret, D., and Cardot, H., "Spotting symbols in line drawing images using graph representations", in "International Workshop on Graphics Recognition", pp. 91–103, Springer 2007

Razavian, A. S., Azizpour, H., Sullivan, J., and Carlsson, S., "CNN features off-the-shelf: an astounding baseline for recognition", in "Computer Vision and Pattern Recognition Workshops", pp. 512–519, IEEE 2014

Razavian, A. S., Sullivan, J., Carlsson, S., and Maki, A., "Visual instance retrieval with deep convolutional networks", *ITE Transactions on Media Technology and Applications*, vol. 4, no. 3, 251–258 2016

Rendek, J., Masini, G., Dosch, P., and Tombre, K., "The search for genericity in graphics recognition applications: Design issues of the qgar software system", in "International Workshop on Document Analysis Systems", pp. 366–377, Springer 2004

Rusinol, M. and Lladós, J., "A region-based hashing approach for symbol spotting in technical documents", in "International Workshop on Graphics Recognition", pp. 104–113, Springer 2007

Rusiñol, M., Borràs, A., and Lladós, J., "Relational indexing of vectorial primitives for symbol spotting in line-drawing images", *Pattern Recognition Letters*, vol. 31, no. 3, 188–201 2010

Russakovsky, O., Deng, J., Su, H., Krause, J., Satheesh, S., Ma, S., Huang, Z., Karpathy, A., Khosla, A., Bernstein, M., Berg, A. C., and Fei-Fei, L., "ImageNet Large Scale Visual Recognition Challenge", *Int. J. Comput. Vision*, vol. 115, no. 3, 211–252 Dec 2015, URL http://dx.doi.org/10.1007/s11263-015-0816-y

S-ROBIN 2018

Sangkloy, P., Burnell, N., Ham, C., and Hays, J., "The sketchy database: learning to retrieve badly drawn bunnies", *ACM Transactions on Graphics (TOG)*, vol. 35, no. 4, 119 2016

Sharma, D., Chattopadhyay, C., and Harit, G., "A Unified Framework for Semantic Matching of Architectural Floorplans", in "ICPR", 2016a

Sharma, D., Chattopadhyay, C., and Harit, G., "Retrieval of Architectural Floor plans based on Layout Semantics", in "WiCV (CVPR)", 2016b

Sharma, D., Gupta, N., Chattopadhyay, C., and Mehta, S., "DANIEL: A deep architecture for automatic analysis and retrieval of building floor plans", in "Document Analysis and Recognition (ICDAR), 2017 14th IAPR International Conference on", vol. 1, pp. 420–425, IEEE 2017

Simonyan, K. and Zisserman, A., "Very deep convolutional networks for large-scale image recognition", *arXiv preprint arXiv:1409.1556* 2014

Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., Rabinovich, A. *et al.*, "Going deeper with convolutions", CVPR 2015

Tabbone, S., Wendling, L., and Tombre, K., "Matching of graphical symbols in line-drawing images using angular signature information", *Document Analysis and Recognition*, vol. 6, no. 2, 115–125 2003

Wan, J., Wang, D., Hoi, S. C. H., Wu, P., Zhu, J., Zhang, Y., and Li, J., "Deep learning for content-based image retrieval: A comprehensive study", in "International conference on Multimedia", pp. 157–166, ACM 2014

Weber, M., Liwicki, M., and Dengel, A., "A. Scatch-a sketch-based retrieval for architectural floor plans", in "ICFHR", 2010

Wessel, R., Blümel, I., and Klein, R., "The room connectivity graph: Shape retrieval in the

architectural domain", 2008

Yamasaki, T., Zhang, J., and Takada, Y., "Apartment Structure Estimation Using Fully Convolutional Networks and Graph Model", in "Proceedings of the 2018 ACM Workshop on Multimedia for Real Estate Tech", pp. 1–6, ACM 2018

Yang, S., "Symbol recognition via statistical integration of pixel-level constraint histograms: A new descriptor", *IEEE transactions on pattern analysis and machine intelligence*, vol. 27, no. 2, 278–281 2005

Yin, H., Gao, Y., Li, B., Zhang, D., Yang, M., Li, Y., Klawonn, F., and Tallón-Ballesteros, A. J., *Intelligent Data Engineering and Automated Learning–IDEAL 2016: 17th International Conference, Yangzhou, China, October 12–14, 2016, Proceedings*, vol. 9937, Springer 2016

Zhang, D. and Lu, G., "Review of shape representation and description techniques", *Pattern recognition*, vol. 37, no. 1, 1–19 2004

Zheng, L., Wang, S., Tian, L., He, F., Liu, Z., and Tian, Q., "Query-adaptive late fusion for image search and person re-identification", in "International conference on Conference on computer vision and pattern recognition", pp. 1741–1750 2015

Zhi, G., Lo, S., and Fang, Z., "A graph-based algorithm for extracting units and loops from architectural floor plans for a building evacuation model", *Computer-Aided Design*, vol. 35, no. 1, 1–14 2003

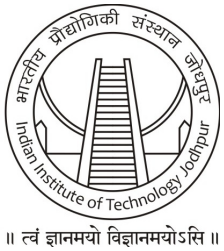
Zhu, J.-Y., Park, T., Isola, P., and Efros, A. A., "Unpaired image-to-image translation using cycle-consistent adversarial networks", *arXiv preprint arXiv:1703.10593* 2017

Zhuang, F., Cheng, X., Luo, P., Pan, S. J., and He, Q., "Supervised Representation Learning: Transfer Learning with Deep Autoencoders.", in "IJCAI", pp. 4119–4125 2015

Content Based Analysis and Retrieval of Architectural Floor plans

A Thesis submitted by
Divya Sharma

in partial fulfillment of the requirements for the award of the degree of
Doctor of Philosophy



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7
Conclusions

Efficient retrieval of similar floor plan images is a critical component of early design phase of construction as it is the primary reason of delay in the design/ conceptualization phase. Traditional CBIR approach is a straight forward and intuitive solution to this problem. The individual techniques proposed in literature, provide an incomplete answer to the question of representing a floor plan in a particular way to facilitate efficient search. Most of the proposed approaches in the literature are tailored for natural images, and some of them for text based document images. However, none of the known results are sufficient to tackle the problem of a unified framework that incorporates multiple modalities of query (sketch and image) for floor plan retrieval. In this chapter the contributions presented during each approach are summarized and the conclusions in the form of strengths and weaknesses of each technique are discussed. Finally, several working lines to be considered in the future are discussed.

In this thesis, it is shown that both the structural and semantic features are important for retrieval of similar floor plan images. It is also argued that hand crafted features are performing better that the learned features (using convolutional neural networks) in case of fine grained retrieval. However, for global similarity (i.e. similarity based on the overall shape of the floor plan), deep features are superior. This thesis also highlights that when sketch is used as a query mode, the traditional techniques (including deep features) fail to scale to this domain shift. Additional processing components need to be incorporated in the framework to tackle such a scenario.

The proposed system can also find application in scenarios where buyers are looking for specific features in house designs, while presenting their requirements to the architect. It can also be deployed in online real estate platforms, where the user can be directed to conveniently query their required floor plans using query as image or sketch and retrieve their required homes from the repository. The novelty of the contribution is that floor plans have been in the past used as graphical documents for analysis and symbol spotting purposes but retrieval based on significant content derived from a floor plan has not been researched. Extensive experimental evaluation on real-world datasets shows that combination of structural and semantic approaches for analysis of floor plans and also employing a combination of traditional and deep learning approaches for feature extraction add value to the overall retrieval accuracy. Both qualitative and quantitative comparisons were made with recent techniques for feature extraction to exhibit the superior performances of the proposed frameworks.

7.1 CONTRIBUTIONS

The key contributions of this thesis are:

- In Chapter 3, a novel approach to perform floor plan retrieval incorporating both symbol spotting and room level analysis is proposed by taking query by example as an image. Inclusion of the room decor arrangement during retrieval is a novel idea that can find application

in scenarios where the user has specific choices in terms of the fully furnished floor plans. Furthermore, a spectral embedding approach to represent graphs obtained from the layout is proposed in the baseline retrieval framework proposed in Chapter 3. This technique helped in reducing the computation time taken during the graph matching stage to a considerable level.

- In Chapter 4, a deep learning framework is proposed to extract semantic features from floor plans and those features are used for matching and retrieval of similar floor plan images. Novelty lies in using convolutional neural networks for the task of feature extraction in floor plans which prove to increase the retrieval performance as compared to hand-crafted feature extraction strategies.

- In Chapter 5, a novel framework for fine grained retrieval of floor plans using features like area, number, type of furniture and also adjacencies between the rooms is proposed. Inclusion of the high level features during retrieval is a novel idea that can find application in scenarios where the user has specific choices in terms of the fully furnished floor plans. Also, weighted feature fusion is proposed which helps in weighing features when a user is querying and the user is fixated over certain features to have a mandatory presence while retrieval. This framework has an edge over the deep learning framework in a respect that priority to certain features can be given while retrieval which in not otherwise possible in a deep learning network where globally the features are extracted.

- In Chapter 6, the introduction of sketch as a query mode while retrieval is proposed. Sketch based query mode is very intuitive because a user can conveniently draw his/her requirements on an interface as if they are drawing with pen and paper. Sketch based querying helps in capturing the content better. Two deep learning approaches, one with the usage of Cyclic GANs and the other with the usage of autoencoders are applied for domain adaptation purposes aiding to sketch to image retrieval. The performance of both the systems were obtained to be impressive while retrieval on benchmark datasets.

- Two datasets namely, ROBIN with floor plan images and S-ROBIN with sketched floor plans have also been proposed to benefit the research community to carry out research in the area of floor plan analysis and retrieval. These two datasets help in overcoming the drawbacks that the existing publicly available datasets had in terms of less variation in samples as well as inadequate number of samples to carry out deep learning and retrieval tasks.

7.2 DISCUSSION ON THE RELATIVE PERFORMANCE OF THE TECHNIQUES PROPOSED

In this section we discuss the comparative quantitative performance of the techniques mentioned in Chapters 3, 4, 5 and 6 of the thesis.

In Chapter 3, handcrafted topological features were proposed to be extracted from floor plan images. The Mean Average Precision (MAP) value obtained during retrieval on the ROBIN dataset was observed to be 0.247 in case only room layout was considered while retrieval, on the other hand incorporating both room layout and room decor features made retrieval quite specific. This was due to the fact that the layout similarity in a category of the ROBIN dataset was higher than the combined layout and decor similarity. Thus, the system gave a MAP value of 0.232 in this case. We observed that although, topology and room decor features help in distinguishing two floor plans and retrieving similar ones, but inclusively are not a good enough measure to analyse a floor plan completely. It is very difficult to curate handcrafted features powerful enough to extract

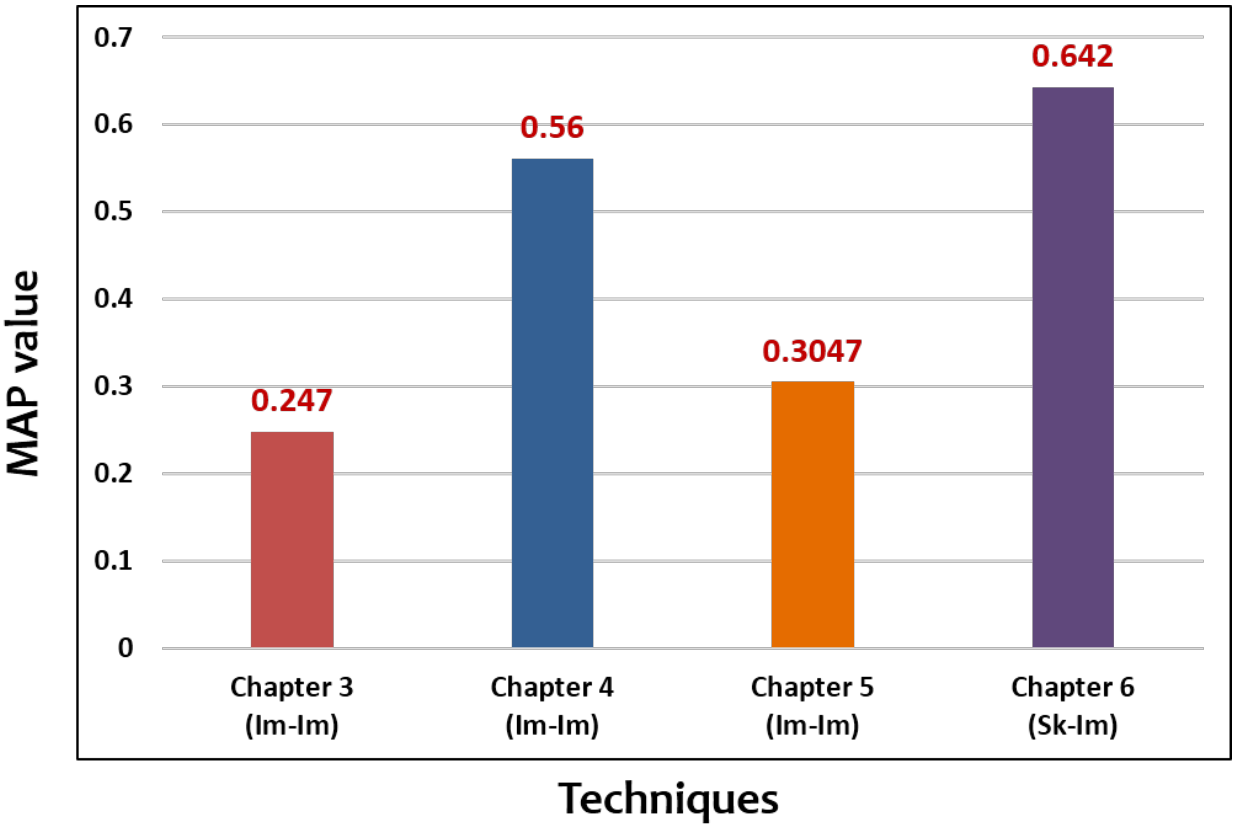

Figure 7.1. : Comparison of MAP values obtained by techniques mentioned in Chapters 3,4,5 and 6.

all the details from a floor plan. Therefore, we explored deep neural networks to extract structural and semantic information from a floor plan.

In Chapter 4, a deep learning framework using CNNs is proposed to extract semantic features from floor plan images and those features are used for matching and retrieval of similar floor plan images. As compared to Chapter 3, the deep learning framework, was able to enhance the retrieval performance considerably. We observed a MAP value of 0.56 on the ROBIN dataset. However, we observed that deep learning is convenient and bypasses the need of curating features manually, the disadvantage is that the layers in the deep learning framework extract features implicitly. This implicit feature extraction at each layer does not give the user the ability to give priorities to certain aspects or features while retrieval.

Therefore, in Chapter 5, high level semantic analysis of floor plans and their subsequent retrieval is proposed. Additionally, a provision to adjust the weights of each feature while retrieval is given in the proposed technique. Such a provision can prove to be more practical to users while looking for similar floor plans sufficing their specific preferences. We observed a MAP value of 0.3047 on the ROBIN dataset using this technique. Although, the MAP value obtained here is less than what was obtained using the deep learning framework for image retrieval, the advantages however are that, weighted feature fusion helps in emphasizing on certain features when a user is querying and has specific feature preferences while retrieval.

In Chapter 6, the querying mode changes and sketch based querying is proposed. Such a query makes the system more convenient for searching from the user's end. The conjunction of

CNN, Cyclic GAN and autoencoders was used in this particular framework for domain adaptation. The MAP value obtained in this case was impressive at a value of 0.642 for ROBIN dataset.

Figure 7.1 presents a bar graph comparing the MAP values obtained by each technique. Im-Im represents, image to image retrieval and Sk-Im represents, sketch to image retrieval as proposed in Chapter 6.

7.3 SCOPE OF FUTURE WORK

The work presented in this thesis, definitely opens the scope of a chain of tasks in many diverse applications, for researchers to solve in future. Some of them are given below.

- In the future, content based retrieval taking a hybrid scheme of querying can be researched. Here, the idea can be that a user can draw/scribble over a floor plan image and in real-time the query modifies according to the addition or removal of the components done by the user in the floor plan image. Such a system can be more interactive to use. Such multimodal inputs would make the process of query representation very convenient to the end-user, and would provide a rich set of specifications to further accurately retrieve similar floor plan samples.

- Future research should concentrate on capturing of better and usable content from floor plans to help in real world scenarios. Researchers may explore on deducing high-level features, which for example can be in the form of accessibility of each room, in terms of shortest path of a room to other rooms or to the exit in a floor plan. This scenario can find application in maintaining safety situations in big architectural buildings and retrieving floor plans abiding by the safety standards set by a regulatory body.

- The main issue with floor plan analysis research is the dearth of publicly available datasets as architects do not prefer to share their designs in the open. Therefore, a good future scope can be to develop a larger dedicated repository of real-world floor plans to foster research in the area of analysis and retrieval in floor plans.

- Acquiring the floor plan images may not always be in digital form. The alternate mode could be through high-resolution scanning. However, scanned documents at times are corrupted with transformations like shear, rotation, etc. Also, during digitization some noise might get added or existing noise in the document might get enhanced. Retrieval of floor plan images under such an uncontrolled situation forms a challenging research problem.

- In Chapter 6, Fig. 6.7, we already propose, a unified multimodal system which can classify query into sketch or image and accordingly perform image to image or sketch to image retrieval. Future research can also explore the combination of feature extraction techniques such as, hand-crafted techniques and deep learning techniques for retrieval. For example, in chapter 4, as we observe that deep learning fails to explicitly give preference to a feature while retrieval which was possible through the technique mentioned in chapter 5. Therefore, a suggested modification can be exploring an approach of incorporating feature importance into neural network learning. The weights connected to an important feature can be biased to have an absolute value larger than the weights connected to the other features.

Annexure A
Floor plan datasets ROBIN and S-ROBIN

Dataset plays a pivotal role in any image processing problem. Availability of a public dataset sometimes makes the life of a researcher easy. However, this is true for only generic problems, and not so for problems which are of some specific nature. Floor plan image analysis is such a problem where a large collection of samples is hardly available. For this reason two public datasets have already been proposed by researchers. They are: (i) SESYD [Delalandre *et al.*, 2010b] and (ii) CVC-FP [de las Heras *et al.*, 2015]. The former has 10 classes of floor plans, where there are 100 samples/class. The latter has 122 scanned floor plan documents divided into 4 categories based on the origin and style. The SESYD dataset is inappropriate for application to a retrieval task due to very less difference between a pair in a class of floor plan images. Also, in the CVC-FP dataset, the samples are insufficient in number for the task of floor plan retrieval.

To train and evaluate the contributions presented in this thesis, two floor plan databases, named Repository Of BuildIng floor plaNs (ROBIN) and Sketched-Repository Of BuildIng floor plaNs (S-ROBIN) are created. S-ROBIN are the sketch replicas of the ROBIN dataset containing floor plan images. These datasets have significantly aided in the content analysis and retrieval of similar floor plans as proposed in this thesis. The primary motivation while creating the ROBIN dataset was to have: i) sufficient number of sample to execute the experiments, and ii) variation (inter and intra class) among the samples. While searching for a dream house, the customer generally has the follwing sequence of requirements in his/her mind; i) the number of rooms inside the house, ii) the global shape of the house, iii) what is the pattern in which the rooms are connected to each other, and iv) if a user is looking for a well furnished home. then which furniture should be placed in the house and in which arrangement. In the following subsections, each of the points mentioned above are elaborated in detail.

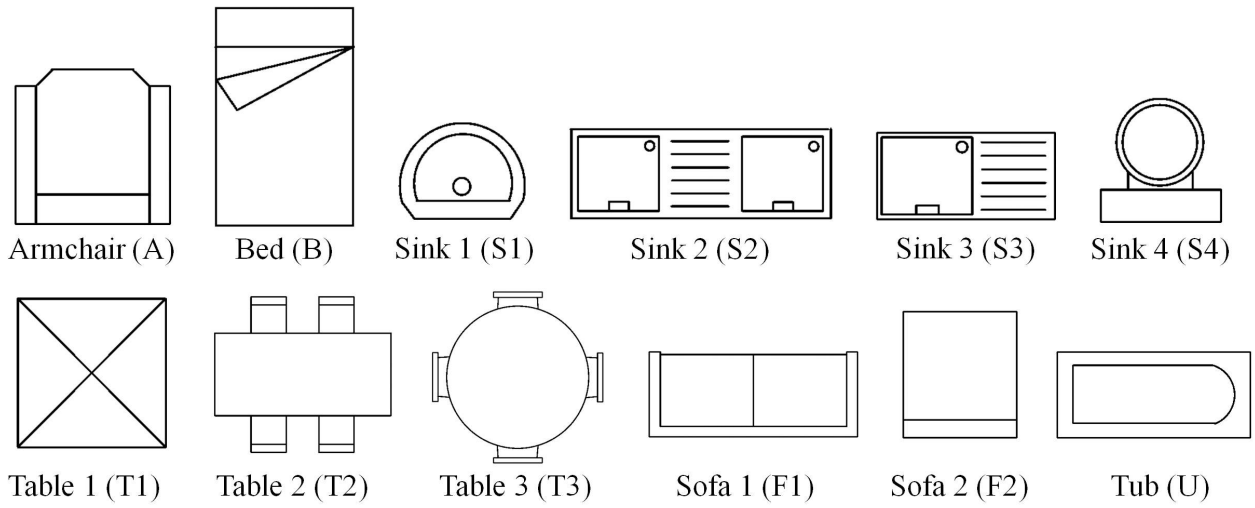

Figure A.1. : Symbol set used as furniture in ROBIN and S-ROBIN dataset.

A.1 NUMBER OF ROOMS

In ROBIN there are three categories of floor plans, depicting the three broad classes of houses. There are houses with 3, 4 and 5 rooms. The purpose and the subsequent label of each room varies depending on the various decors present in them. To be consistent with the existing floor plan datasets, the furniture symbols used in ROBIN dataset are similar to the ones used in SESYD dataset [Delalandre *et al.*, 2010a] as shown in Fig. A.1.

A.2 GLOBAL SHAPE OF THE HOUSE

The second important criteria kept in mind while designing the dataset was the global or overall shape of the layout. To design the floor plans the rooms were arranged either in the horizontal or the vertical direction with respect to each other. While designing, due consideration was given to the fact that, there is no arbitrary shaped floor plan image and also that curvilinear structures in the layout walls are avoided. Moreover, to increase the variability among the samples, difference in arrangement of rooms across 3, 4 and 5 room layout categories was also introduced. Figure A.2 (a), (b) and (c) depict the example of variation in global shapes across a layout category.

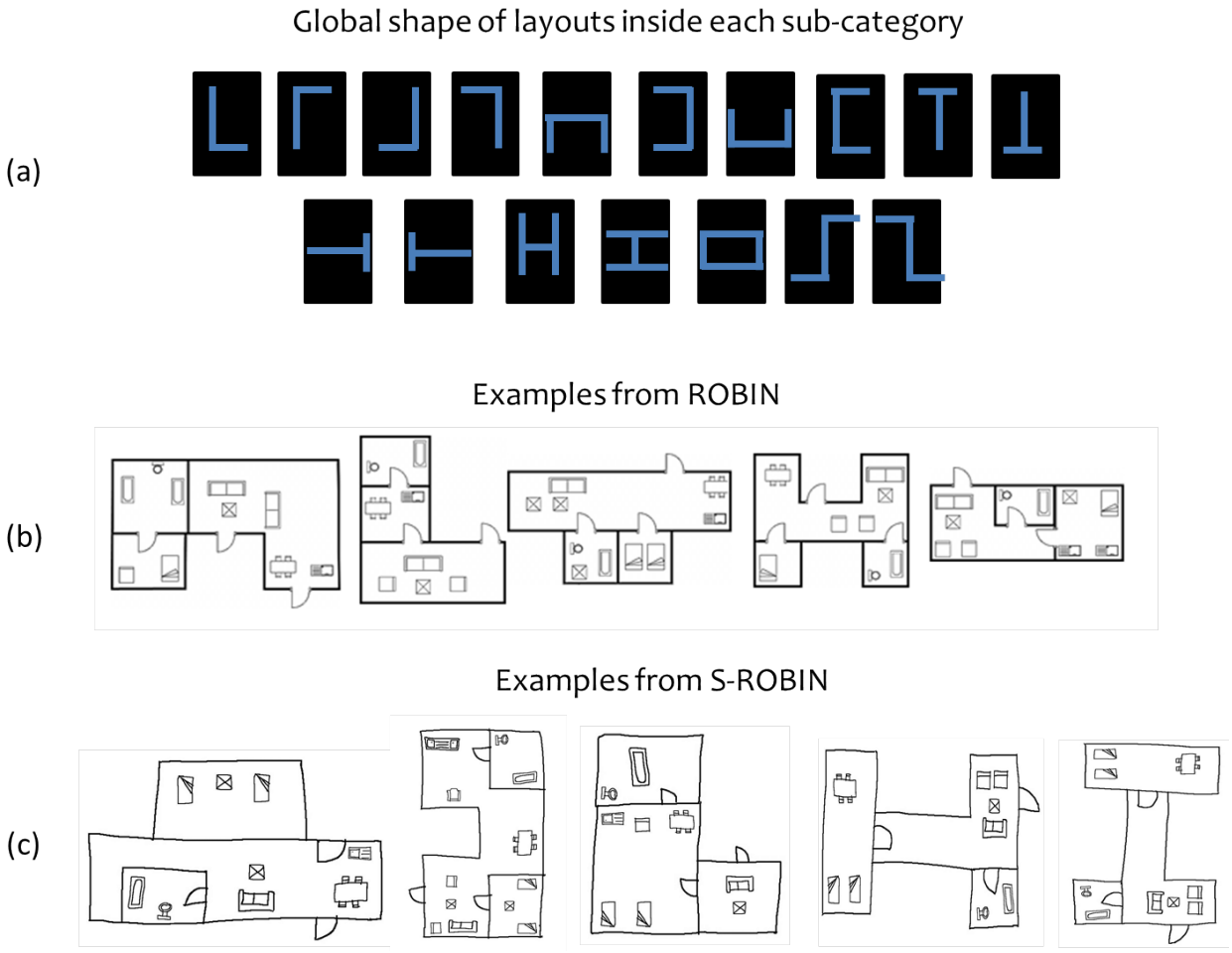

Figure A.2. : 3-Room layouts in the ROBIN and S-ROBIN dataset.

A.3 CONNECTIVITY WITHIN A HOUSE

Doors are placed at the desired places in the floor plans to ensure connectivity between the rooms. Orientation of each door is given special consideration to ensure proper direction of movement within the house. For every floor plan, there is atleast one door lying on the outer wall of the floor plan that opens inside the house and provides entry/exit for the house. Windows are appropriately placed on the walls of the floor plan to ensure proper ventilation and aesthetics. This makes the floor plan more realistic. Figure A.3 (b) can be observed for the arc-type door symbols satisfying entry-exit criteria in the floor plans.

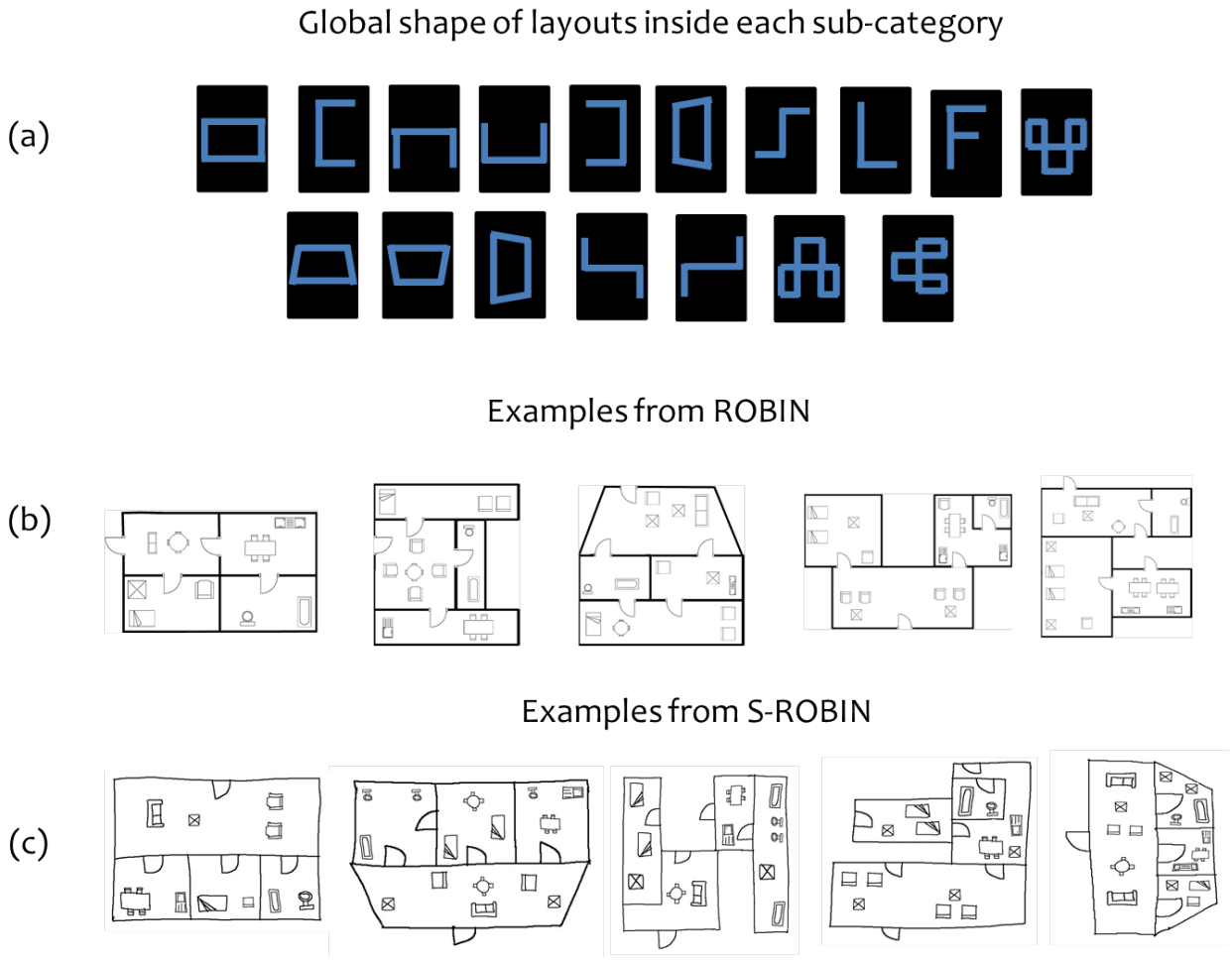

Figure A.3. : 4-Room layouts in the ROBIN and S-ROBIN dataset.

A.4 PLACEMENT OF FURNITURE

Furnitures or decors are inseparable part of a floor plan. To make the floor plans of ROBIN realistic, decor symbols as proposed by Delalandre *et al.* [2010a] were utilized. As shown in Fig. A.1, there are 12 different floor plan symbols. In each floor plan a subset of the entire symbol set was chosen and placed inside every room to give a definition to the room. As the furnitures are indicators of the purpose of the rooms, therefore, while designing, sufficient care has been taken to ensure that a house contains a i) kitchen, ii) bedroom, iv) washroom and a v) lobby. Moreover,

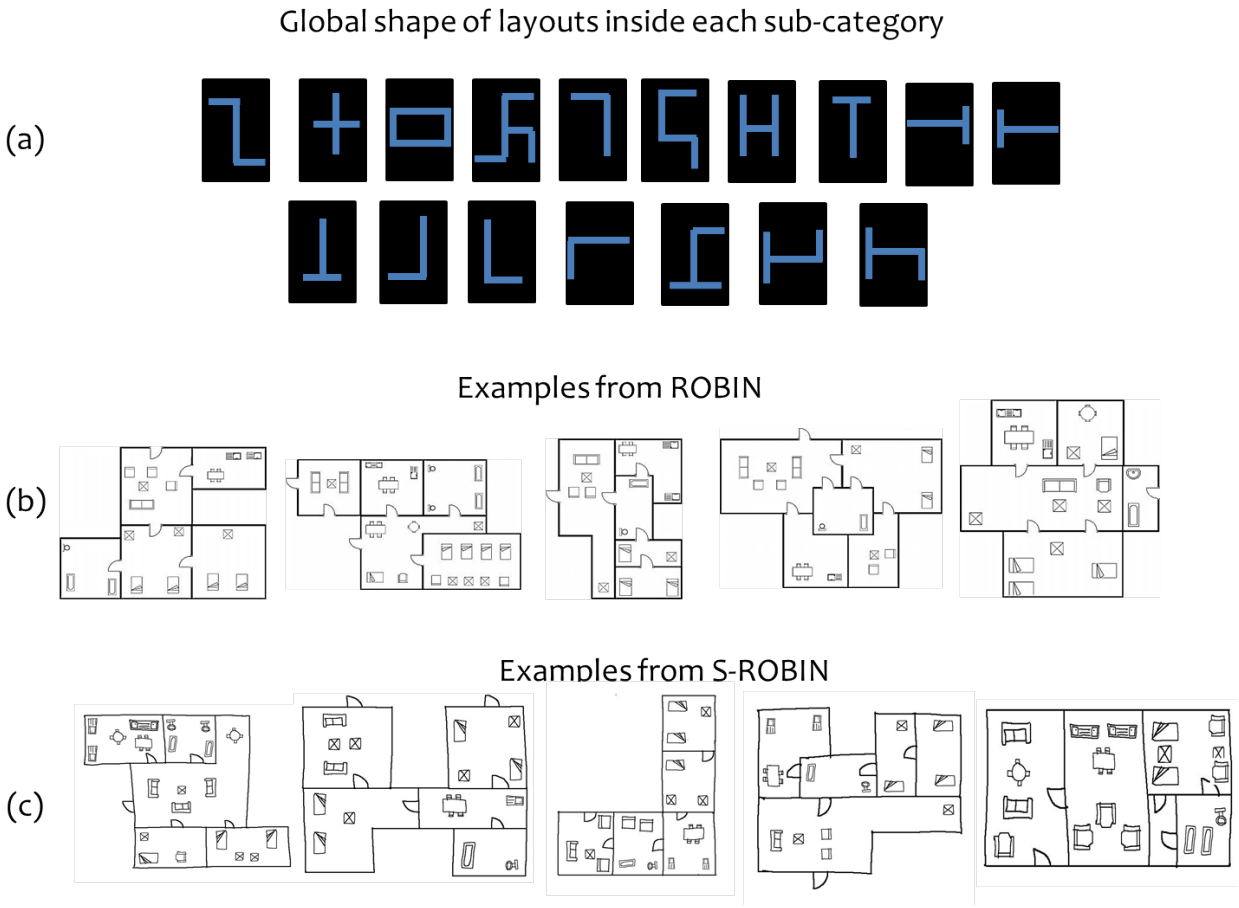

Figure A.4. : 5-Room layouts in the ROBIN and S-ROBIN dataset.

special consideration was given to the fact that the furniture is not placed in an arbitrary manner. For example, furniture should not be placed in a way that it blocks the door. Also, sufficient area should be left vacant after the furniture placement for navigation purposes.

A.5 DATASET CREATION PROCESS

ROBIN dataset was created on similar lines as the SESYD dataset [Delalandre *et al.*, 2010a]. Basic room structure of the floor plan was built using rectangular shape in Microsoft powerpoint. Thickness of each wall was kept uniform to be around 40 pixels. The decor symbols were inserted inside the rooms from the symbol library and door symbols were carefully placed on the room walls. On the other hand, to capture hand drawn sketched floor plans in S-ROBIN, volunteers were asked to draw the floor plans on a digital platform using a Wacom tablet. Wacom tablet is an interactive tool, where the user can draw on a board like tablet using a pen stylus conveniently and the resultant floor plan sketch is captured on the Microsoft paint software by adjusting the thickness of the strokes of the pen stylus. Since, offline mode is being used, the hand drawn floor plans are stored as an image and used for experimentation.

A.6 STATISTICS OF THE DATASET

Both the datasets contain 510 floor plans divided into 3 categories of layouts each with 17 sub categories. The unique characteristic of the datasets is that they are designed keeping in mind the need of a potential buyer. Every prospective buyer wants to have certain amenities and functionalities in their house. Thus, in the dataset, there are three broad categories, which are different from each other in terms of the number and type of rooms present in a floor plan. The broad categories are layouts with 3-rooms, 4-rooms and 5-rooms each which differ in the global shape of layout across their sub-categories. Each broad category is further classified into 17 sub-categories depending upon the global layout shape of the floor plan. Standard notations as mentioned by Delalandre *et al.* [2007], are followed for all the floor plans so that the datasets can be used for other floor plan analysis tasks as well, and not only for retrieval. The dataset has 12 different categories of furnitures inside each layout. These datasets help in better visualization of the floor plans and aid in efficient capturing of various high-level features while fine-grained retrieval. Illustrations of the various sub-categories in the floor plans and how their shapes are modeled are shown in Fig. A.2, Fig. A.3 and Fig. A.4. The size of the floor plans in ROBIN vary from 825×708 to 2916×1845 whereas the resolution in S-ROBIN dataset ranges from 591×517 pixels to 1483×884.

A.7 CONCLUSION

The rationale behind the creation of ROBIN and S-ROBIN dataset was to create a floor plan repository that can be used for research purposes by the Document Analysis and Retrieval (DAR) community. Both ROBIN and S-ROBIN dataset are sufficient for carrying out tasks like, retrieval, semantic segmentation, door to door navigation, etc. The entire dataset along with the annotations can be downloaded from [S-ROBIN, 2018].